MATHEMATICAL PHYSICS COMPENDIUM

A Treatise on the Analytical Dynamics of Particles and Rigid Bodies

Based on the Classic Work by

E. T. Whittaker

F.R.S., Late Professor of Mathematics in the University of Edinburgh

Explicit Scalar Formulations & Problem Invariants

Featuring complete coordinate-based dynamics, pure algebraic derivations, and Victorian Tripos conversions without matrix or tensor notation.

SOLUBLE PROBLEMS IN RIGID DYNAMICS

Chapter VI and Subsequent Problem Sets

October 03,2026

Editorial and Mathematical Conventions

This compendium establishes a rigorous, coordinate-explicit formulation of the classical dynamics problems found in E. T. Whittaker's *Treatise on the Analytical Dynamics of Particles and Rigid Bodies*.

Coordinate-Only Framework (Non-Matrix Formulation)

In accordance with classical scalar analytical mechanics:

- **Inertial Parameters:** Specified strictly through scalar principal moments of inertia (A, B, C) about the principal axes of inertia, or scalar products of inertia (D, E, F) relative to Cartesian frames fixed in the rigid body:

$$I = Al^2 + Bm^2 + Cn^2 - 2Dmn - 2Enl - 2Flm.$$

- **Kinematics:** Direction cosines (l, m, n) satisfy the fundamental identity $l^2 + m^2 + n^2 = 1$. Angular velocities about body-fixed axes are resolved as scalar components (p, q, r), governed by Euler's kinematic equations in terms of Eulerian angles (θ, ϕ, ψ).

- **Dynamical Equations:** Evaluated exclusively as scalar differential equations via Euler's dynamic equations:

$$\begin{aligned} A\frac{dp}{dt} - (B - C)qr &= L, \\ B\frac{dq}{dt} - (C - A)rp &= M, \\ C\frac{dr}{dt} - (A - B)pq &= N, \end{aligned}$$

  where (L, M, N) denote the scalar couples about the instantaneous principal axes.

- **Constraint Surfaces:** Invariants such as the Poinsot momental ellipsoid, herpolhode cylinder, invariable plane, and kinetic energy manifolds are treated algebraically via pure scalar coordinate geometry.

Contents

Editorial and Mathematical Conventions **1**

1 Chapter VI: Soluble Problems in Rigid Dynamics **3**

Chapter 1

Chapter VI: Soluble Problems in Rigid Dynamics

Whittaker Chapter VI — Problem 1

Problem Statement

A uniform solid sphere of radius b rolls without slipping down the inside of a fixed cycloidal track whose parametric equations are

$$x = a(\theta + \sin\theta), \qquad y = a(1 - \cos\theta).$$

The sphere starts from rest with its centre lying on the horizontal line $y = 2a$. Prove that the velocity V of its centre when at the lowest point of the track is given by

$$V^2 = \frac{10}{7} g(2a - b).$$

1. Geometric Setup & Coordinate Definitions

Consider the vertical Cartesian plane Oxy, where the positive x-axis is horizontal and the positive y-axis points vertically upward against gravity g.

Geometry of the Cycloidal Track. The profile of the fixed curve is given parametrically by

$$x(\theta) = a(\theta + \sin\theta), \qquad y(\theta) = a(1 - \cos\theta).$$

Differentiating with respect to the parameter $\theta \in [-\pi, \pi]$:

$$\frac{dx}{d\theta} = a(1 + \cos\theta) = 2a\cos^2\left(\frac{\theta}{2}\right),$$
$$\frac{dy}{d\theta} = a\sin\theta = 2a\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right).$$

The differential arc length ds along the cycloid is

$$ds = \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta = 2a\cos\left(\frac{\theta}{2}\right) d\theta.$$

The inclination angle ψ of the unit tangent relative to the positive horizontal axis satisfies

$$\tan\psi = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin(\theta/2)}{\cos(\theta/2)} = \tan\left(\frac{\theta}{2}\right) \implies \psi = \frac{\theta}{2}.$$

The unit tangent components (t_x, t_y) and the inward (upward-pointing) unit normal components (n_x, n_y) at any point θ are

$$t_x = \cos\psi = \cos\left(\frac{\theta}{2}\right), \qquad t_y = \sin\psi = \sin\left(\frac{\theta}{2}\right),$$
$$n_x = -\sin\psi = -\sin\left(\frac{\theta}{2}\right), \qquad n_y = \cos\psi = \cos\left(\frac{\theta}{2}\right).$$

Coordinates of the Sphere's Centre of Mass. Let (x_c, y_c) denote the Cartesian coordinates of the centre of mass of the sphere of radius b. Because the sphere remains in contact with the concave side of the curve:

$$x_c = x(\theta) + b\,n_x = a(\theta + \sin\theta) - b\sin\left(\frac{\theta}{2}\right),$$
$$y_c = y(\theta) + b\,n_y = a(1 - \cos\theta) + b\cos\left(\frac{\theta}{2}\right).$$

At the vertex (the lowest point of the cycloidal track), $\theta = 0$:

$$x_c(0) = a(0 + 0) - b(0) = 0,$$
$$y_c(0) = a(1 - 1) + b(1) = b.$$

Scalar Moment of Inertia. For a uniform solid sphere of mass m and radius b, the scalar moment of inertia about any diametral axis through its centre of mass is

$$I = \frac{2}{5}mb^2.$$

2. Lagrangian / Dynamical Derivation

Kinematics and the No-Slip Rolling Condition. Let ω denote the scalar angular velocity of the sphere about the axis perpendicular to the plane of motion. The instantaneous material velocity of the contact point P is expressed in scalar components as

$$v_{Px} = \dot{x}_c + \omega b\cos\psi,$$
$$v_{Py} = \dot{y}_c + \omega b\sin\psi.$$

Because the sphere rolls without slipping on a stationary track, the instantaneous velocity of the contact point vanishes identically ($v_{Px} = 0$, $v_{Py} = 0$). Consequently:

$$\dot{x}_c = -\omega b\cos\psi, \qquad \dot{y}_c = -\omega b\sin\psi.$$

The scalar translational speed V of the centre of mass satisfies

$$V = \sqrt{\dot{x}_c^2 + \dot{y}_c^2} = b\,|\omega| \implies |\omega| = \frac{V}{b}.$$

Workless Constraints & Conservation of Energy. The external reaction forces acting at the instantaneous point of contact P consist of the normal force N and the tangential friction force F. The instantaneous power developed by the contact reactions is

$$\mathcal{P} = F_x v_{Px} + F_y v_{Py} = 0.$$

Because the contact constraints are entirely workless and gravity is conservative, the total mechanical energy $E = T + U$ is an invariant of the motion:

$$\frac{dE}{dt} = 0.$$

Kinetic Energy Expansion. The total kinetic energy consists of translational and rotational components:

$$\begin{aligned} T &= T_{\text{trans}} + T_{\text{rot}} \\ &= \frac{1}{2}mV^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mV^2 + \frac{1}{2}\left(\frac{2}{5}mb^2\right)\left(\frac{V}{b}\right)^2 \\ &= \frac{1}{2}mV^2 + \frac{1}{5}mV^2 \\ &= \frac{7}{10}mV^2. \end{aligned}$$

Gravitational Potential Energy. Taking the horizontal datum $y = 0$ as the reference plane for gravitational potential energy:

$$U(y_c) = mgy_c.$$

3. Proof & Final Scalar Form

Evaluating the initial mechanical energy at release from rest ($V_0 = 0$) with the centre of the sphere at height $y_c = 2a$:

$$E_{\text{initial}} = T_0 + U_{\text{initial}} = 0 + mg(2a) = 2mga.$$

Evaluating the final mechanical energy at the lowest point, where $y_c = b$ and the velocity of the centre is V:

$$E_{\text{final}} = T + U_{\text{final}} = \frac{7}{10}mV^2 + mgb.$$

Equating the initial and final mechanical energy via the conservation theorem:

$$\begin{aligned} \frac{7}{10}mV^2 + mgb &= 2mga \\ \frac{7}{10}mV^2 &= mg(2a - b) \\ V^2 &= \frac{10}{7}g(2a - b). \end{aligned}$$

This completes the analytical proof in scalar coordinate form. ■

Whittaker Chapter VI — Problem 2

Problem Statement

Prove that for a planar disc free to turn about a horizontal axis perpendicular to its plane, the locus on the disc of the centres of suspension for which the simple equivalent pendulum has a given length L consists of two concentric circles.
Furthermore, if A and B are two points, one on each circle, and L' is the length of the simple equivalent pendulum when the centre of suspension is chosen at the midpoint of AB, prove that the radius of gyration k of the disc about its centre of inertia satisfies

$$k^2L'^2 = \left(\frac{1}{2}L^2 - c^2\right)\left(L'^2 - \frac{1}{2}L^2 + c^2\right),$$

where $2c$ is the distance between A and B.

1. Geometric Setup & Coordinate Definitions

Let the plane of the disc be defined by Cartesian coordinates (x, y) fixed in the body, with the origin placed at the centre of inertia (centre of mass) G, such that $(x_G, y_G) = (0, 0)$.

Inertial Parameters. Let M be the total mass of the disc. The scalar moment of inertia of the disc about a perpendicular axis passing through its centre of inertia G is

$$I_G = Mk^2,$$

where k is the radius of gyration about G.

Suspension Geometry. Let O with coordinates (x, y) be an arbitrary point on the disc serving as the centre of suspension. The rotation axis is horizontal and perpendicular to the xy-plane. The distance between the centre of suspension O and the centre of inertia G is

$$h = \sqrt{x^2 + y^2}.$$

By the parallel axis theorem, the scalar moment of inertia I_O about the horizontal axis of suspension through O is

$$I_O = I_G + Mh^2 = M(k^2 + h^2).$$

2. Lagrangian / Dynamical Derivation

Equation of Motion. Let θ denote the angular deflection of the line OG from the downward vertical. The kinetic energy T and the gravitational potential energy U (taking the horizontal datum at O) are

$$T = \frac{1}{2} I_O \dot{\theta}^2 = \frac{1}{2} M(k^2 + h^2)\dot{\theta}^2, \qquad U = -Mgh\cos\theta.$$

The Euler-Lagrange equation for the single generalized coordinate θ is

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}}\right) - \frac{\partial T}{\partial \theta} = -\frac{\partial U}{\partial \theta},$$

which yields the exact scalar equation of motion:

$$M(k^2 + h^2)\ddot{\theta} + Mgh\sin\theta = 0 \implies \ddot{\theta} + \frac{gh}{k^2 + h^2}\sin\theta = 0.$$

For small oscillations ($\sin\theta \approx \theta$), this reduces to simple harmonic motion with period

$$\mathcal{T} = 2\pi\sqrt{\frac{k^2 + h^2}{gh}}.$$

Simple Equivalent Pendulum Length. The length L of the simple equivalent pendulum having period $\mathcal{T} = 2\pi\sqrt{L/g}$ is

$$L = \frac{k^2 + h^2}{h}.$$

Clearing denominators yields a quadratic equation in h:

$$h^2 - Lh + k^2 = 0.$$

For real physical solutions, the length must satisfy $L \geq 2k$. The two distinct positive roots are

$$h_1 = \frac{L - \sqrt{L^2 - 4k^2}}{2}, \qquad h_2 = \frac{L + \sqrt{L^2 - 4k^2}}{2}.$$

By Vieta's formulas:

$$h_1 + h_2 = L,$$
$$h_1 h_2 = k^2.$$

Because $h = \sqrt{x^2 + y^2}$, the locus of the points of suspension on the disc consists of the two concentric circles:

$$x^2 + y^2 = h_1^2 \qquad \text{and} \qquad x^2 + y^2 = h_2^2.$$

3. Proof & Final Scalar Form

Let A be a point on the circle of radius h_1, and B be a point on the circle of radius h_2:

$$GA = h_1, \qquad GB = h_2.$$

Let O' be the midpoint of the segment AB, and let $2c$ denote the length of AB, so that

$$AO' = O'B = c.$$

Let $h' = GO'$ be the distance from the centre of inertia G to the midpoint O'.

Application of Apollonius' Theorem. In the triangle formed by G, A, and B, GO' is the median from vertex G to the base AB. By Apollonius' theorem:

$$GA^2 + GB^2 = 2\left(GO'^2 + AO'^2\right).$$

Substituting the scalar lengths:

$$h_1^2 + h_2^2 = 2\left(h'^2 + c^2\right).$$

Using the identities $h_1 + h_2 = L$ and $h_1 h_2 = k^2$:

$$h_1^2 + h_2^2 = (h_1 + h_2)^2 - 2h_1 h_2 = L^2 - 2k^2.$$

Equating the two expressions:

$$2\left(h'^2 + c^2\right) = L^2 - 2k^2$$
$$2h'^2 + 2c^2 = L^2 - 2k^2$$
$$h'^2 = \frac{1}{2}L^2 - k^2 - c^2.$$

Pendulum Length for Suspension at O'. When the disc is suspended from the midpoint O', the distance to G is h'. The length L' of the simple equivalent pendulum is

$$L' = \frac{k^2 + h'^2}{h'}.$$

Multiplying by h':

$$h'L' = k^2 + h'^2.$$

Substitute the expression for h'^2:

$$k^2 + h'^2 = k^2 + \left(\frac{1}{2}L^2 - k^2 - c^2\right)$$
$$= \frac{1}{2}L^2 - c^2.$$

Therefore:

$$h'L' = \frac{1}{2}L^2 - c^2.$$

Squaring both sides:

$$h'^2L'^2 = \left(\frac{1}{2}L^2 - c^2\right)^2.$$

Substitute $h'^2 = \frac{1}{2}L^2 - c^2 - k^2$:

$$\left(\frac{1}{2}L^2 - c^2 - k^2\right)L'^2 = \left(\frac{1}{2}L^2 - c^2\right)^2$$
$$\left(\frac{1}{2}L^2 - c^2\right)L'^2 - k^2L'^2 = \left(\frac{1}{2}L^2 - c^2\right)^2.$$

Isolating $k^2L'^2$:

$$k^2L'^2 = \left(\frac{1}{2}L^2 - c^2\right)L'^2 - \left(\frac{1}{2}L^2 - c^2\right)^2$$
$$k^2L'^2 = \left(\frac{1}{2}L^2 - c^2\right)\left[L'^2 - \left(\frac{1}{2}L^2 - c^2\right)\right]$$
$$k^2L'^2 = \left(\frac{1}{2}L^2 - c^2\right)\left(L'^2 - \frac{1}{2}L^2 + c^2\right).$$

This completes the analytical proof. ■

Whittaker Chapter VI — Problem 3

Problem Statement

A heavy rigid body can turn freely about a fixed horizontal axis. If a fixed point O in the body is given through which the horizontal axis must pass, determine the orientation of the axis in the body such that the simple equivalent pendulum has a prescribed length L. Show that the axes which satisfy this condition form the generators of a quartic cone whose vertex is at O.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point in the body through which the axis of suspension passes. Define a body-fixed rectangular Cartesian coordinate system $Oxyz$ with origin at O.

Inertial Parameters at Origin O. Let M be the total mass of the body. Relative to the axes $Oxyz$, the moments and products of inertia are defined by the scalar integrals:

$$A = \int (y^2 + z^2)\, dm, \qquad D = \int yz\, dm,$$
$$B = \int (z^2 + x^2)\, dm, \qquad E = \int zx\, dm,$$
$$C = \int (x^2 + y^2)\, dm, \qquad F = \int xy\, dm.$$

Let the coordinates of the centre of gravity G of the body relative to O be $(\bar{x}, \bar{y}, \bar{z})$.

Direction Cosines of the Axis. Let (l, m, n) be the direction cosines of the horizontal rotation axis passing through O relative to the body-fixed axes $Oxyz$. Because they are direction cosines:

$$l^2 + m^2 + n^2 = 1.$$

The scalar moment of inertia I of the body about this axis is given by the quadratic form:

$$I = Al^2 + Bm^2 + Cn^2 - 2Dmn - 2Enl - 2Flm.$$

Distance from Centre of Gravity to Axis. Because the axis passes through O with unit direction cosines (l, m, n), the projection of the displacement vector from O to G along the axis is

$$d_{\parallel} = l\bar{x} + m\bar{y} + n\bar{z}.$$

The square of the distance from O to G is $OG^2 = \bar{x}^2 + \bar{y}^2 + \bar{z}^2$. The perpendicular distance h from the centre of gravity G to the axis of rotation satisfies

$$h^2 = OG^2 - d_{\parallel}^2 = (\bar{x}^2 + \bar{y}^2 + \bar{z}^2)(l^2 + m^2 + n^2) - (l\bar{x} + m\bar{y} + n\bar{z})^2.$$

By Lagrange's identity in scalar algebra:

$$h^2 = (m\bar{z} - n\bar{y})^2 + (n\bar{x} - l\bar{z})^2 + (l\bar{y} - m\bar{x})^2.$$

2. Lagrangian / Dynamical Derivation

Dynamics of Compound Motion. Because the axis of rotation is constrained to remain horizontal in space, the downward gravitational acceleration vector g is perpendicular to the rotation axis at all times.

Let θ denote the angular displacement of the normal from the axis to G measured from the downward vertical. The scalar equation of motion is

$$I\ddot{\theta} = -Mgh\sin\theta.$$

For small vibrations around the stable equilibrium configuration:

$$I\ddot{\theta} + Mgh\theta = 0.$$

The period of small oscillations is $\mathcal{T} = 2\pi\sqrt{I/(Mgh)}$.

Condition for Prescribed Equivalent Length. The length L of the simple equivalent pendulum is defined by

$$L = \frac{I}{Mh}.$$

Rewriting this scalar relation:

$$MLh = I.$$

Squaring both sides eliminates square roots:

$$M^2L^2h^2 = I^2.$$

Substituting the explicit scalar expressions for I and h^2:

$$M^2L^2\Big[(m\bar{z} - n\bar{y})^2 + (n\bar{x} - l\bar{z})^2 + (l\bar{y} - m\bar{x})^2\Big] = \left(Al^2 + Bm^2 + Cn^2 - 2Dmn - 2Enl - 2Flm\right)^2.$$

3. Proof & Final Scalar Form

To determine the surface in space traced by these axes, let (x, y, z) be the Cartesian coordinates of any arbitrary point lying along a permissible rotation axis passing through O.

The direction cosines (l, m, n) are related to the coordinates (x, y, z) by

$$l = \frac{x}{\sqrt{x^2+y^2+z^2}}, \qquad m = \frac{y}{\sqrt{x^2+y^2+z^2}}, \qquad n = \frac{z}{\sqrt{x^2+y^2+z^2}}.$$

Substituting these into the squared dynamical condition:

$$M^2L^2\left[\frac{(y\bar{z} - z\bar{y})^2 + (z\bar{x} - x\bar{z})^2 + (x\bar{y} - y\bar{x})^2}{x^2+y^2+z^2}\right] = \left[\frac{Ax^2 + By^2 + Cz^2 - 2Dyz - 2Ezx - 2Fxy}{x^2+y^2+z^2}\right]^2.$$

Multiplying both sides by the denominator $(x^2+y^2+z^2)^2$:

$$M^2L^2(x^2+y^2+z^2)\left[(y\bar{z} - z\bar{y})^2 + (z\bar{x} - x\bar{z})^2 + (x\bar{y} - y\bar{x})^2\right] = \left(Ax^2 + By^2 + Cz^2 - 2Dyz - 2Ezx - 2Fxy\right)^2.$$

Geometric Degree Analysis. Examining the polynomial degree of each term:

1. On the left-hand side, the factor $(x^2 + y^2 + z^2)$ is homogeneous of degree 2, and the bracketed factor $(y\bar{z} - z\bar{y})^2 + (z\bar{x} - x\bar{z})^2 + (x\bar{y} - y\bar{x})^2$ is homogeneous of degree 2. Their product is homogeneous of degree $2 + 2 = 4$.
2. On the right-hand side, the quadratic moment of inertia form $(Ax^2 + By^2 + Cz^2 - 2Dyz - 2Ezx - 2Fxy)$ is homogeneous of degree 2. Its square is homogeneous of degree $2 \times 2 = 4$.

Every term in the expanded equation is therefore a homogeneous polynomial of the fourth degree in (x, y, z).

In three-dimensional analytical geometry, any homogeneous equation $\Phi(x, y, z) = 0$ of degree n represents a conical surface with its vertex at the coordinate origin. Since the degree is $n = 4$, the locus of the axes passing through O for which the simple equivalent pendulum has length L is a cone of the fourth degree.

The allowable axes are precisely the rectilinear **generators of a quartic cone**. ■

Whittaker Chapter VI — Problem 4

Problem Statement

A uniform smooth cube of edge $2a$ and mass M rests symmetrically on two shelves each of breadth b and mass m attached to walls at a distance $2c$ apart. Show that, if one of the shelves gives way and begins to turn about the edge where it is attached to the wall, the initial angular acceleration of the cube is

$$\frac{Mg(c-a)^2(c-b) + \frac{1}{2}mgb(c-a)(c-b+a)}{M(c-a)^2\{k^2 + (c-b)^2\} + I(c-b+a)^2},$$

where Mk^2 and I are respectively the moments of inertia of the cube about its centre and of the shelf about its edge.

1. Geometric Setup & Coordinate Definitions

Consider a vertical Cartesian plane with horizontal axis x and vertical axis y.

Shelf and Wall Geometry. The two vertical walls are fixed at $x = -c$ and $x = c$.

- The left shelf is hinged along the wall at $(-c, 0)$ and extends horizontally inward to $x = -(c - b)$. It remains fixed horizontally.
- The right shelf is hinged along the wall at $(c, 0)$ and extends horizontally inward to $x = c - b$. Its hinge is at $O_2(c, 0)$.

Cube Geometry and Initial Contacts. The cube has mass M, edge $2a$, and rests symmetrically between the walls. Initially:

- Its centre of mass G is at $(0, a)$.
- Its bottom horizontal face lies along $y = 0$, spanning $x \in [-a, a]$.
- The left inner edge of the fixed shelf, $P_1(-(c - b), 0)$, supports the bottom face of the cube.
- The lower-right corner of the cube, $C_2(a, 0)$, rests on the upper face of the right shelf. The distance from the hinge $O_2(c, 0)$ to this contact point is $c - a$.
- The horizontal distance from P_1 to C_2 is $a - (-(c - b)) = c - b + a$.

Kinematic Coordinates. Let θ denote the clockwise angle of tilt of the cube, and ϕ the downward angle of rotation of the falling shelf. Initially, $\theta = 0$, $\phi = 0$, and all velocities vanish.

2. Dynamical Derivation

Kinematics of the Initial Acceleration. Because all contact surfaces are smooth, the reaction force R_1 exerted by the corner P_1 is normal to the bottom face of the cube (initially vertical), and the reaction force R_2 between the falling shelf and corner C_2 is normal to the shelf (initially vertical).

Since gravity is vertical and no horizontal forces act on the system at $t = 0$:

$$\ddot{x}_G = 0.$$

The bottom face of the cube maintains contact with the fixed corner $P_1(-(c-b), 0)$. For a small rotation θ, the vertical position of the point on the base directly above P_1 is

$$y_{P_1} = y_G - a + (c-b)\theta = 0.$$

Differentiating twice gives the vertical acceleration of the centre of mass:

$$\ddot{y}_G = -(c-b)\ddot{\theta}.$$

The downward acceleration of the lower-right corner C_2 (located at horizontal offset $+a$ from G) is

$$a_{C_2} = -\ddot{y}_G + a\ddot{\theta} = (c-b)\ddot{\theta} + a\ddot{\theta} = (c-b+a)\ddot{\theta}.$$

The downward acceleration of the shelf at contact point C_2, which is at a distance $(c-a)$ from the hinge O_2, is

$$a_{\text{shelf}} = (c-a)\ddot{\phi}.$$

Since contact is maintained ($a_{C_2} = a_{\text{shelf}}$):

$$(c-b+a)\ddot{\theta} = (c-a)\ddot{\phi} \implies \ddot{\phi} = \frac{c-b+a}{c-a}\ddot{\theta}.$$

Equations of Motion for the Shelf. The falling shelf has mass m, length b, moment of inertia I about its hinge, and its centre of gravity is at distance $b/2$ from the wall. The downward force exerted by the corner of the cube on the shelf is R_2 at distance $(c-a)$ from the hinge. Taking moments about the hinge:

$$I\ddot{\phi} = mg\left(\frac{b}{2}\right) + R_2(c-a).$$

Solving for the reaction R_2:

$$R_2 = \frac{I\ddot{\phi} - \frac{1}{2}mgb}{c-a} = \frac{I(c-b+a)}{(c-a)^2}\ddot{\theta} - \frac{\frac{1}{2}mgb}{c-a}.$$

Equations of Motion for the Cube. The vertical equation of motion for the cube is

$$M\ddot{y}_G = R_1 + R_2 - Mg \implies R_1 = Mg - R_2 - M(c-b)\ddot{\theta}.$$

Taking moments about the centre of mass G for the clockwise angular acceleration $\ddot{\theta}$:

$$Mk^2\ddot{\theta} = R_1(c-b) - R_2 a.$$

Substituting R_1:

$$\begin{aligned} Mk^2\ddot{\theta} &= \left[Mg - R_2 - M(c-b)\ddot{\theta}\right](c-b) - R_2 a \\ &= Mg(c-b) - M(c-b)^2\ddot{\theta} - R_2(c-b+a). \end{aligned}$$

Collecting the inertial terms for the cube:

$$M\left[k^2 + (c-b)^2\right]\ddot{\theta} = Mg(c-b) - R_2(c-b+a).$$

3. Proof & Final Scalar Form

Substitute the scalar expression for R_2 into the moment equation of the cube:

$$M\left[k^2+(c-b)^2\right]\ddot{\theta} = Mg(c-b) - (c-b+a)\left[\frac{I(c-b+a)}{(c-a)^2}\ddot{\theta} - \frac{\frac{1}{2}mgb}{c-a}\right].$$

Multiplying the entire equation by $(c-a)^2$:

$$M(c-a)^2\left[k^2+(c-b)^2\right]\ddot{\theta} = Mg(c-a)^2(c-b) - I(c-b+a)^2\ddot{\theta} + \frac{1}{2}mgb(c-a)(c-b+a).$$

Grouping all terms containing $\ddot{\theta}$ on the left-hand side:

$$\left\{M(c-a)^2\left[k^2+(c-b)^2\right] + I(c-b+a)^2\right\}\ddot{\theta} = Mg(c-a)^2(c-b) + \frac{1}{2}mgb(c-a)(c-b+a).$$

Dividing by the bracketed coefficient yields the initial angular acceleration:

$$\ddot{\theta} = \frac{Mg(c-a)^2(c-b) + \frac{1}{2}mgb(c-a)(c-b+a)}{M(c-a)^2\{k^2+(c-b)^2\} + I(c-b+a)^2}.$$

■

Whittaker Chapter VI — Problem 5

Problem Statement

A homogeneous rod of mass M and length $2a$ moves on a smooth horizontal plane, one end being constrained to slide without friction in a fixed straight line. The rod is initially perpendicular to the line, and is struck at the free end by a blow I parallel to the line. Show that after time t the perpendicular distance y of the middle point of the rod from the line is given by the equation

$$\int_{y/a}^{1} (1 - \tfrac{3}{4}x^2)^{\frac{1}{2}} (1 - x^2)^{-\frac{1}{2}} dx = \frac{3It}{2Ma}.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed straight line be the Cartesian x-axis on the horizontal plane.

Coordinates. Let the constrained end A have coordinates $(x_A, 0)$. Let θ be the angle between the rod and the x-axis.

- The middle point G of the rod has coordinates:

$$x_G = x_A + a\cos\theta, \qquad y_G = a\sin\theta.$$

- The free end B has coordinates:

$$x_B = x_A + 2a\cos\theta, \qquad y_B = 2a\sin\theta.$$

Initially, $\theta = \pi/2$, so the perpendicular distance of the middle point from the line is $y = a$.

Scalar Moment of Inertia. The radius of gyration of the uniform rod about its centre of mass is given by $k^2 = \frac{1}{3}a^2$.

2. Lagrangian / Dynamical Derivation

Kinetic Energy. Differentiating the position of G:

$$\dot{x}_G = \dot{x}_A - a\sin\theta\,\dot{\theta}, \qquad \dot{y}_G = a\cos\theta\,\dot{\theta}.$$

The kinetic energy T of the rod is

$$\begin{aligned} T &= \frac{1}{2}M\left(\dot{x}_G^2 + \dot{y}_G^2\right) + \frac{1}{2}Mk^2\dot{\theta}^2 \\ &= \frac{1}{2}M\left(\dot{x}_A - a\sin\theta\,\dot{\theta}\right)^2 + \frac{1}{2}Ma^2\cos^2\theta\,\dot{\theta}^2 + \frac{1}{6}Ma^2\dot{\theta}^2 \\ &= \frac{1}{2}M\dot{x}_A^2 - Ma\sin\theta\,\dot{x}_A\dot{\theta} + \frac{2}{3}Ma^2\dot{\theta}^2. \end{aligned}$$

Impulsive Initial Motion. At $t = 0$, $\theta = \pi/2$ ($\cos\theta = 0$, $\sin\theta = 1$). The blow of impulse I is applied at B along the positive x-direction:

$$\delta W_{\text{imp}} = I\,\delta x_B = I(\delta x_A - 2a\sin\theta\,\delta\theta) = I\,\delta x_A - 2aI\,\delta\theta.$$

The generalised impulses are $Q_{x_A} = I$ and $Q_\theta = -2aI$. Using the Lagrangian equations of impulsive motion from rest:

$$\frac{\partial T}{\partial \dot{x}_A} = M\dot{x}_A - Ma\dot{\theta} = I,$$
$$\frac{\partial T}{\partial \dot{\theta}} = -Ma\dot{x}_A + \frac{4}{3}Ma^2\dot{\theta} = -2aI.$$

Multiplying the first equation by a and adding to the second gives

$$\frac{1}{3}Ma^2\dot{\theta}_0 = -aI \implies \dot{\theta}_0 = -\frac{3I}{Ma}.$$

Substituting $\dot{\theta}_0$ into the first equation:

$$\dot{x}_{A0} = a\left(-\frac{3I}{Ma}\right) + \frac{I}{M} = -\frac{2I}{M}.$$

The total kinetic energy immediately after the impulse is

$$E = T_0 = \frac{1}{2}M\left(-\frac{2I}{M}\right)^2 - Ma\left(-\frac{2I}{M}\right)\left(-\frac{3I}{Ma}\right) + \frac{2}{3}Ma^2\left(-\frac{3I}{Ma}\right)^2 = \frac{2I^2}{M}.$$

Subsequent Motion. For $t > 0$, no forces act in the x-direction, so the coordinate x_A is cyclic:

$$\frac{\partial T}{\partial \dot{x}_A} = M(\dot{x}_A - a\sin\theta\,\dot{\theta}) = I \implies \dot{x}_A = \frac{I}{M} + a\sin\theta\,\dot{\theta}.$$

Substituting this expression for $\dot{x}_A$ into T:

$$\begin{aligned} T &= \frac{1}{2}M\left(\frac{I}{M}\right)^2 + \frac{1}{2}Ma^2\left(\frac{1}{3} + \cos^2\theta\right)\dot{\theta}^2 \\ &= \frac{I^2}{2M} + \frac{1}{2}Ma^2\left(\frac{4}{3} - \sin^2\theta\right)\dot{\theta}^2. \end{aligned}$$

By conservation of energy, $T = E = \frac{2I^2}{M}$:

$$\begin{aligned} \frac{1}{2}Ma^2\left(\frac{4}{3} - \sin^2\theta\right)\dot{\theta}^2 &= \frac{2I^2}{M} - \frac{I^2}{2M} = \frac{3I^2}{2M} \\ \frac{4}{3}\left(1 - \frac{3}{4}\sin^2\theta\right)\dot{\theta}^2 &= \frac{3I^2}{M^2a^2} \\ \left(1 - \frac{3}{4}\sin^2\theta\right)\dot{\theta}^2 &= \frac{9I^2}{4M^2a^2}. \end{aligned}$$

3. Proof & Final Scalar Form

Taking the square root (with $\dot{\theta} < 0$ as the angle decreases from $\pi/2$):

$$\left(1 - \frac{3}{4}\sin^2\theta\right)^{\frac{1}{2}}\left(-\frac{d\theta}{dt}\right) = \frac{3I}{2Ma}.$$

Let $x = \sin\theta = y/a$. Differentiating gives

$$dx = \cos\theta\,d\theta = \sqrt{1-x^2}\,d\theta \implies -d\theta = \frac{-dx}{\sqrt{1-x^2}}.$$

As the time runs from 0 to t, θ decreases from $\pi/2$ to $\arcsin(y/a)$, so x decreases from 1 to y/a. Integrating both sides:

$$\int_{y/a}^{1} \frac{(1-\frac{3}{4}x^2)^{\frac{1}{2}}}{\sqrt{1-x^2}}\,dx = \frac{3It}{2Ma},$$

which is identically

$$\int_{y/a}^{1} (1-\tfrac{3}{4}x^2)^{\frac{1}{2}}(1-x^2)^{-\frac{1}{2}}dx = \frac{3It}{2Ma}.$$

■

Whittaker Chapter VI — Problem 6

Problem Statement

Four equal uniform rods, each of length $2a$ and mass m, are smoothly jointed to form a rhombus $ABCD$. The joint A is fixed, whilst C is free to move on a smooth vertical rod through A. Initially C coincides with A and the whole system is rotating about the vertical with angular velocity ω. Prove that, if in the subsequent motion 2α is the least angle between the upper rods,

$$a\omega^2 \cos\alpha = 3g \sin^2\alpha.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed joint A be the origin, with the z-axis directed vertically downwards along the smooth fixed rod.

Coordinates. Let θ denote the inclination of each of the two upper rods (AB and AD) to the vertical downward axis. The total angle between the two upper rods is 2θ. Let ϕ denote the azimuthal angle of the rhombus about the vertical axis.

- **Upper rods (AB, AD):** The centres of mass are at distance a from A:

$$z_1 = a\cos\theta, \qquad r_1 = a\sin\theta.$$

- **Lower rods (BC, DC):** The joint B is at depth $2a\cos\theta$, and C is at depth $4a\cos\theta$. The centres of mass are:

$$z_2 = 3a\cos\theta, \qquad r_2 = a\sin\theta.$$

Initially, C coincides with A, meaning the rods are horizontal: $\theta = \pi/2$.

2. Lagrangian / Dynamical Derivation

Potential Energy. Taking the datum at A ($z = 0$), the gravitational potential energy of the four rods of mass m is

$$V = -mg(2z_1 + 2z_2) = -mg\,(2a\cos\theta + 6a\cos\theta) = -8mga\cos\theta.$$

Kinetic Energy. For each uniform rod of length $2a$, the moment of inertia about the axis through its centre of mass perpendicular to the rod is $\frac{1}{3}ma^2$.

For an upper rod:

$$T_1 = \frac{1}{2}m\left(\dot{z}_1^2 + \dot{r}_1^2 + r_1^2\dot{\phi}^2\right) + \frac{1}{6}ma^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right) = \frac{2}{3}ma^2\dot{\theta}^2 + \frac{2}{3}ma^2\sin^2\theta\,\dot{\phi}^2.$$

For a lower rod:

$$\begin{aligned} T_2 &= \frac{1}{2}m\left(\dot{z}_2^2 + \dot{r}_2^2 + r_2^2\dot{\phi}^2\right) + \frac{1}{6}ma^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right) \\ &= \frac{1}{2}m\left(9a^2\sin^2\theta\,\dot{\theta}^2 + a^2\cos^2\theta\,\dot{\theta}^2 + a^2\sin^2\theta\,\dot{\phi}^2\right) + \frac{1}{6}ma^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right) \\ &= \frac{2}{3}ma^2\left(1 + 6\sin^2\theta\right)\dot{\theta}^2 + \frac{2}{3}ma^2\sin^2\theta\,\dot{\phi}^2. \end{aligned}$$

Summing over all four rods (2 upper and 2 lower):

$$\begin{aligned} T &= 2T_1 + 2T_2 \\ &= \frac{4}{3}ma^2\dot{\theta}^2 + \frac{4}{3}ma^2\left(1 + 6\sin^2\theta\right)\dot{\theta}^2 + \frac{8}{3}ma^2\sin^2\theta\,\dot{\phi}^2 \\ &= \frac{8}{3}ma^2\left(1 + 3\sin^2\theta\right)\dot{\theta}^2 + \frac{8}{3}ma^2\sin^2\theta\,\dot{\phi}^2. \end{aligned}$$

Integrals of Motion. Since ϕ is ignorable, the angular momentum about the vertical axis is conserved:

$$\frac{\partial T}{\partial \dot{\phi}} = \frac{16}{3}ma^2\sin^2\theta\,\dot{\phi} = \text{constant}.$$

Using the initial condition at $\theta = \pi/2$ with $\dot{\phi} = \omega$:

$$\frac{16}{3}ma^2\sin^2\theta\,\dot{\phi} = \frac{16}{3}ma^2\omega \implies \dot{\phi} = \frac{\omega}{\sin^2\theta}.$$

The total mechanical energy $E = T + V$ is also conserved. Initially ($\theta = \pi/2$, $\dot{\theta} = 0$, $\dot{\phi} = \omega$):

$$E = \frac{8}{3}ma^2\omega^2 - 8mga\cos\left(\frac{\pi}{2}\right) = \frac{8}{3}ma^2\omega^2.$$

At any subsequent configuration:

$$\frac{8}{3}ma^2\left(1 + 3\sin^2\theta\right)\dot{\theta}^2 + \frac{8}{3}ma^2\sin^2\theta\left(\frac{\omega}{\sin^2\theta}\right)^2 - 8mga\cos\theta = \frac{8}{3}ma^2\omega^2.$$

Dividing across by $\frac{8}{3}ma$:

$$a\left(1 + 3\sin^2\theta\right)\dot{\theta}^2 + \frac{a\omega^2}{\sin^2\theta} - 3g\cos\theta = a\omega^2.$$

3. Proof & Final Scalar Form

At the configuration of least angle between the upper rods, the angle attains its minimum value $2\theta = 2\alpha$, so $\theta = \alpha$ and $\dot{\theta} = 0$. Substituting these values into the energy equation gives

$$\frac{a\omega^2}{\sin^2\alpha} - 3g\cos\alpha = a\omega^2.$$

Rearranging terms:

$$\begin{aligned} a\omega^2\left(\frac{1}{\sin^2\alpha} - 1\right) &= 3g\cos\alpha \\ a\omega^2\left(\frac{1 - \sin^2\alpha}{\sin^2\alpha}\right) &= 3g\cos\alpha \\ a\omega^2\frac{\cos^2\alpha}{\sin^2\alpha} &= 3g\cos\alpha. \end{aligned}$$

Dividing both sides by $\cos\alpha$ (since $\alpha \neq \pi/2$):

$$a\omega^2\frac{\cos\alpha}{\sin^2\alpha} = 3g \implies a\omega^2\cos\alpha = 3g\sin^2\alpha.$$

■

Whittaker Chapter VI — Problem 7

Problem Statement

A particle of mass m is placed in a smooth uniform tube which can rotate in a vertical plane about its middle point. The system starts from rest when the tube is horizontal. If θ is the angle the tube makes with the vertical when its angular velocity is a maximum and equal to ω, prove that

$$4(mr^2 + Mk^2)\omega^4 - 8mgr\omega^2 \cos\theta + mg^2 \sin^2\theta = 0,$$

where Mk^2 is the moment of inertia of the tube about its centre and r is the distance of the particle from the centre of the tube.

1. Geometric Setup & Coordinate Definitions

Let the fixed midpoint of the tube be the origin O in the vertical plane.

Generalized Coordinates. Specify the configuration of the system at any instant t by two scalar polar coordinates:

- θ: the angle the tube makes with the downward vertical (initially $\theta_0 = \frac{\pi}{2}$, since the tube is horizontal).
- r: the distance of the particle of mass m from the midpoint O measured along the tube.

Inertial Parameters. The uniform tube has mass M, its centre of gravity remains fixed at O, and its scalar moment of inertia about O is Mk^2.

2. Lagrangian / Dynamical Derivation

Kinetic and Potential Energies. The velocity squared of the particle of mass m in polar coordinates (r, θ) is $\dot{r}^2 + r^2\dot{\theta}^2$. The total kinetic energy T of the tube and particle is

$$T = \frac{1}{2}Mk^2\dot{\theta}^2 + \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) = \frac{1}{2}\left(mr^2 + Mk^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{r}^2.$$

Taking the horizontal line through O as the zero datum for gravitational potential energy, the potential energy V of the system is

$$V = -mgr\cos\theta.$$

The Lagrangian $L = T - V$ is therefore

$$L = \frac{1}{2}\left(mr^2 + Mk^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{r}^2 + mgr\cos\theta.$$

Euler-Lagrange Equation for θ. Computing the derivatives with respect to θ and $\dot{\theta}$:

$$\frac{\partial L}{\partial \dot{\theta}} = \left(mr^2 + Mk^2\right)\dot{\theta}, \qquad \frac{\partial L}{\partial \theta} = -mgr\sin\theta.$$

The Euler-Lagrange equation $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$ gives

$$\left(mr^2 + Mk^2\right)\ddot{\theta} + 2mr\dot{r}\dot{\theta} + mgr\sin\theta = 0.$$

Condition for Maximum Angular Velocity. When the angular velocity $\dot{\theta}$ reaches its maximum value ω, its time derivative vanishes ($\ddot{\theta} = 0$). Setting $\ddot{\theta} = 0$ and $\dot{\theta} = \omega$ in the θ-equation of motion yields

$$2mr\dot{r}\omega + mgr\sin\theta = 0 \implies \dot{r} = -\frac{g\sin\theta}{2\omega}.$$

3. Proof & Final Scalar Form

Since the system starts from rest ($\dot{r} = 0, \dot{\theta} = 0$) when the tube is horizontal ($\theta = \frac{\pi}{2}$), the initial mechanical energy is $E = T_0 + V_0 = 0$. By conservation of energy, $T + V = 0$ throughout the motion:

$$\frac{1}{2}\left(mr^2 + Mk^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{r}^2 - mgr\cos\theta = 0.$$

At the instant of maximum angular velocity, substitute $\dot{\theta} = \omega$ and $\dot{r} = -\frac{g\sin\theta}{2\omega}$:

$$\frac{1}{2}\left(mr^2 + Mk^2\right)\omega^2 + \frac{1}{2}m\left(-\frac{g\sin\theta}{2\omega}\right)^2 - mgr\cos\theta = 0$$

$$\frac{1}{2}\left(mr^2 + Mk^2\right)\omega^2 + \frac{mg^2\sin^2\theta}{8\omega^2} - mgr\cos\theta = 0.$$

Multiplying the entire equation by $8\omega^2$ produces the required scalar relation:

$$4\left(mr^2 + Mk^2\right)\omega^4 - 8mgr\omega^2\cos\theta + mg^2\sin^2\theta = 0.$$

■

Whittaker Chapter VI — Problem 8

Problem Statement

Two homogeneous rough spheres of equal radii a and of masses m, m' rest on a smooth horizontal plane with m' at the highest point of m. If the system is disturbed, show that the inclination of their common normal θ to the vertical is given by the equation

$$a\dot{\theta}^2\left(7m + 5m'\sin^2\theta\right) = 5g(m+m')(1-\cos\theta).$$

1. Geometric Setup & Coordinate Definitions

Let the motion take place in a vertical Cartesian plane Oxy, where Ox lies along the smooth horizontal plane and Oy points vertically upward.

Coordinates and Rolling Constraint.

- The lower sphere of mass m and radius a remains in contact with the smooth horizontal plane, so its centre C_1 has coordinates (x_1, a).
- The upper sphere of mass m' and radius a stays in contact with the lower sphere. Let θ be the angle made by the line of centres C_1C_2 with the upward vertical. The centre C_2 has coordinates:
$$x_2 = x_1 + 2a\sin\theta, \qquad y_2 = a + 2a\cos\theta.$$
- Let ϕ and ϕ' denote the clockwise angles of rotation in space of the lower sphere m and upper sphere m', respectively. Relative to the rotating line of centres C_1C_2, the two spheres turn through angles $\phi - \theta$ and $\phi' - \theta$. Since the spheres roll without slipping on each other, the arc lengths rolled along their common contact point must be equal and opposite:
$$a(\phi-\theta) + a(\phi'-\theta) = 0 \implies \dot{\phi} + \dot{\phi}' = 2\dot{\theta} \implies \dot{\phi}' = 2\dot{\theta} - \dot{\phi}.$$

Moments of Inertia. The scalar moments of inertia of the two uniform spheres about their centres are

$$I = \frac{2}{5}ma^2, \qquad I' = \frac{2}{5}m'a^2.$$

2. Lagrangian / Dynamical Derivation

Lagrangian Construction. Differentiating the coordinates of C_2 with respect to t:

$$\dot{x}_2 = \dot{x}_1 + 2a\cos\theta\,\dot{\theta}, \qquad \dot{y}_2 = -2a\sin\theta\,\dot{\theta}.$$

Using $\dot{\phi}' = 2\dot{\theta} - \dot{\phi}$, the total kinetic energy T in terms of the generalized coordinates (x_1, ϕ, θ) is

$$\begin{aligned} T = {} & \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m'\left[\left(\dot{x}_1 + 2a\cos\theta\,\dot{\theta}\right)^2 + 4a^2\sin^2\theta\,\dot{\theta}^2\right] \\ & + \frac{1}{5}ma^2\dot{\phi}^2 + \frac{1}{5}m'a^2\left(2\dot{\theta} - \dot{\phi}\right)^2. \end{aligned}$$

The gravitational potential energy (measured from the initial configuration at $\theta = 0$) is

$$V = -2m'ga(1-\cos\theta).$$

Thus the Lagrangian is $L = T - V$.

Elimination of Cyclic Coordinates x_1 and ϕ. Since x_1 and ϕ do not appear explicitly in L, both coordinates are cyclic, and their conjugate momenta are conserved. Because the system starts from rest ($\dot{x}_1 = 0, \dot{\phi} = 0, \dot{\theta} = 0$):

1. From $\frac{\partial L}{\partial \dot{x}_1} = 0$:

$$m\dot{x}_1 + m'\left(\dot{x}_1 + 2a\cos\theta\,\dot{\theta}\right) = 0 \implies \dot{x}_1 = -\frac{2m'a\cos\theta}{m+m'}\dot{\theta}.$$

2. From $\frac{\partial L}{\partial \dot{\phi}} = 0$:

$$\frac{2}{5}ma^2\dot{\phi} - \frac{2}{5}m'a^2\left(2\dot{\theta} - \dot{\phi}\right) = 0 \implies \dot{\phi} = \frac{2m'}{m+m'}\dot{\theta}.$$

3. Proof & Final Scalar Form

Substitute $\dot{x}_1$ and $\dot{\phi}$ back into the kinetic energy T:

$$\begin{aligned} T &= \frac{1}{2}\frac{mm'}{m+m'}\left(4a^2\cos^2\theta\,\dot{\theta}^2\right) + 2m'a^2\sin^2\theta\,\dot{\theta}^2 + \frac{4}{5}\frac{mm'}{m+m'}a^2\dot{\theta}^2 \\ &= \frac{m'a^2\dot{\theta}^2}{5(m+m')}\left[10m\cos^2\theta + 10(m+m')\sin^2\theta + 4m\right] \\ &= \frac{m'a^2\dot{\theta}^2}{5(m+m')}\left[10m\left(\cos^2\theta + \sin^2\theta\right) + 4m + 10m'\sin^2\theta\right] \\ &= \frac{2m'a^2\dot{\theta}^2}{5(m+m')}\left(7m + 5m'\sin^2\theta\right). \end{aligned}$$

By conservation of mechanical energy, $T + V = 0$:

$$\frac{2m'a^2\dot{\theta}^2}{5(m+m')}\left(7m + 5m'\sin^2\theta\right) = 2m'ga(1-\cos\theta).$$

Dividing both sides by $\frac{2m'a}{5(m+m')}$ yields

$$a\dot{\theta}^2\left(7m + 5m'\sin^2\theta\right) = 5g(m+m')(1-\cos\theta).$$

■

Whittaker Chapter VI — Problem 9

Problem Statement

A uniform rod AB is of length $2a$ and is attached at one end to a light inextensible string of length c. The other end of this string is fixed at O to a point in a smooth horizontal plane on which the rod moves. Initially OAB is a straight line and the rod is projected without rotation with velocity V in the direction perpendicular to its length. Prove that the cosine of the greatest subsequent angle between the rod and string is $1 - a/6c$.

1. Geometric Setup & Coordinate Definitions

Let O be the origin of a Cartesian coordinate system Oxy in the smooth horizontal plane, with Ox lying along the initial straight line OAB.

Generalized Coordinates.

- Let θ be the angle the taut string OA (of length c) makes with Ox.
- Let ψ be the angle the rod AB (of length $2a$ and mass m) makes with the string OA produced, so that the angle made by the rod AB with Ox is $\phi = \theta + \psi$.

The coordinates of the centre of mass G of the rod AB are

$$x_G = c\cos\theta + a\cos(\theta+\psi), \qquad y_G = c\sin\theta + a\sin(\theta+\psi).$$

2. Lagrangian / Dynamical Derivation

Lagrangian and Kinetic Energy. Differentiating (x_G, y_G) with respect to time t:

$$\dot{x}_G = -c\sin\theta\,\dot{\theta} - a\sin(\theta+\psi)(\dot{\theta}+\dot{\psi}),$$
$$\dot{y}_G = c\cos\theta\,\dot{\theta} + a\cos(\theta+\psi)(\dot{\theta}+\dot{\psi}).$$

Squaring and adding gives the translational speed squared of G:

$$\dot{x}_G^2 + \dot{y}_G^2 = c^2\dot{\theta}^2 + a^2(\dot{\theta}+\dot{\psi})^2 + 2ac\cos\psi\,\dot{\theta}(\dot{\theta}+\dot{\psi}).$$

Since the moment of inertia of the uniform rod about G is $\frac{1}{3}ma^2$ and the angular velocity of the rod is $\dot{\theta}+\dot{\psi}$, the Lagrangian $L = T$ is

$$L = \frac{1}{2}m\left[c^2\dot{\theta}^2 + \frac{4}{3}a^2(\dot{\theta}+\dot{\psi})^2 + 2ac\cos\psi\,\dot{\theta}(\dot{\theta}+\dot{\psi})\right].$$

First Integrals of Motion. Initially, $\theta = 0$ and $\psi = 0$. The rod is projected without rotation ($\dot{\theta}_0 + \dot{\psi}_0 = 0$) with velocity V perpendicular to its length, so the end A also has transverse velocity V:

$$c\dot{\theta}_0 = V \implies \dot{\theta}_0 = \frac{V}{c}, \qquad \dot{\psi}_0 = -\frac{V}{c}.$$

1. **Conservation of Angular Momentum (θ is cyclic):**

$$\frac{\partial L}{\partial\dot{\theta}} = m\left[c^2\dot{\theta} + \frac{4}{3}a^2(\dot{\theta}+\dot{\psi}) + ac\cos\psi\,(2\dot{\theta}+\dot{\psi})\right] = \text{constant}.$$

Evaluating at $t = 0$ ($\psi = 0$, $\dot{\theta}_0 + \dot{\psi}_0 = 0$, $\dot{\theta}_0 = V/c$):

$$m\left[c^2\dot{\theta} + \frac{4}{3}a^2(\dot{\theta}+\dot{\psi}) + ac\cos\psi\,(2\dot{\theta}+\dot{\psi})\right] = m(c+a)V.$$

2. **Conservation of Energy:**

$$T = \frac{1}{2}m\left[c^2\dot{\theta}^2 + \frac{4}{3}a^2(\dot{\theta}+\dot{\psi})^2 + 2ac\cos\psi\,\dot{\theta}(\dot{\theta}+\dot{\psi})\right] = \frac{1}{2}mV^2.$$

3. Proof & Final Scalar Form

When the angle ψ between the rod and string reaches its greatest value, $\dot{\psi} = 0$. Setting $\dot{\psi} = 0$ in the angular momentum and energy integrals gives

$$\left(c^2 + 2ac\cos\psi + \frac{4}{3}a^2\right)\dot{\theta} = (c+a)V,$$
$$\left(c^2 + 2ac\cos\psi + \frac{4}{3}a^2\right)\dot{\theta}^2 = V^2.$$

Squaring the first equation and dividing by the second eliminates $\dot{\theta}$ and V:

$$c^2 + 2ac\cos\psi + \frac{4}{3}a^2 = (c+a)^2$$
$$c^2 + 2ac\cos\psi + \frac{4}{3}a^2 = c^2 + 2ac + a^2$$
$$2ac\cos\psi = 2ac - \frac{1}{3}a^2$$
$$\cos\psi = 1 - \frac{a}{6c}.$$

■

Whittaker Chapter VI — Problem 10

Problem Statement

To a fixed point are smoothly jointed two uniform rods of length $2a$, and upon them slides, by means of a smooth ring at each end, a third rod similar in all respects. Initially the three rods are in a horizontal line with the ends of the third rod at the middle points of the other two and, on the application of an impulse, the rods begin to rotate with angular velocity Ω in a horizontal plane. Show that the third rod will slide right off the other two unless

$$\Omega^2 > \frac{2g}{a\sqrt{3}}.$$

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point in space, and let the z-axis be directed vertically downward from O.

Geometry of the Three Rods. Let each of the three uniform rods have mass m and length $2a$.

- The two pivoted rods OA and OB are smoothly jointed at O and move symmetrically in a vertical plane that rotates about the vertical axis Oz with azimuthal angle ϕ. Let θ be the angle each of OA and OB makes with the downward vertical Oz.
- The third rod CD of length $2a$ remains horizontal by symmetry, with its rings C and D sliding along OA and OB at equal distances $r = OC = OD$ from O. From the right triangle of half-base a and hypotenuse r:

$$r\sin\theta = a \implies r = a\csc\theta.$$

- Initially, the rods lie in a horizontal line ($\theta = \frac{\pi}{2}$), so $r = a\csc\frac{\pi}{2} = a$ (the midpoints of OA and OB).
- The rings C and D reach the lower ends A and B of the two rods when $r = 2a$, which corresponds to

$$\sin\theta = \frac{a}{2a} = \frac{1}{2} \implies \theta = \frac{\pi}{6}, \qquad \cos\theta = \frac{\sqrt{3}}{2}, \qquad \cot\theta = \sqrt{3}.$$

2. Lagrangian / Dynamical Derivation

Kinetic Energy.

1. **Pivoted rods OA and OB:** Each rod has moment of inertia $\frac{4}{3}ma^2$ about O for changes in θ, and $\frac{4}{3}ma^2\sin^2\theta$ about the vertical axis Oz. For both rods combined:

$$T_{1,2} = \frac{4}{3}ma^2\left(\dot\theta^2 + \sin^2\theta\,\dot\phi^2\right).$$

2. **Sliding rod CD:** Its midpoint G lies on the vertical axis Oz at depth $z_G = r\cos\theta = a\cot\theta$, so its vertical velocity is $\dot z_G = -a\csc^2\theta\,\dot\theta$. Its moment of inertia about the vertical axis Oz through its midpoint is $\frac{1}{3}ma^2$. Thus:

$$T_3 = \frac{1}{2}ma^2\csc^4\theta\,\dot\theta^2 + \frac{1}{6}ma^2\dot\phi^2.$$

Summing these contributions gives the total kinetic energy:

$$T = \frac{1}{2}ma^2\left(\frac{8}{3} + \csc^4\theta\right)\dot\theta^2 + \frac{1}{2}ma^2\left(\frac{8}{3}\sin^2\theta + \frac{1}{3}\right)\dot\phi^2.$$

Potential Energy. The centres of mass of OA and OB are at depth $a\cos\theta$, and that of CD is at depth $a\cot\theta$. Taking the horizontal plane through O as datum:

$$V = -mg\left(2a\cos\theta + a\cot\theta\right) = -mga\left(2\cos\theta + \cot\theta\right).$$

Conservation Laws. Since ϕ is cyclic in $L = T - V$, angular momentum about the vertical axis Oz is conserved. Using the initial condition $\theta_0 = \frac{\pi}{2}$ and $\dot{\phi}_0 = \Omega$:

$$\frac{\partial L}{\partial \dot{\phi}} = ma^2\left(\frac{8}{3}\sin^2\theta + \frac{1}{3}\right)\dot{\phi} = ma^2\left(\frac{8}{3} + \frac{1}{3}\right)\Omega = 3ma^2\Omega \implies \dot{\phi} = \frac{9\Omega}{8\sin^2\theta + 1}.$$

At $t = 0$ ($\theta_0 = \frac{\pi}{2}, \dot{\theta}_0 = 0, \dot{\phi}_0 = \Omega$), the total energy is

$$E = T_0 + V_0 = \frac{3}{2}ma^2\Omega^2.$$

At any subsequent angle θ, conservation of energy $T + V = E$ gives

$$\frac{1}{2}ma^2\left(\frac{8}{3} + \csc^4\theta\right)\dot{\theta}^2 + \frac{27ma^2\Omega^2}{2(8\sin^2\theta + 1)} - mga(2\cos\theta + \cot\theta) = \frac{3}{2}ma^2\Omega^2.$$

3. Proof & Final Scalar Form

The third rod will slide right off the ends of the other two rods if $\dot{\theta}$ remains non-zero when θ decreases to $\theta = \frac{\pi}{6}$ ($r = 2a$). Conversely, the third rod will come to rest before reaching the ends only if the turning point ($\dot{\theta} = 0$) occurs for $\theta > \frac{\pi}{6}$.

Evaluating the energy equation at the boundary angle $\theta = \frac{\pi}{6}$ (where $\sin\theta = \frac{1}{2}$, $8\sin^2\theta + 1 = 3$, $\cos\theta = \frac{\sqrt{3}}{2}$, and $\cot\theta = \sqrt{3}$), the condition that the rod does not slide off is that the potential + azimuthal kinetic energy at $\theta = \frac{\pi}{6}$ exceeds the initial energy E:

$$\begin{aligned}
\frac{27ma^2\Omega^2}{2(3)} - mga\left(2\cdot\frac{\sqrt{3}}{2} + \sqrt{3}\right) &> \frac{3}{2}ma^2\Omega^2 \\
\frac{9}{2}ma^2\Omega^2 - 2\sqrt{3}\,mga &> \frac{3}{2}ma^2\Omega^2 \\
3ma^2\Omega^2 &> 2\sqrt{3}\,mga \\
\Omega^2 &> \frac{2g}{a\sqrt{3}}.
\end{aligned}$$

■

Whittaker Chapter VI — Problem 11

Problem Statement

A hollow thin cylinder of radius a and mass M is maintained at rest in a horizontal position on a rough plane whose inclination is α, and contains an insect of mass m at rest on the line of contact with the plane. The cylinder is released as the insect starts off with velocity V: if this relative velocity be maintained and the cylinder roll up hill, show that it will come to instantaneous rest when the radius through the insect makes an angle θ with the vertical given by

$$V^2\{1-\cos(\theta-\alpha)\}+ag(\cos\alpha-\cos\theta)=\left(1+\frac{M}{m}\right)ag(\theta-\alpha)\sin\alpha.$$

1. Geometric Setup & Coordinate Definitions

Consider the cross-sectional vertical plane perpendicular to the axis of the cylinder. Let Ox be directed uphill along the rough inclined plane (inclined at angle α to the horizontal), and let Oy be directed upward perpendicular to the inclined plane.

Coordinates.

- Let ψ be the angle through which the cylinder rotates as it rolls uphill without slipping, so its centre C has coordinates $x_C = a\psi$, $y_C = a$.
- Let θ be the angle that the radius CP from the centre C to the insect P makes with the downward vertical (measured in the uphill direction, so that at the line of contact I with the plane, $\theta = \alpha$).
- Relative to the contact point $I(a\psi, 0)$, the coordinates of the insect P along and perpendicular to the inclined plane are:

$$x_P - x_I = a\sin(\theta-\alpha), \qquad y_P - y_I = a\left[1-\cos(\theta-\alpha)\right].$$

2. Lagrangian / Dynamical Derivation

Kinetic Energy and Lagrangian Form. For the thin hollow cylinder of mass M and radius a, its moment of inertia about its axis is Ma^2, so its kinetic energy is

$$T_{\text{cyl}} = \frac{1}{2}M(a\dot\psi)^2 + \frac{1}{2}(Ma^2)\dot\psi^2 = Ma^2\dot\psi^2.$$

For the insect of mass m, its absolute coordinates are $x_P = a\psi + a\sin(\theta-\alpha)$ and $y_P = a - a\cos(\theta-\alpha)$, giving velocity components:

$$\dot x_P = a\dot\psi + a\dot\theta\cos(\theta-\alpha), \qquad \dot y_P = a\dot\theta\sin(\theta-\alpha).$$

Thus the kinetic energy of the insect is

$$T_{\text{insect}} = \frac{1}{2}ma^2\left[\dot\psi^2 + 2\dot\psi\dot\theta\cos(\theta-\alpha) + \dot\theta^2\right].$$

The total gravitational potential energy of the cylinder and insect (with datum at the initial position $\psi = 0, \theta = \alpha$) is

$$U = (M+m)ga\psi\sin\alpha + mga(\cos\alpha - \cos\theta).$$

Thus the total Lagrangian of the system is

$$L = Ma^2\dot\psi^2 + \frac{1}{2}ma^2\left[\dot\psi^2 + 2\dot\psi\dot\theta\cos(\theta-\alpha) + \dot\theta^2\right] - (M+m)ga\psi\sin\alpha - mga(\cos\alpha-\cos\theta).$$

Elimination of the Internal Tangential Force. Let S be the internal tangential force exerted by the insect along the inner wall of the cylinder to maintain its relative motion. In a virtual displacement $\delta\psi$ of the cylinder alone, work $-Sa\,\delta\psi$ is done, whereas in a virtual displacement $\delta\theta$ of the insect, work $+Sa\,\delta\theta$ is done. The Euler-Lagrange equations for ψ and θ are therefore

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\psi}}\right) - \frac{\partial L}{\partial \psi} = -Sa,$$
$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = +Sa.$$

Adding these two equations eliminates the internal force S:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\psi}} + \frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \psi} - \frac{\partial L}{\partial \theta} = 0.$$

Evaluating each derivative explicitly:

$$\frac{\partial L}{\partial \dot{\psi}} = (2M+m)a^2\dot{\psi} + ma^2\dot{\theta}\cos(\theta-\alpha),$$
$$\frac{\partial L}{\partial \dot{\theta}} = ma^2\dot{\theta} + ma^2\dot{\psi}\cos(\theta-\alpha),$$
$$\frac{\partial L}{\partial \psi} = -(M+m)ga\sin\alpha,$$
$$\frac{\partial L}{\partial \theta} = -ma^2\dot{\psi}\dot{\theta}\sin(\theta-\alpha) - mga\sin\theta.$$

Differentiating with respect to t and substituting into the sum gives

$$2Ma^2\ddot{\psi} + ma^2\left[1+\cos(\theta-\alpha)\right](\ddot{\psi}+\ddot{\theta}) - ma^2\dot{\theta}^2\sin(\theta-\alpha) + (M+m)ga\sin\alpha - mga\sin\theta = 0.$$

(Here ψ is measured uphill while θ increases uphill; with relative velocity maintained at V, $\ddot{\psi}+\ddot{\theta}=0$ and $a\dot{\theta}=V$, reducing the inertial term to $2Ma^2\ddot{\psi}$).

3. Proof & Final Scalar Form

Rearranging the dynamical equation with $a\dot{\theta}=V$ ($\dot{\theta}=V/a$):

$$2Ma^2\ddot{\psi} = mV^2\sin(\theta-\alpha) + mga\sin\theta - (M+m)ga\sin\alpha.$$

Multiply both sides by $d\theta = \frac{V}{a}dt$ and integrate from the initial state at $t=0$ (where $\theta=\alpha$ and $\dot{\psi}=0$) to the instant when the cylinder comes to instantaneous rest (where $\dot{\psi}=0$):

$$2MaV\int_0^t \ddot{\psi}\,dt = \int_\alpha^\theta \left[mV^2\sin(\theta-\alpha) + mga\sin\theta - (M+m)ga\sin\alpha\right]d\theta.$$

Since $\int_0^t \ddot{\psi}\,dt = \dot{\psi}(t) - \dot{\psi}(0) = 0 - 0 = 0$, the left-hand side vanishes:

$$0 = mV^2\Big[-\cos(\theta-\alpha)\Big]_\alpha^\theta + mga\Big[-\cos\theta\Big]_\alpha^\theta - (M+m)ga\sin\alpha\,\Big[\theta\Big]_\alpha^\theta$$
$$0 = mV^2\left\{1-\cos(\theta-\alpha)\right\} + mga(\cos\alpha-\cos\theta) - (M+m)ga(\theta-\alpha)\sin\alpha.$$

Dividing through by m yields the required equation:

$$V^2\left\{1-\cos(\theta-\alpha)\right\} + ag(\cos\alpha-\cos\theta) = \left(1+\frac{M}{m}\right)ag(\theta-\alpha)\sin\alpha.$$

■

Whittaker Chapter VI — Problem 12

Problem Statement

A uniform smooth plane tube can turn smoothly about a fixed axis of rotation lying in its plane and intersecting it: the moment of inertia of the tube about the axis is I. Initially the tube is rotating with angular velocity Ω about the axis, and a particle of mass m is projected with velocity V within the tube from the point of intersection of the tube with the axis. The system then moves under no external forces. Prove that, when the particle is at a distance r from the axis, the square of its velocity relative to the tube is

$$V^2 + \frac{Ir^2}{I + mr^2}\Omega^2.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed axis of rotation be the z-axis, and let O be the point of intersection of the tube with this axis.

Coordinates.

- Let θ be the angle turned through by the plane of the tube about the fixed axis of rotation.
- Let s be the arc length measured along the tube from O to the particle of mass m, so that $\dot{s}$ is the velocity of the particle relative to the tube.
- In the plane of the tube, let (r, z) be the perpendicular distance from the axis and the coordinate along the axis. Since $dr^2 + dz^2 = ds^2$ along the curve of the tube, the translational velocity squared of the particle in space is

$$v^2 = \dot{r}^2 + \dot{z}^2 + r^2\dot{\theta}^2 = \dot{s}^2 + r^2\dot{\theta}^2.$$

2. Lagrangian / Dynamical Derivation

Lagrangian Function. Since no external forces act on the system ($U = 0$), the Lagrangian L equals the total kinetic energy T of the tube and the particle:

$$L = T = \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}m\left(\dot{s}^2 + r^2\dot{\theta}^2\right) = \frac{1}{2}\left(I + mr^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{s}^2.$$

Integrals of Motion. Initially, at $r = 0$, we have $\dot{\theta}_0 = \Omega$ and $\dot{s}_0 = V$.

1. **Conservation of Angular Momentum (θ is cyclic):**

$$\frac{\partial L}{\partial \dot{\theta}} = \left(I + mr^2\right)\dot{\theta} = \text{constant}.$$

   Evaluating at $t = 0$ ($r = 0$, $\dot{\theta}_0 = \Omega$):

$$\left(I + mr^2\right)\dot{\theta} = I\Omega \implies \dot{\theta} = \frac{I\Omega}{I + mr^2}.$$

2. **Conservation of Energy:**

$$\frac{1}{2}\left(I + mr^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{s}^2 = \frac{1}{2}I\Omega^2 + \frac{1}{2}mV^2.$$

3. Proof & Final Scalar Form

Substitute $\dot{\theta} = \frac{I\Omega}{I+mr^2}$ into the energy equation and multiply by 2:

$$\begin{aligned}
\left(I + mr^2\right)\left(\frac{I\Omega}{I+mr^2}\right)^2 + m\dot{s}^2 &= I\Omega^2 + mV^2 \\
\frac{I^2\Omega^2}{I+mr^2} + m\dot{s}^2 &= I\Omega^2 + mV^2 \\
m\dot{s}^2 &= mV^2 + I\Omega^2\left(1 - \frac{I}{I+mr^2}\right) \\
m\dot{s}^2 &= mV^2 + \frac{Imr^2}{I+mr^2}\Omega^2.
\end{aligned}$$

Dividing through by m gives the square of the relative velocity $\dot{s}^2$:

$$\dot{s}^2 = V^2 + \frac{Ir^2}{I+mr^2}\Omega^2.$$

■

Whittaker Chapter VI — Problem 13

Problem Statement

A uniform straight rod of mass M is laid across two smooth horizontal pegs so that each of its ends projects beyond the corresponding peg. A second uniform rod of mass m and length $2l$ is fastened to the first at some point between the pegs by a universal joint. This rod is initially held horizontal and in contact with the first rod; and then let go, so as to oscillate in the vertical plane through the first rod. Prove that if θ be the angle which the second rod makes with the vertical at any instant, and x the distance through which the first rod has moved from rest,

$$(M+m)x + ml\sin\theta = ml,$$

and

$$\left(\frac{4}{3} - \frac{m}{m+M}\cos^2\theta\right) l\dot{\theta}^2 = 2g\cos\theta.$$

1. Geometric Setup & Coordinate Definitions

Consider the vertical Cartesian plane Oxy containing the two rods, with the x-axis horizontal along the line of the first rod and the y-axis directed vertically downwards.

Coordinates.

- Let x denote the horizontal distance through which the first rod of mass M has moved from its initial position of rest ($x_0 = 0$).
- Let θ denote the angle the second rod (of mass m and length $2l$) makes with the downward vertical (initially $\theta_0 = \frac{\pi}{2}$).
- The centre of mass G of the second rod is at distance l from the joint, so its coordinates are

$$x_G = x + l\sin\theta, \qquad y_G = l\cos\theta.$$

The scalar moment of inertia of the second rod about its centre of mass G is $\frac{1}{3}ml^2$.

2. Lagrangian / Dynamical Derivation

Kinetic and Potential Energies. Differentiating the coordinates of G gives

$$\dot{x}_G = \dot{x} + l\cos\theta\,\dot{\theta}, \qquad \dot{y}_G = -l\sin\theta\,\dot{\theta}.$$

The total kinetic energy T of the two rods is

$$\begin{aligned} T &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left[\left(\dot{x} + l\cos\theta\,\dot{\theta}\right)^2 + l^2\sin^2\theta\,\dot{\theta}^2\right] + \frac{1}{2}\left(\frac{1}{3}ml^2\right)\dot{\theta}^2 \\ &= \frac{1}{2}(M+m)\dot{x}^2 + ml\cos\theta\,\dot{x}\dot{\theta} + \frac{2}{3}ml^2\dot{\theta}^2. \end{aligned}$$

With the horizontal line of the first rod as the datum, the gravitational potential energy is $V = -mgl\cos\theta$, so the Lagrangian is

$$L = \frac{1}{2}(M+m)\dot{x}^2 + ml\cos\theta\,\dot{x}\dot{\theta} + \frac{2}{3}ml^2\dot{\theta}^2 + mgl\cos\theta.$$

Conservation of Horizontal Momentum. Since x is cyclic in L:

$$\frac{\partial L}{\partial \dot{x}} = (M+m)\dot{x} + ml\cos\theta\,\dot{\theta} = \text{constant}.$$

Since the system starts from rest ($\dot{x} = 0, \dot{\theta} = 0$), the constant is zero:

$$(M+m)\dot{x} + ml\cos\theta\,\dot{\theta} = 0.$$

Integrating with respect to time t, using the initial conditions $x = 0$ and $\theta = \frac{\pi}{2}$:

$$(M+m)x + ml\sin\theta = ml\sin\left(\frac{\pi}{2}\right) = ml.$$

3. Proof & Final Scalar Form

From the momentum relation, express $\dot{x}$ in terms of $\dot{\theta}$:

$$\dot{x} = -\frac{ml\cos\theta}{M+m}\dot{\theta}.$$

Substitute this into the kinetic energy T:

$$\begin{aligned} T &= \frac{1}{2}(M+m)\left(-\frac{ml\cos\theta}{M+m}\dot{\theta}\right)^2 + ml\cos\theta\left(-\frac{ml\cos\theta}{M+m}\dot{\theta}\right)\dot{\theta} + \frac{2}{3}ml^2\dot{\theta}^2 \\ &= \frac{1}{2}ml^2\left(\frac{4}{3} - \frac{m}{m+M}\cos^2\theta\right)\dot{\theta}^2. \end{aligned}$$

By conservation of mechanical energy ($T + V = 0$, since at $t = 0$, $\theta = \frac{\pi}{2}$ and velocities vanish):

$$\frac{1}{2}ml^2\left(\frac{4}{3} - \frac{m}{m+M}\cos^2\theta\right)\dot{\theta}^2 - mgl\cos\theta = 0.$$

Dividing through by $\frac{1}{2}ml$ yields

$$\left(\frac{4}{3} - \frac{m}{m+M}\cos^2\theta\right)l\dot{\theta}^2 = 2g\cos\theta.$$

■

Whittaker Chapter VI — Problem 14

Problem Statement

A plane body is free to rotate in its plane about a fixed point, and a second plane body is free to slide along a smooth straight groove in the first body, its motion being in the same plane; show that the relation between the relative advance x along the groove and the angle of rotation θ (no external forces being supposed to act on the system) is of the form

$$\left(\frac{dx}{d\theta}\right)^2 + P\frac{dx}{d\theta} + Q = 0,$$

where P and Q are respectively linear and quadratic functions of x^2.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point of rotation in the plane.

Coordinates and Inertial Parameters.

- Let θ be the angle of rotation of the first body in the plane, and let I_1 be its moment of inertia about O.
- As the second body (of mass m and moment of inertia mk^2 about its centre of mass G) slides along the straight groove, G describes a straight line fixed in the first body. Let p be the perpendicular distance from O to this line, and let x be the advance of G along this line measured from the foot of the perpendicular from O.
- Because the second body slides along the groove without rotating relative to the first body, its angular velocity in space is also $\dot{\theta}$.

2. Lagrangian / Dynamical Derivation

Lagrangian Construction. Relative to axes rotating with the first body at angular velocity $\dot{\theta}$ (one axis parallel to the groove and the other perpendicular to it), the coordinates of G are (x, p). The scalar velocity components of G parallel and perpendicular to the groove are

$$v_{\parallel} = \dot{x} - p\dot{\theta}, \qquad v_{\perp} = x\dot{\theta}.$$

Since no external forces act ($U = 0$), the Lagrangian L equals the total kinetic energy T:

$$\begin{aligned} L = T &= \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}m\left[\left(\dot{x} - p\dot{\theta}\right)^2 + x^2\dot{\theta}^2\right] + \frac{1}{2}mk^2\dot{\theta}^2 \\ &= \frac{1}{2}m\dot{x}^2 - mp\dot{x}\dot{\theta} + \frac{1}{2}\left(I_0 + mx^2\right)\dot{\theta}^2, \end{aligned}$$

where $I_0 = I_1 + m(p^2 + k^2)$ is a constant.

Integrals of Motion. Since θ is cyclic in L and the system is conservative:

1. **Angular Momentum Integral:**

$$\frac{\partial L}{\partial \dot{\theta}} = \left(I_0 + mx^2\right)\dot{\theta} - mp\dot{x} = h \quad \text{(constant)}.$$

2. **Energy Integral:**

$$\frac{1}{2}m\dot{x}^2 - mp\dot{x}\dot{\theta} + \frac{1}{2}\left(I_0 + mx^2\right)\dot{\theta}^2 = E \quad \text{(constant)}.$$

3. Proof & Final Scalar Form

Substitute $\dot{x} = \frac{dx}{d\theta}\dot{\theta}$ into both integrals:

$$\dot{\theta}\left[\left(I_0 + mx^2\right) - mp\frac{dx}{d\theta}\right] = h,$$
$$\dot{\theta}^2\left[\frac{1}{2}m\left(\frac{dx}{d\theta}\right)^2 - mp\frac{dx}{d\theta} + \frac{1}{2}\left(I_0 + mx^2\right)\right] = E.$$

Squaring the first equation and dividing by the second eliminates $\dot{\theta}$:

$$\frac{\left[\left(I_0 + mx^2\right) - mp\frac{dx}{d\theta}\right]^2}{\frac{1}{2}m\left(\frac{dx}{d\theta}\right)^2 - mp\frac{dx}{d\theta} + \frac{1}{2}\left(I_0 + mx^2\right)} = \frac{h^2}{E}.$$

Expanding and collecting powers of $\frac{dx}{d\theta}$:

$$\left(m^2p^2 - \frac{mh^2}{2E}\right)\left(\frac{dx}{d\theta}\right)^2 + mp\left[\frac{h^2}{E} - 2\left(I_0 + mx^2\right)\right]\frac{dx}{d\theta} + \left(I_0 + mx^2\right)^2 - \frac{h^2}{2E}\left(I_0 + mx^2\right) = 0.$$

Dividing through by the constant coefficient $\left(m^2p^2 - \frac{mh^2}{2E}\right)$ yields

$$\left(\frac{dx}{d\theta}\right)^2 + P\frac{dx}{d\theta} + Q = 0,$$

where

$$P = \frac{mp\left[\frac{h^2}{E} - 2\left(I_0 + mx^2\right)\right]}{m^2p^2 - \frac{mh^2}{2E}}, \qquad Q = \frac{\left(I_0 + mx^2\right)^2 - \frac{h^2}{2E}\left(I_0 + mx^2\right)}{m^2p^2 - \frac{mh^2}{2E}}$$

are respectively linear and quadratic functions of x^2. ■

Whittaker Chapter VI — Problem 15

Problem Statement

A pendulum is formed of a straight rod and a hollow circular bob, and fitting inside the bob is a smooth vertical lamina in the shape of a segment of a circle, the distances of the centre (C) of the bob from the point of suspension (O) and from the centre of gravity (G) of the lamina being l and c respectively. Prove that if M, m are the masses of the pendulum and lamina, k and k' their respective radii of gyration about O and G, θ and ϕ the angles which OC and CG make with the vertical, then twice the work done by gravity on the system during its motion from rest is equal to

$$\left(Mk^2 + ml^2\right)\dot{\theta}^2 + m\left(k'^2 + c^2\right)\dot{\phi}^2 + 2mcl\cos(\theta - \phi)\dot{\theta}\dot{\phi}.$$

1. Geometric Setup & Coordinate Definitions

Let the point of suspension O be the origin of Cartesian coordinates (x, y) in the vertical plane, with Ox directed vertically downwards and Oy horizontally.

Coordinates.

- The pendulum of mass M turns about O with angular coordinate θ. The centre C of the circular bob is at distance l from O:

$$x_C = l\cos\theta, \qquad y_C = l\sin\theta.$$

- The circular-segment lamina of mass m slides smoothly inside the circular bob, so its centre of curvature remains at C. Its centre of gravity G is at distance c from C, and CG makes an angle ϕ with the downward vertical:

$$x_G = l\cos\theta + c\cos\phi, \qquad y_G = l\sin\theta + c\sin\phi.$$

2. Lagrangian / Dynamical Derivation

Kinetic Energy of the System. The kinetic energy of the pendulum of mass M and radius of gyration k about O is

$$T_{\text{pend}} = \frac{1}{2}Mk^2\dot{\theta}^2.$$

Differentiating the coordinates of the centre of gravity G of the lamina with respect to t:

$$\dot{x}_G = -l\sin\theta\,\dot{\theta} - c\sin\phi\,\dot{\phi}, \qquad \dot{y}_G = l\cos\theta\,\dot{\theta} + c\cos\phi\,\dot{\phi}.$$

Squaring and adding gives the velocity squared of G:

$$\begin{aligned}\dot{x}_G^2 + \dot{y}_G^2 &= l^2\dot{\theta}^2 + c^2\dot{\phi}^2 + 2cl\left(\sin\theta\sin\phi + \cos\theta\cos\phi\right)\dot{\theta}\dot{\phi} \\ &= l^2\dot{\theta}^2 + c^2\dot{\phi}^2 + 2cl\cos(\theta - \phi)\dot{\theta}\dot{\phi}.\end{aligned}$$

Since the angular velocity of the lamina about G is $\dot{\phi}$ and its radius of gyration about G is k', the kinetic energy of the lamina is

$$T_{\text{lam}} = \frac{1}{2}m\left(\dot{x}_G^2 + \dot{y}_G^2\right) + \frac{1}{2}mk'^2\dot{\phi}^2 = \frac{1}{2}m\left[l^2\dot{\theta}^2 + \left(k'^2 + c^2\right)\dot{\phi}^2 + 2cl\cos(\theta - \phi)\dot{\theta}\dot{\phi}\right].$$

3. Proof & Final Scalar Form

Summing both contributions gives the total kinetic energy T of the system:

$$T = \frac{1}{2}\left(Mk^2 + ml^2\right)\dot{\theta}^2 + \frac{1}{2}m\left(k'^2 + c^2\right)\dot{\phi}^2 + mcl\cos(\theta - \phi)\dot{\theta}\dot{\phi}.$$

Since all contacts are smooth and the system starts from rest ($T_0 = 0$), by the conservation of energy ($T - T_0 = W$), the work W done by gravity equals T. Therefore, twice the work done by gravity is

$$2W = 2T = \left(Mk^2 + ml^2\right)\dot{\theta}^2 + m\left(k'^2 + c^2\right)\dot{\phi}^2 + 2mcl\cos(\theta - \phi)\dot{\theta}\dot{\phi}.$$

■

Whittaker Chapter VI — Problem 16

Problem Statement

A particle of mass m is attached to the end of a fine string which passes round the circumference of a wheel of mass M, the other end of the string being attached to a point in that circumference, a length l of the string being straight initially, and the wheel (radius a and radius of gyration k) being free to move about a fixed vertical axis through its centre; the particle, which lies on a smooth horizontal plane, is projected at right angles to the string, so that the string begins to wrap round the wheel; prove that, if the string eventually unwinds from the wheel, the shortest length of the straight portion is

$$\left(l^2 - a^2 - Mk^2/m\right)^{\frac{1}{2}}.$$

1. Geometric Setup & Coordinate Definitions

Let the centre of the wheel O be the origin of Cartesian coordinates (x, y) on the smooth horizontal plane.

Coordinates and Winding Constraint.

- Let θ be the angle through which the wheel has rotated from its initial position.
- Let ϕ be the angle made with the fixed x-axis by the radius OA to the point of tangency A where the straight portion of the string leaves the wheel.
- Since a relative rotation $\phi - \theta$ wraps a length $a(\phi - \theta)$ of string onto the wheel, the length r of the straight tangent portion AP of the string at any instant is

$$r = l - a(\phi - \theta) \implies \dot{r} = -a(\dot{\phi} - \dot{\theta}).$$

The Cartesian coordinates of the particle P are

$$x = a\cos\phi - r\sin\phi, \qquad y = a\sin\phi + r\cos\phi.$$

2. Lagrangian / Dynamical Derivation

Velocity of the Particle and Lagrangian. Differentiating (x, y) with respect to time t and using $\dot{r} + a\dot{\phi} = a\dot{\theta}$:

$$\begin{aligned}\dot{x} &= -(\dot{r} + a\dot{\phi})\sin\phi - r\dot{\phi}\cos\phi = -a\dot{\theta}\sin\phi - r\dot{\phi}\cos\phi,\\ \dot{y} &= (\dot{r} + a\dot{\phi})\cos\phi - r\dot{\phi}\sin\phi = a\dot{\theta}\cos\phi - r\dot{\phi}\sin\phi.\end{aligned}$$

Squaring and adding gives the speed squared of the particle:

$$\dot{x}^2 + \dot{y}^2 = a^2\dot{\theta}^2 + r^2\dot{\phi}^2.$$

Substituting $\dot{\phi} = \dot{\theta} - \frac{\dot{r}}{a}$, the Lagrangian $L = T$ in terms of the generalized coordinates (θ, r) is

$$L = T = \frac{1}{2}Mk^2\dot{\theta}^2 + \frac{1}{2}m\left[a^2\dot{\theta}^2 + r^2\left(\dot{\theta} - \frac{\dot{r}}{a}\right)^2\right].$$

Integrals of Motion. At $t = 0$, the wheel is at rest ($\dot{\theta}_0 = 0$), $r_0 = l$, and the particle is projected perpendicular to the string with speed V, so $l\dot{\phi}_0 = V$.

1. **Conservation of Angular Momentum (θ is cyclic):**

$$\frac{\partial L}{\partial \dot{\theta}} = \left(Mk^2 + ma^2\right)\dot{\theta} + mr^2\left(\dot{\theta} - \frac{\dot{r}}{a}\right) = \text{constant} = mlV.$$

2. **Conservation of Energy:**

$$\frac{1}{2}\left(Mk^2 + ma^2\right)\dot{\theta}^2 + \frac{1}{2}mr^2\left(\dot{\theta} - \frac{\dot{r}}{a}\right)^2 = \frac{1}{2}mV^2.$$

3. Proof & Final Scalar Form

At the instant when the straight portion of the string reaches its shortest length r and begins to unwind, $\dot{r} = 0$. Setting $\dot{r} = 0$ in the angular momentum and energy integrals gives

$$\left(Mk^2 + ma^2 + mr^2\right)\dot{\theta} = mlV,$$
$$\left(Mk^2 + ma^2 + mr^2\right)\dot{\theta}^2 = mV^2.$$

Squaring the first equation and dividing by the second eliminates $\dot{\theta}$ and V:

$$Mk^2 + ma^2 + mr^2 = ml^2$$
$$r^2 = l^2 - a^2 - \frac{Mk^2}{m}$$
$$r = \left(l^2 - a^2 - Mk^2/m\right)^{\frac{1}{2}}.$$

■

Whittaker Chapter VI — Problem 12

Problem Statement

A uniform smooth plane tube can turn smoothly about a fixed axis of rotation lying in its plane and intersecting it: the moment of inertia of the tube about the axis is I. Initially the tube is rotating with angular velocity Ω about the axis, and a particle of mass m is projected with velocity V within the tube from the point of intersection of the tube with the axis. The system then moves under no external forces. Prove that, when the particle is at a distance r from the axis, the square of its velocity relative to the tube is

$$V^2 + \frac{Ir^2}{I + mr^2}\Omega^2.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed axis of rotation be the z-axis, and let O be the fixed point where the tube intersects this axis[cite: 10, 11].

Coordinates.

- Let θ denote the azimuthal angle of the plane of the tube about the fixed z-axis.
- Let s denote the arc length measured along the tube from the intersection point O to the particle of mass m. The velocity of the particle relative to the tube is $\dot{s}$[cite: 10, 11].
- Let r denote the perpendicular distance of the particle from the axis of rotation[cite: 10, 11]. Since the tube is a fixed curve in the rotating plane, r is a known geometric function of the arc length s, with $dr/ds \leq 1$.

Velocity in Space. Because the plane of the tube rotates about the z-axis with angular speed $\dot{\theta}$, the particle has:

1. A velocity component $\dot{s}$ directed along the tangent to the tube in the plane of the tube.
2. A velocity component $r\dot{\theta}$ directed perpendicular to the plane of the tube.

These two directions are mutually perpendicular, so the square of the particle's speed in space is

$$v^2 = \dot{s}^2 + r^2\dot{\theta}^2.$$

2. Lagrangian / Dynamical Derivation

Lagrangian Construction. The moment of inertia of the tube about the axis of rotation is I[cite: 10, 11]. In the absence of external forces ($U = 0$), the Lagrangian of the system equals its total kinetic energy[cite: 10, 11]:

$$L = T = \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}m\left(\dot{s}^2 + r^2\dot{\theta}^2\right) = \frac{1}{2}\left(I + mr^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{s}^2.$$

First Integrals. At $t = 0$, the particle is at the origin ($r = 0$), projected with relative speed $\dot{s} = V$, and the tube has initial angular velocity $\dot{\theta} = \Omega$[cite: 10, 11].

1. **Angular Momentum (θ is cyclic):** Since θ does not appear in L, its conjugate momentum is conserved:
$$\frac{\partial L}{\partial \dot{\theta}} = \left(I + mr^2\right)\dot{\theta} = \text{constant}.$$
Evaluating at the initial instant ($r = 0, \dot{\theta} = \Omega$):
$$\left(I + mr^2\right)\dot{\theta} = I\Omega \implies \dot{\theta} = \frac{I\Omega}{I + mr^2}.$$
2. **Conservation of Energy:** Since no external forces act, the total kinetic energy is constant[cite: 10, 11]:
$$\frac{1}{2}\left(I + mr^2\right)\dot{\theta}^2 + \frac{1}{2}m\dot{s}^2 = \frac{1}{2}I\Omega^2 + \frac{1}{2}mV^2.$$

3. Proof & Final Scalar Form

Substitute the expression for $\dot{\theta}$ into the energy conservation equation:

$$\frac{1}{2}\left(I + mr^2\right)\left(\frac{I\Omega}{I + mr^2}\right)^2 + \frac{1}{2}m\dot{s}^2 = \frac{1}{2}I\Omega^2 + \frac{1}{2}mV^2.$$

Multiplying throughout by 2:

$$\frac{I^2\Omega^2}{I+mr^2} + m\dot{s}^2 = I\Omega^2 + mV^2.$$

Isolating $m\dot{s}^2$:

$$\begin{aligned} m\dot{s}^2 &= mV^2 + I\Omega^2 - \frac{I^2\Omega^2}{I+mr^2} \\ &= mV^2 + I\Omega^2\left(1 - \frac{I}{I+mr^2}\right) \\ &= mV^2 + I\Omega^2\left(\frac{mr^2}{I+mr^2}\right) \\ &= mV^2 + \frac{Imr^2}{I+mr^2}\Omega^2. \end{aligned}$$

Dividing both sides by the particle mass m yields the square of the velocity relative to the tube[cite: 10, 11]:

$$\dot{s}^2 = V^2 + \frac{Ir^2}{I+mr^2}\Omega^2.$$

■

Whittaker Chapter VI — Problem 13

Problem Statement

A uniform straight rod of mass M is laid across two smooth horizontal pegs so that each of its ends projects beyond the corresponding peg. A second uniform rod of mass m and length $2l$ is fastened to the first at some point between the pegs by a universal joint. This rod is initially held horizontal and in contact with the first rod; and then let go, so as to oscillate in the vertical plane through the first rod. Prove that if θ be the angle which the second rod makes with the vertical at any instant, and x the distance through which the first rod has moved from rest,

$$(M+m)x + ml\sin\theta = ml,$$

and

$$\left(\frac{4}{3} - \frac{m}{m+M}\cos^2\theta\right) l\dot{\theta}^2 = 2g\cos\theta.$$

1. Geometric Setup & Coordinate Definitions

Consider a vertical Cartesian plane with the horizontal axis along the straight line of the pegs and the vertical axis pointing downwards[cite: 8, 12].

Coordinates.

- Let x denote the horizontal displacement of the first rod of mass M from its initial position of rest ($x = 0$ at $t = 0$)[cite: 8, 12].
- Let θ denote the angle the oscillating second rod (of mass m and length $2l$) makes with the downward vertical[cite: 8, 12].
- Initially, the second rod is held horizontal in contact with the first rod, so $\theta_0 = \frac{\pi}{2}$[cite: 8, 12].
- The joint J has horizontal position x and vertical position $y = 0$. The centre of mass G of the second rod is at distance l from J, giving coordinates

$$x_G = x + l\sin\theta, \qquad y_G = l\cos\theta.$$

Inertial Parameters. The scalar moment of inertia of the second uniform rod about its centre of mass G is $I_G = \frac{1}{3}ml^2$.

2. Lagrangian / Dynamical Derivation

Kinetic Energy. Differentiating the position of G:

$$\dot{x}_G = \dot{x} + l\cos\theta\,\dot{\theta}, \qquad \dot{y}_G = -l\sin\theta\,\dot{\theta}.$$

The total kinetic energy T of the two rods is

$$\begin{aligned} T &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}_G^2 + \dot{y}_G^2\right) + \frac{1}{2}I_G\dot{\theta}^2 \\ &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left[\left(\dot{x} + l\cos\theta\,\dot{\theta}\right)^2 + l^2\sin^2\theta\,\dot{\theta}^2\right] + \frac{1}{6}ml^2\dot{\theta}^2 \\ &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}^2 + 2l\cos\theta\,\dot{x}\dot{\theta} + l^2\dot{\theta}^2\right) + \frac{1}{6}ml^2\dot{\theta}^2 \\ &= \frac{1}{2}(M+m)\dot{x}^2 + ml\cos\theta\,\dot{x}\dot{\theta} + \frac{2}{3}ml^2\dot{\theta}^2. \end{aligned}$$

Potential Energy and Lagrangian. Taking the horizontal line of the first rod as the reference datum, the gravitational potential energy is

$$V = -mgy_G = -mgl\cos\theta.$$

The Lagrangian $L = T - V$ is therefore

$$L = \frac{1}{2}(M+m)\dot{x}^2 + ml\cos\theta\,\dot{x}\dot{\theta} + \frac{2}{3}ml^2\dot{\theta}^2 + mgl\cos\theta.$$

Conservation of Horizontal Linear Momentum. Because x is cyclic in L:

$$\frac{\partial L}{\partial \dot{x}} = (M+m)\dot{x} + ml\cos\theta\,\dot{\theta} = \text{constant}.$$

Since the system is released from rest ($\dot{x} = 0, \dot{\theta} = 0$), this constant is zero[cite: 8, 12]:

$$(M+m)\dot{x} + ml\cos\theta\,\dot{\theta} = 0.$$

Integrating directly with respect to t:

$$(M+m)x + ml\sin\theta = C.$$

Using the initial condition at $t = 0$ ($x = 0$ and $\theta = \frac{\pi}{2}$)[cite: 8, 12]:

$$C = (M+m)(0) + ml\sin\left(\frac{\pi}{2}\right) = ml.$$

Hence, the first relation is established[cite: 8, 12]:

$$(M+m)x + ml\sin\theta = ml.$$

3. Proof & Final Scalar Form

From the momentum equation, express $\dot{x}$ explicitly in terms of $\dot{\theta}$:

$$\dot{x} = -\frac{ml\cos\theta}{M+m}\dot{\theta}.$$

Substitute this relation into the expression for T:

$$\begin{aligned} T &= \frac{1}{2}(M+m)\left(-\frac{ml\cos\theta}{M+m}\dot{\theta}\right)^2 + ml\cos\theta\left(-\frac{ml\cos\theta}{M+m}\dot{\theta}\right)\dot{\theta} + \frac{2}{3}ml^2\dot{\theta}^2 \\ &= \frac{1}{2}\frac{m^2l^2\cos^2\theta}{M+m}\dot{\theta}^2 - \frac{m^2l^2\cos^2\theta}{M+m}\dot{\theta}^2 + \frac{2}{3}ml^2\dot{\theta}^2 \\ &= \frac{2}{3}ml^2\dot{\theta}^2 - \frac{1}{2}\frac{m^2l^2\cos^2\theta}{M+m}\dot{\theta}^2 \\ &= \frac{1}{2}ml^2\left(\frac{4}{3} - \frac{m}{M+m}\cos^2\theta\right)\dot{\theta}^2. \end{aligned}$$

By conservation of mechanical energy, $T + V = E$. Initially, all velocities vanish and $\theta = \frac{\pi}{2}$, so $T_0 = 0$ and $V_0 = -mgl\cos(\frac{\pi}{2}) = 0$, giving $E = 0$[cite: 8, 12]. Thus:

$$\frac{1}{2}ml^2\left(\frac{4}{3} - \frac{m}{M+m}\cos^2\theta\right)\dot{\theta}^2 - mgl\cos\theta = 0.$$

Dividing both sides by $\frac{1}{2}ml$ gives the second relation[cite: 8, 12]:

$$\left(\frac{4}{3} - \frac{m}{m+M}\cos^2\theta\right)l\dot{\theta}^2 = 2g\cos\theta.$$

■

Whittaker Chapter VI — Problem 14

Problem Statement

A plane body is free to rotate in its plane about a fixed point, and a second plane body is free to slide along a smooth straight groove in the first body, its motion being in the same plane; show that the relation between the relative advance x along the groove and the angle of rotation θ (no external forces being supposed to act on the system) is of the form

$$\left(\frac{dx}{d\theta}\right)^2 + P\frac{dx}{d\theta} + Q = 0,$$

where P and Q are respectively linear and quadratic functions of x^2.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point of rotation in the plane[cite: 9, 13].

Coordinates.

- Let θ denote the rotation angle of the first body relative to a fixed direction in space[cite: 9, 13]. Let I_1 denote the moment of inertia of the first body about O.
- The second body (of mass m) is constrained to slide along a smooth straight groove fixed in the first body[cite: 9, 13]. Its centre of mass G traces a straight line in the first body.
- Let p be the perpendicular distance from the fixed point O to this line of motion.
- Let x denote the position of G along this straight line, measured from the foot of the perpendicular from O[cite: 9, 13].
- Because the second body translates along the groove without independent rotation relative to the first body, its angular velocity in space is also $\dot{\theta}$[cite: 9, 13]. Let mk^2 be its moment of inertia about G.

2. Lagrangian / Dynamical Derivation

Kinetic Energy. In a rotating Cartesian frame $O\xi\eta$ fixed in the first body (with $O\xi$ parallel to the groove and $O\eta$ perpendicular to it), the coordinates of G are (x, p).

The velocity components of G resolved along $O\xi$ and $O\eta$ are

$$v_\xi = \dot{x} - p\dot{\theta}, \qquad v_\eta = x\dot{\theta}.$$

The square of the speed of G is

$$v_G^2 = v_\xi^2 + v_\eta^2 = (\dot{x} - p\dot{\theta})^2 + x^2\dot{\theta}^2 = \dot{x}^2 - 2p\dot{x}\dot{\theta} + (x^2 + p^2)\dot{\theta}^2.$$

Since no external forces act ($U = 0$), the Lagrangian is the total kinetic energy[cite: 9, 13]:

$$\begin{aligned} L = T &= \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}m\left[\dot{x}^2 - 2p\dot{x}\dot{\theta} + (x^2 + p^2)\dot{\theta}^2\right] + \frac{1}{2}mk^2\dot{\theta}^2 \\ &= \frac{1}{2}m\dot{x}^2 - mp\dot{x}\dot{\theta} + \frac{1}{2}\left(I_0 + mx^2\right)\dot{\theta}^2, \end{aligned}$$

where $I_0 = I_1 + m(p^2 + k^2)$ is a constant.

Conservation Laws. Because θ does not appear in L and the system is conservative:

1. **Conservation of Angular Momentum:**

$$\frac{\partial L}{\partial \dot{\theta}} = \left(I_0 + mx^2\right)\dot{\theta} - mp\dot{x} = h \quad \text{(constant)}.$$

2. **Conservation of Energy:**

$$T = \frac{1}{2}m\dot{x}^2 - mp\dot{x}\dot{\theta} + \frac{1}{2}\left(I_0 + mx^2\right)\dot{\theta}^2 = E \quad \text{(constant)}.$$

3. Proof & Final Scalar Form

Express the velocities in terms of the relative advance derivative $x' = \frac{dx}{d\theta}$ using $\dot{x} = x'\dot{\theta}$[cite: 9, 13]:

$$\begin{aligned}\dot{\theta}\left[\left(I_0 + mx^2\right) - mpx'\right] &= h,\\ \dot{\theta}^2\left[mx'^2 - 2mpx' + \left(I_0 + mx^2\right)\right] &= 2E.\end{aligned}$$

Squaring the first equation and dividing by the second eliminates $\dot{\theta}$:

$$\frac{\left[\left(I_0 + mx^2\right) - mpx'\right]^2}{mx'^2 - 2mpx' + (I_0 + mx^2)} = \frac{h^2}{2E}.$$

Cross-multiplying and expanding both sides:

$$2E\left[m^2p^2x'^2 - 2mp(I_0 + mx^2)x' + (I_0 + mx^2)^2\right] = h^2\left[mx'^2 - 2mpx' + (I_0 + mx^2)\right].$$

Collecting powers of $x' = \frac{dx}{d\theta}$[cite: 9, 13]:

$$\begin{aligned}\left(2Em^2p^2 - mh^2\right)x'^2 - 2mp\left[2E(I_0 + mx^2) - h^2\right]x' \\ + \left[2E(I_0 + mx^2)^2 - h^2(I_0 + mx^2)\right] = 0.\end{aligned}$$

Dividing across by the constant coefficient $(2Em^2p^2 - mh^2)$ produces

$$\left(\frac{dx}{d\theta}\right)^2 + P\frac{dx}{d\theta} + Q = 0,$$

where the coefficients are given explicitly by

$$P = -\frac{2mp\left[2E(I_0 + mx^2) - h^2\right]}{2Em^2p^2 - mh^2},$$

and

$$Q = \frac{2E(I_0 + mx^2)^2 - h^2(I_0 + mx^2)}{2Em^2p^2 - mh^2}.$$

Inspection confirms that P is a linear polynomial in x^2, while Q is a quadratic polynomial in x^2[cite: 9, 13]. ■

Whittaker Chapter VI — Problem 15

Problem Statement

A pendulum is formed of a straight rod and a hollow circular bob, and fitting inside the bob is a smooth vertical lamina in the shape of a segment of a circle, the distances of the centre (C) of the bob from the point of suspension (O) and from the centre of gravity (G) of the lamina being l and c respectively. Prove that if M, m are the masses of the pendulum and lamina, k and k' their respective radii of gyration about O and G, θ and ϕ the angles which OC and CG make with the vertical, then twice the work done by gravity on the system during its motion from rest is equal to

$$\left(Mk^2 + ml^2\right)\dot{\theta}^2 + m\left(k'^2 + c^2\right)\dot{\phi}^2 + 2mcl\cos(\theta - \phi)\dot{\theta}\dot{\phi}.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed point of suspension O be the coordinate origin in the vertical plane, with the x-axis directed vertically downwards and the y-axis horizontal[cite: 9, 13].

Coordinates.

- The pendulum of mass M swings about O through angle θ[cite: 9, 13]. The centre C of the circular bob is at distance l from O[cite: 9, 13]:
$$x_C = l\cos\theta, \qquad y_C = l\sin\theta.$$
- The smooth circular lamina of mass m fits inside the circular bob, so its circular rim slides along the inner boundary of the bob; thus its centre of curvature remains coincident with C[cite: 9, 13].
- The centre of gravity G of the lamina is at distance c from C, and the line CG makes an angle ϕ with the downward vertical[cite: 9, 13]:
$$x_G = l\cos\theta + c\cos\phi, \qquad y_G = l\sin\theta + c\sin\phi.$$
- Because the lamina executes a pure rotation in space as its angle ϕ changes, its instantaneous angular velocity in space is $\dot{\phi}$.

2. Lagrangian / Dynamical Derivation

Kinetic Energy of the Pendulum. The pendulum has mass M and radius of gyration k about the suspension point O[cite: 9, 13]. Its kinetic energy is

$$T_{\text{pend}} = \frac{1}{2}Mk^2\dot{\theta}^2.$$

Kinetic Energy of the Lamina. Differentiating the position coordinates of G with respect to time t:

$$\dot{x}_G = -l\sin\theta\,\dot{\theta} - c\sin\phi\,\dot{\phi}, \qquad \dot{y}_G = l\cos\theta\,\dot{\theta} + c\cos\phi\,\dot{\phi}.$$

Squaring and adding gives the square of the translational velocity of G:

$$\begin{aligned}\dot{x}_G^2 + \dot{y}_G^2 &= (-l\sin\theta\,\dot{\theta} - c\sin\phi\,\dot{\phi})^2 + (l\cos\theta\,\dot{\theta} + c\cos\phi\,\dot{\phi})^2 \\ &= l^2\left(\sin^2\theta + \cos^2\theta\right)\dot{\theta}^2 + c^2\left(\sin^2\phi + \cos^2\phi\right)\dot{\phi}^2 \\ &\quad + 2cl\left(\sin\theta\sin\phi + \cos\theta\cos\phi\right)\dot{\theta}\dot{\phi} \\ &= l^2\dot{\theta}^2 + c^2\dot{\phi}^2 + 2cl\cos(\theta - \phi)\dot{\theta}\dot{\phi}.\end{aligned}$$

The lamina has mass m and radius of gyration k' about its centre of gravity G[cite: 9, 13]. By Koenig's theorem, its kinetic energy is

$$\begin{aligned} T_{\text{lam}} &= \frac{1}{2}m\left(\dot{x}_G^2 + \dot{y}_G^2\right) + \frac{1}{2}mk'^2\dot{\phi}^2 \\ &= \frac{1}{2}m\left[l^2\dot{\theta}^2 + c^2\dot{\phi}^2 + 2cl\cos(\theta-\phi)\dot{\theta}\dot{\phi}\right] + \frac{1}{2}mk'^2\dot{\phi}^2 \\ &= \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}m\left(k'^2 + c^2\right)\dot{\phi}^2 + mcl\cos(\theta-\phi)\dot{\theta}\dot{\phi}. \end{aligned}$$

3. Proof & Final Scalar Form

Adding the two components gives the total kinetic energy of the system:

$$T = T_{\text{pend}} + T_{\text{lam}} = \frac{1}{2}\left(Mk^2 + ml^2\right)\dot{\theta}^2 + \frac{1}{2}m\left(k'^2 + c^2\right)\dot{\phi}^2 + mcl\cos(\theta-\phi)\dot{\theta}\dot{\phi}.$$

Because all internal contacts and constraints are smooth, the constraint forces do no work[cite: 9, 13]. By the work-energy theorem, the change in kinetic energy from rest ($T_0 = 0$) equals the total work W performed by gravity[cite: 9, 13]:

$$W = T - T_0 = T.$$

Multiplying this relation by 2:

$$2W = 2T = \left(Mk^2 + ml^2\right)\dot{\theta}^2 + m\left(k'^2 + c^2\right)\dot{\phi}^2 + 2mcl\cos(\theta-\phi)\dot{\theta}\dot{\phi}.$$

■

Whittaker Chapter VI — Problem 16

Problem Statement

A particle of mass m is attached to the end of a fine string which passes round the circumference of a wheel of mass M, the other end of the string being attached to a point in that circumference, a length l of the string being straight initially, and the wheel (radius a and radius of gyration k) being free to move about a fixed vertical axis through its centre; the particle, which lies on a smooth horizontal plane, is projected at right angles to the string, so that the string begins to wrap round the wheel; prove that, if the string eventually unwinds from the wheel, the shortest length of the straight portion is

$$\left(l^2 - a^2 - Mk^2/m\right)^{\frac{1}{2}}.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed vertical axis of rotation pass through the centre O of the wheel, taken as the origin of a polar coordinate system (r, ϕ) in the horizontal plane[cite: 9, 13].

Coordinates.

- Let θ be the angle of rotation of the wheel from its initial position[cite: 9, 13].
- Let A be the point where the straight tangent portion of the string leaves the circumference of the wheel. Let ϕ be the angle that the radius OA makes with a fixed reference line.
- Let r denote the length of the straight tangent portion AP at any instant[cite: 9, 13].
- Initially, $\theta_0 = 0$ and $r_0 = l$[cite: 9, 13]. When the wheel turns through θ and the contact point advances by ϕ, the length of string wrapped onto the wheel is $a(\phi - \theta)$. By inextensibility[cite: 9, 13]:

$$r = l - a(\phi - \theta) \implies \dot{r} = -a(\dot{\phi} - \dot{\theta}) \implies \dot{\phi} = \dot{\theta} - \frac{\dot{r}}{a}.$$

Particle Velocity. The position of the particle P relative to O is given by resolving along the radius OA and along the perpendicular tangent AP:

$$x = a\cos\phi - r\sin\phi, \qquad y = a\sin\phi + r\cos\phi.$$

Differentiating with respect to t:

$$\dot{x} = -(a\dot{\phi} + \dot{r})\sin\phi - r\dot{\phi}\cos\phi,$$
$$\dot{y} = (a\dot{\phi} + \dot{r})\cos\phi - r\dot{\phi}\sin\phi.$$

Since $a\dot{\phi} + \dot{r} = a\dot{\theta}$, this simplifies to

$$\dot{x} = -a\dot{\theta}\sin\phi - r\dot{\phi}\cos\phi, \qquad \dot{y} = a\dot{\theta}\cos\phi - r\dot{\phi}\sin\phi.$$

The square of the speed of the particle is therefore

$$v_P^2 = \dot{x}^2 + \dot{y}^2 = a^2\dot{\theta}^2 + r^2\dot{\phi}^2.$$

2. Lagrangian / Dynamical Derivation

Lagrangian of the System. The moment of inertia of the wheel about the vertical axis is Mk^2[cite: 9, 13]. Since the plane is smooth and horizontal ($U = 0$), the Lagrangian is[cite: 9, 13]

$$L = T = \frac{1}{2}Mk^2\dot{\theta}^2 + \frac{1}{2}m\left(a^2\dot{\theta}^2 + r^2\dot{\phi}^2\right) = \frac{1}{2}\left(Mk^2 + ma^2\right)\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2.$$

Angular Momentum and Energy Conservation. The total angular momentum of the system about the fixed vertical axis O is

$$\begin{aligned} H &= Mk^2\dot{\theta} + m\left(x\dot{y} - y\dot{x}\right) \\ &= Mk^2\dot{\theta} + m\left(a^2\dot{\theta} + r^2\dot{\phi}\right) \\ &= \left(Mk^2 + ma^2\right)\dot{\theta} + mr^2\dot{\phi}. \end{aligned}$$

Because θ does not appear in L, H is strictly conserved.

At $t = 0$, the wheel is at rest ($\dot{\theta}_0 = 0$), $r = l$, and the particle is projected perpendicular to the string with velocity V, giving $l\dot{\phi}_0 = V$[cite: 9, 13]. Thus:

1. **Angular Momentum:**
$$\left(Mk^2 + ma^2\right)\dot{\theta} + mr^2\dot{\phi} = mlV.$$

2. **Energy:**
$$\frac{1}{2}\left(Mk^2 + ma^2\right)\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 = \frac{1}{2}mV^2.$$

3. Proof & Final Scalar Form

At the instant when the straight portion reaches its minimum length r and is about to unwind, $\dot{r} = 0$[cite: 9, 13].

From the kinematic relation $\dot{r} = -a(\dot{\phi} - \dot{\theta})$, the condition $\dot{r} = 0$ implies that the angular velocities of the string and wheel are equal:

$$\dot{\phi} = \dot{\theta} = \omega.$$

Substituting $\dot{\phi} = \dot{\theta} = \omega$ into the conservation equations gives

$$\begin{aligned} \left(Mk^2 + ma^2 + mr^2\right)\omega &= mlV, \\ \left(Mk^2 + ma^2 + mr^2\right)\omega^2 &= mV^2. \end{aligned}$$

Squaring the first equation and dividing by the second eliminates ω:

$$\begin{aligned} \frac{\left[\left(Mk^2 + ma^2 + mr^2\right)\omega\right]^2}{\left(Mk^2 + ma^2 + mr^2\right)\omega^2} &= \frac{(mlV)^2}{mV^2} \\ Mk^2 + ma^2 + mr^2 &= ml^2. \end{aligned}$$

Dividing across by m:

$$\frac{Mk^2}{m} + a^2 + r^2 = l^2.$$

Isolating r:

$$r^2 = l^2 - a^2 - \frac{Mk^2}{m} \implies r = \left(l^2 - a^2 - Mk^2/m\right)^{\frac{1}{2}}.$$

■

Whittaker Chapter VI — Problem 17

Problem Statement

A carriage is placed on an inclined plane making an angle α with the horizon and rolls straight down without any slipping between the wheels and the plane. The floor of the carriage is parallel to the plane and a perfectly rough ball is placed freely on it. Show that the acceleration of the carriage down the plane is

$$\frac{14M + 4M' + 14m}{14M + 4M' + 21m} g \sin\alpha,$$

where M is the mass of the carriage excluding the wheels, m the sum of the masses of the wheels, which are uniform discs, and M' that of the ball. The friction between the wheels and the axes is neglected.

1. Geometric Setup & Coordinate Definitions

Consider a vertical plane perpendicular to the inclined plane[cite: 14]. Let the Cartesian axis x be directed straight down the inclined plane, parallel to the line of greatest slope[cite: 14].

Kinematic Coordinates.

- Let x denote the displacement of the carriage down the plane from a fixed reference position[cite: 14].
- Let y denote the displacement of the centre of the rough ball down the carriage floor relative to the carriage, so that the absolute displacement of the ball's centre parallel to the plane is $x + y$[cite: 14].
- Let R denote the radius of the carriage wheels. Since the wheels roll without slipping on the fixed inclined plane, each wheel turns through an angle $\theta_w = x/R$, giving an angular speed $\omega_w = \dot{x}/R$[cite: 14].
- Let b denote the radius of the ball[cite: 14]. The floor of the carriage is parallel to the inclined plane and moves with speed $\dot{x}$[cite: 14]. Since the ball rolls without slipping on the floor, the speed of its point of contact with the floor is $\dot{x}$[cite: 14]. The velocity of its centre is $\dot{x} + \dot{y}$, which requires the angular velocity ω_b of the ball to satisfy $(\dot{x} + \dot{y}) - b\omega_b = \dot{x} \implies \omega_b = \dot{y}/b$[cite: 14].

Inertial Parameters.

- The carriage body has mass M and translates with speed $\dot{x}$[cite: 14].
- The wheels are uniform discs of total mass m[cite: 14]. The sum of their scalar moments of inertia about their respective axle centres is $\frac{1}{2}mR^2$[cite: 14].
- The ball is a uniform solid sphere of mass M' and radius b[cite: 14]. Its scalar moment of inertia about its centre of mass is $I_b = \frac{2}{5}M'b^2$.

2. Lagrangian / Dynamical Derivation

Kinetic Energy. The kinetic energy of the carriage body is

$$T_{\text{body}} = \frac{1}{2}M\dot{x}^2.$$

The kinetic energy of the wheels (translation plus rotation without slipping) is

$$T_{\text{wheels}} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\left(\frac{1}{2}mR^2\right)\left(\frac{\dot{x}}{R}\right)^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{4}m\dot{x}^2 = \frac{3}{4}m\dot{x}^2.$$

The kinetic energy of the rolling ball is

$$T_{\text{ball}} = \frac{1}{2}M'(\dot{x}+\dot{y})^2 + \frac{1}{2}\left(\frac{2}{5}M'b^2\right)\left(\frac{\dot{y}}{b}\right)^2 = \frac{1}{2}M'(\dot{x}+\dot{y})^2 + \frac{1}{5}M'\dot{y}^2.$$

Summing these contributions gives the total kinetic energy:

$$T = \frac{1}{2}\left(M + \frac{3}{2}m\right)\dot{x}^2 + \frac{1}{2}M'(\dot{x}+\dot{y})^2 + \frac{1}{5}M'\dot{y}^2.$$

Potential Energy. Taking the vertical position at $x = 0, y = 0$ as datum, the vertical depth fallen by the carriage (mass $M + m$) is $x\sin\alpha$, and by the ball (mass M') is $(x + y)\sin\alpha$[cite: 14]. The gravitational potential energy is therefore

$$V = -(M+m)gx\sin\alpha - M'g(x+y)\sin\alpha = -(M+m+M')gx\sin\alpha - M'gy\sin\alpha.$$

The Lagrangian of the system is $L = T - V$:

$$L = \frac{1}{2}\left(M + \frac{3}{2}m\right)\dot{x}^2 + \frac{1}{2}M'(\dot{x}+\dot{y})^2 + \frac{1}{5}M'\dot{y}^2 + (M+m+M')gx\sin\alpha + M'gy\sin\alpha.$$

Equations of Motion. For the relative coordinate y:

$$\frac{\partial L}{\partial \dot{y}} = M'(\dot{x}+\dot{y}) + \frac{2}{5}M'\dot{y} = M'\dot{x} + \frac{7}{5}M'\dot{y}, \qquad \frac{\partial L}{\partial y} = M'g\sin\alpha.$$

The Euler-Lagrange equation $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} = 0$ yields

$$M'\ddot{x} + \frac{7}{5}M'\ddot{y} = M'g\sin\alpha \implies \ddot{x} + \frac{7}{5}\ddot{y} = g\sin\alpha.$$

Isolating the relative acceleration $\ddot{y}$:

$$\ddot{y} = \frac{5}{7}\left(g\sin\alpha - \ddot{x}\right).$$

For the carriage coordinate x:

$$\frac{\partial L}{\partial \dot{x}} = \left(M + \frac{3}{2}m\right)\dot{x} + M'(\dot{x}+\dot{y}) = \left(M + \frac{3}{2}m + M'\right)\dot{x} + M'\dot{y},$$

$$\frac{\partial L}{\partial x} = (M+m+M')g\sin\alpha.$$

The Euler-Lagrange equation for x becomes

$$\left(M + \frac{3}{2}m + M'\right)\ddot{x} + M'\ddot{y} = (M+m+M')g\sin\alpha.$$

3. Proof & Final Scalar Form

Substitute the expression for $\ddot{y}$ into the carriage equation of motion:

$$\left(M + \frac{3}{2}m + M'\right)\ddot{x} + M'\left[\frac{5}{7}(g\sin\alpha - \ddot{x})\right] = (M + m + M')g\sin\alpha.$$

Expanding and grouping terms in $\ddot{x}$:

$$\left(M + \frac{3}{2}m + M' - \frac{5}{7}M'\right)\ddot{x} = \left(M + m + M' - \frac{5}{7}M'\right)g\sin\alpha,$$

which simplifies to

$$\left(M + \frac{3}{2}m + \frac{2}{7}M'\right)\ddot{x} = \left(M + m + \frac{2}{7}M'\right)g\sin\alpha.$$

Multiplying the entire equation by 14 eliminates the fractional denominators:

$$\left(14M + 21m + 4M'\right)\ddot{x} = \left(14M + 14m + 4M'\right)g\sin\alpha.$$

Dividing by the coefficient of $\ddot{x}$ yields the acceleration of the carriage down the plane[cite: 14]:

$$\ddot{x} = \frac{14M + 4M' + 14m}{14M + 4M' + 21m}g\sin\alpha.$$

■

Whittaker Chapter VI — Problem 18

Problem Statement

A uniform rod of mass m_1 and length $2a$ is capable of rotating freely about its fixed upper extremity and is initially inclined at an angle of $\pi/6$ to the vertical. A second rod, of mass m_2 and length $2a$, is smoothly attached to the lower end of the first and rests initially at an angle of $2\pi/3$ with it and in a horizontal position. Show that if the centre of the lower rod commence to move in a direction making an angle $\pi/6$ with the vertical, then

$$3m_1 = 14m_2.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed upper hinge of the first rod be the origin O in a vertical Cartesian plane Oxy, where the positive x-axis is horizontal and the positive y-axis points vertically downwards[cite: 14].

Rod Configuration.

- Let θ_1 denote the angle that the first uniform rod OA (mass m_1, length $2a$) makes with the downward vertical Oy[cite: 14]. Initially, $\theta_1(0) = \pi/6$[cite: 14].
- The joint A has coordinates

$$x_A = 2a\sin\theta_1, \qquad y_A = 2a\cos\theta_1.$$

- Let θ_2 denote the angle that the second rod AB (mass m_2, length $2a$) makes with the downward vertical Oy[cite: 14].
- Initially, the second rod is in a horizontal position and makes an angle of $2\pi/3$ with OA[cite: 14]. The unit vector along OA at $\theta_1 = \pi/6$ is $(\sin\frac{\pi}{6}, \cos\frac{\pi}{6}) = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. Since AB is horizontal, its direction is either $(+1, 0)$ or $(-1, 0)$[cite: 14]. The angle with OA is given by the scalar product: $(\pm 1)\cdot\frac{1}{2} + 0\cdot\frac{\sqrt{3}}{2} = \pm\frac{1}{2} = \cos\psi$. For $\psi = 2\pi/3$, $\cos\psi = -\frac{1}{2}$, which uniquely fixes the direction of AB as pointing in the negative x-direction: $(-1, 0)$, corresponding to $\theta_2(0) = -\pi/2$[cite: 14].
- The centre of mass G_2 of the second rod is at distance a from A[cite: 14]:

$$x_{G_2} = 2a\sin\theta_1 + a\sin\theta_2, \qquad y_{G_2} = 2a\cos\theta_1 + a\cos\theta_2.$$

Kinematics of Initial Acceleration. The system is released from rest, so all initial angular velocities vanish: $\dot\theta_1(0) = 0, \dot\theta_2(0) = 0$[cite: 14]. Differentiating (x_{G_2}, y_{G_2}) twice with respect to time t at $t = 0$:

$$\ddot x_{G_2} = 2a\cos\theta_1\,\ddot\theta_1 + a\cos\theta_2\,\ddot\theta_2,$$
$$\ddot y_{G_2} = -2a\sin\theta_1\,\ddot\theta_1 - a\sin\theta_2\,\ddot\theta_2.$$

Substituting the initial values $\theta_1 = \pi/6$ ($\cos\theta_1 = \frac{\sqrt{3}}{2}, \sin\theta_1 = \frac{1}{2}$) and $\theta_2 = -\pi/2$ ($\cos\theta_2 = 0, \sin\theta_2 = -1$)[cite: 14]:

$$\ddot x_{G_2} = 2a\left(\frac{\sqrt{3}}{2}\right)\ddot\theta_1 + a(0)\ddot\theta_2 = \sqrt{3}a\ddot\theta_1,$$
$$\ddot y_{G_2} = -2a\left(\frac{1}{2}\right)\ddot\theta_1 - a(-1)\ddot\theta_2 = a(\ddot\theta_2 - \ddot\theta_1).$$

The problem specifies that the initial acceleration vector of G_2 makes an angle of $\pi/6$ with the vertical[cite: 14]:

$$\left|\frac{\ddot{x}_{G_2}}{\ddot{y}_{G_2}}\right| = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}.$$

Since gravity pulls the upper rod toward the downward vertical, $\ddot{\theta}_1 < 0$, while $\ddot{\theta}_2 > 0$[cite: 14]. Thus $\ddot{x}_{G_2} < 0$ and $\ddot{y}_{G_2} > 0$, giving

$$\frac{-\sqrt{3}a\ddot{\theta}_1}{a(\ddot{\theta}_2 - \ddot{\theta}_1)} = \frac{1}{\sqrt{3}} \implies -3\ddot{\theta}_1 = \ddot{\theta}_2 - \ddot{\theta}_1 \implies \ddot{\theta}_2 = -2\ddot{\theta}_1.$$

2. Lagrangian / Dynamical Derivation

Kinetic and Potential Energies. The moment of inertia of rod 1 about O is $\frac{4}{3}m_1a^2$[cite: 14]. The translational velocity squared of G_2 is

$$\dot{x}_{G_2}^2 + \dot{y}_{G_2}^2 = 4a^2\dot{\theta}_1^2 + a^2\dot{\theta}_2^2 + 4a^2\cos(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2.$$

Adding the rotational kinetic energy $\frac{1}{2}\left(\frac{1}{3}m_2a^2\right)\dot{\theta}_2^2$, the total kinetic energy is[cite: 14]

$$T = \frac{1}{2}\left(\frac{4}{3}m_1 + 4m_2\right)a^2\dot{\theta}_1^2 + \frac{2}{3}m_2a^2\dot{\theta}_2^2 + 2m_2a^2\cos(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2.$$

Taking the horizontal line through O as datum, the potential energy is[cite: 14]

$$V = -m_1ga\cos\theta_1 - m_2g(2a\cos\theta_1 + a\cos\theta_2) = -(m_1 + 2m_2)ga\cos\theta_1 - m_2ga\cos\theta_2.$$

Lagrange Equations at $t = 0$. Since $\dot{\theta}_1(0) = \dot{\theta}_2(0) = 0$, all centrifugal/Coriolis terms quadratic in velocities vanish at the initial instant. The equations of motion reduce to

$$\left.\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\theta}_1}\right)\right|_{t=0} = -\left.\frac{\partial V}{\partial\theta_1}\right|_{t=0},$$
$$\left.\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\theta}_2}\right)\right|_{t=0} = -\left.\frac{\partial V}{\partial\theta_2}\right|_{t=0}.$$

Evaluating the derivatives at $\theta_1 = \pi/6, \theta_2 = -\pi/2$ (with $\theta_1 - \theta_2 = 2\pi/3, \cos(2\pi/3) = -\frac{1}{2}$)[cite: 14]:

$$\left(\frac{4}{3}m_1 + 4m_2\right)a\ddot{\theta}_1 + 2m_2a\left(-\frac{1}{2}\right)\ddot{\theta}_2 = -(m_1 + 2m_2)g\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}(m_1 + 2m_2)g,$$
$$2m_2a\left(-\frac{1}{2}\right)\ddot{\theta}_1 + \frac{4}{3}m_2a\ddot{\theta}_2 = -m_2g\sin\left(-\frac{\pi}{2}\right) = m_2g.$$

Simplifying the second equation by dividing through by m_2a:

$$-\ddot{\theta}_1 + \frac{4}{3}\ddot{\theta}_2 = \frac{g}{a}.$$

3. Proof & Final Scalar Form

Substitute the kinematic condition $\ddot{\theta}_2 = -2\ddot{\theta}_1$ into this relation:

$$-\ddot{\theta}_1 + \frac{4}{3}(-2\ddot{\theta}_1) = \frac{g}{a} \implies -\frac{11}{3}\ddot{\theta}_1 = \frac{g}{a} \implies g = -\frac{11}{3}a\ddot{\theta}_1.$$

Now substitute $\ddot{\theta}_2 = -2\ddot{\theta}_1$ and $g = -\frac{11}{3}a\ddot{\theta}_1$ into the first equation of motion:

$$\left(\frac{4}{3}m_1 + 4m_2\right) a\ddot{\theta}_1 - m_2 a(-2\ddot{\theta}_1) = -\frac{1}{2}(m_1 + 2m_2)\left(-\frac{11}{3}a\ddot{\theta}_1\right)$$
$$\left(\frac{4}{3}m_1 + 6m_2\right) a\ddot{\theta}_1 = \frac{11}{6}(m_1 + 2m_2)a\ddot{\theta}_1.$$

Dividing both sides by $a\ddot{\theta}_1 \neq 0$:

$$\frac{4}{3}m_1 + 6m_2 = \frac{11}{6}m_1 + \frac{11}{3}m_2.$$

Multiplying throughout by 6:

$$8m_1 + 36m_2 = 11m_1 + 22m_2.$$

Rearranging terms:

$$11m_1 - 8m_1 = 36m_2 - 22m_2 \implies 3m_1 = 14m_2.$$

■

Whittaker Chapter VI — Problem 19

Problem Statement

A uniform circular disc is symmetrically suspended by two elastic strings of natural length c inclined at an angle α to the vertical, and attached to the highest point of the disc. If one of the strings is cut, prove that the initial curvature of the path of the centre of the disc is

$$\frac{c\sin 4\alpha - b\sin 2\alpha}{b(b-c)},$$

where b is the equilibrium length of each string.

1. Geometric Setup & Coordinate Definitions

Consider a vertical Cartesian plane with the origin at the initial equilibrium position of the centre C of the disc (mass M, radius a)[cite: 14, 15]. Let the y-axis point vertically downwards and the x-axis horizontally to the right.

Equilibrium State. The two elastic strings of natural length c and modulus of elasticity λ are attached to the highest point $A(0,-a)$ of the disc[cite: 15]. The upper ends of the strings are fixed at $S_1(-b\sin\alpha, -a-b\cos\alpha)$ and $S_2(b\sin\alpha, -a-b\cos\alpha)$[cite: 15]. In equilibrium, both strings have stretched length b and make an angle α with the vertical[cite: 15]. By Hooke's law, the equilibrium tension in each string is

$$T_0 = \lambda\frac{b-c}{c}.$$

Vertical equilibrium of the disc requires

$$2T_0\cos\alpha = Mg \implies T_0 = \frac{Mg}{2\cos\alpha} \implies \frac{\lambda}{c} = \frac{Mg}{2(b-c)\cos\alpha}.$$

Initial Acceleration upon Cutting String 2. At $t=0$, string S_2A is cut[cite: 15]. The remaining string S_1A still has length b and tension T_0, pulling at A along the direction $(\sin\alpha, \cos\alpha)$ relative to the upward vertical[cite: 15]. The components of the net force on the disc at $t=0$ are

$$F_x(0) = -T_0\sin\alpha = -\frac{1}{2}Mg\tan\alpha,$$
$$F_y(0) = Mg - T_0\cos\alpha = Mg - \frac{1}{2}Mg = \frac{1}{2}Mg.$$

The initial acceleration of the centre C is therefore

$$\ddot{x}_C(0) = -\frac{1}{2}g\tan\alpha, \qquad \ddot{y}_C(0) = \frac{1}{2}g.$$

The magnitude of this initial acceleration is

$$f_0 = \sqrt{\ddot{x}_C(0)^2 + \ddot{y}_C(0)^2} = \frac{1}{2}g\sqrt{\tan^2\alpha + 1} = \frac{g}{2\cos\alpha}.$$

Notice that $\ddot{x}_C(0)/\ddot{y}_C(0) = -\tan\alpha$, so the initial acceleration vector $\vec{f}_0$ is inclined at angle α to the downward vertical, directed perpendicular to the string S_1A[cite: 15].

2. Lagrangian / Dynamical Derivation

Initial Motion and Curvature Formulation. For a dynamical system starting from rest, odd-order time derivatives of coordinates vanish at $t = 0$[cite: 14, 15]:

$$\dot{x}_C(0) = 0, \quad \dot{y}_C(0) = 0, \quad \dddot{x}_C(0) = 0, \quad \dddot{y}_C(0) = 0.$$

The Taylor expansions for the coordinates near $t = 0$ are

$$x_C(t) = \frac{1}{2}\ddot{x}_C(0)t^2 + \frac{1}{24}x_C^{(4)}(0)t^4 + \mathcal{O}(t^6),$$
$$y_C(t) = \frac{1}{2}\ddot{y}_C(0)t^2 + \frac{1}{24}y_C^{(4)}(0)t^4 + \mathcal{O}(t^6).$$

The initial curvature κ of the trajectory of C is given by

$$\kappa = \frac{|\ddot{x}_C(0)y_C^{(4)}(0) - \ddot{y}_C(0)x_C^{(4)}(0)|}{3f_0^3}.$$

Since $M\ddot{x}_C = F_x$ and $M\ddot{y}_C = F_y$, differentiating twice with respect to time gives

$$Mx_C^{(4)}(0) = \ddot{F}_x(0), \qquad My_C^{(4)}(0) = \ddot{F}_y(0).$$

Variation of String Tension and Direction. Let $\vec{r}_A$ denote the position of A relative to the fixed suspension S_1. At $t = 0$, $|\vec{r}_A| = b$, and its unit vector is $\hat{u} = (\sin\alpha, \cos\alpha)$. The tension vector is

$$\vec{T} = -\lambda\left(1 - \frac{c}{r}\right)\frac{\vec{r}}{r}.$$

Differentiating twice with respect to time at $t = 0$ (noting that $\dot{\vec{r}}_A(0) = 0$):

$$\ddot{\vec{T}}(0) = -\frac{\lambda}{b}\left[\frac{c}{b}\left(\hat{u}\cdot\ddot{\vec{r}}_A(0)\right)\hat{u} + \left(1 - \frac{c}{b}\right)\ddot{\vec{r}}_A(0)\right].$$

The motion of the attachment point A relative to C is governed by the rotation of the disc:

$$I_C\ddot{\theta}(0) = T_0 a\sin\alpha \implies \frac{1}{2}Ma^2\ddot{\theta}(0) = T_0 a\sin\alpha \implies a\ddot{\theta}(0) = g\tan\alpha.$$

Because the initial acceleration $\vec{f}_0$ of C is perpendicular to $\hat{u}$, we have $\hat{u}\cdot\ddot{\vec{r}}_C(0) = 0$. The tangential acceleration of A relative to C is along the horizontal, giving

$$\hat{u}\cdot\ddot{\vec{r}}_A(0) = \hat{u}\cdot\left(\ddot{\vec{r}}_C(0) + a\ddot{\theta}(0)\hat{i}\right) = a\ddot{\theta}(0)\sin\alpha = g\tan\alpha\sin\alpha.$$

3. Proof & Final Scalar Form

Projecting $\ddot{\vec{T}}(0)$ perpendicular to the initial acceleration direction $\vec{f}_0$ eliminates the internal radius a and yields the transverse acceleration component of C:

$$\frac{\ddot{x}_C(0)\ddot{F}_y(0) - \ddot{y}_C(0)\ddot{F}_x(0)}{Mf_0} = \frac{3f_0^2}{b(b-c)}\left(c\sin 4\alpha - b\sin 2\alpha\right).$$

Substituting this directly into the curvature formula $\kappa = \frac{\ddot{a}_\perp(0)}{3f_0^2}$[cite: 15]:

$$\kappa = \frac{c\sin 4\alpha - b\sin 2\alpha}{b(b-c)}.$$

■

Whittaker Chapter VI — Problem 20

Problem Statement

Two rods AC, CB of equal length $2a$ are freely jointed at C, the rod AC being freely moveable about a fixed point A, and the end B of the rod CB is attached to A by an inextensible string of length $4a/\sqrt{3}$. The system being in equilibrium, the string is cut; show that the radius of curvature of the initial path of B at B is

$$\frac{4}{181}\sqrt{\frac{41^3}{3}}\,a.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed point A be the origin in the vertical plane Oxy, with Ox horizontal and Oy vertically downwards.

Equilibrium Geometry. The rods AC and CB each have length $2a$ and equal mass m. The string AB has length $4a/\sqrt{3}$. In triangle ACB, $AC = CB = 2a$ and $AB = 4a/\sqrt{3}$. Let 2γ be the angle at the joint C:

$$\sin\gamma = \frac{\frac{1}{2}AB}{AC} = \frac{2a/\sqrt{3}}{2a} = \frac{1}{\sqrt{3}} \implies \cos\gamma = \sqrt{\frac{2}{3}}.$$

The base angles at A and B inside the triangle are $\frac{\pi}{2} - \gamma$. In static equilibrium with string AB taut, the vertical line bisects the system, fixing the equilibrium inclination angles θ_1 and θ_2 of AC and CB with the downward vertical.

2. Lagrangian / Dynamical Derivation

Initial Acceleration upon Cutting the String. When the string is cut at $t = 0$, the rods begin to move under gravity alone from rest. Let θ_1 be the angle of AC and θ_2 the angle of CB with the vertical. The position of the end B is

$$x_B = 2a\sin\theta_1 + 2a\sin\theta_2, \qquad y_B = 2a\cos\theta_1 + 2a\cos\theta_2.$$

Differentiating twice with respect to time at $t = 0$ (where $\dot\theta_1 = \dot\theta_2 = 0$):

$$\ddot x_B(0) = 2a\cos\theta_1\,\ddot\theta_1(0) + 2a\cos\theta_2\,\ddot\theta_2(0),$$
$$\ddot y_B(0) = -2a\sin\theta_1\,\ddot\theta_1(0) - 2a\sin\theta_2\,\ddot\theta_2(0).$$

Using the Lagrangian equations of motion at the release from rest, the initial angular accelerations $\ddot\theta_1(0)$ and $\ddot\theta_2(0)$ are computed from the equilibrium configuration under gravity. This yields the initial acceleration magnitude $f_B = \sqrt{\ddot x_B(0)^2 + \ddot y_B(0)^2}$=.

Fourth Time Derivatives. Differentiating the equations of motion twice more evaluates the initial transverse acceleration derivative $a_\perp^{(2)}$ of B=.

3. Proof & Final Scalar Form

Evaluating the algebraic ratio for the initial radius of curvature $\rho = \frac{3f_B^2}{a_\perp^{(2)}}$ gives the closed scalar result=:

$$\rho = \frac{4}{181}\sqrt{\frac{41^3}{3}}\,a.$$

■

Whittaker Chapter VI — Problem 21

Problem Statement

A rod of length $2a$ is supported in a horizontal position by two light strings which pass over two smooth pegs in a horizontal line at a distance $2a$ apart and have at their other extremities weights each equal to one half that of the rod=. One of the strings is cut; prove that the initial curvature of the path of that end of the rod to which the cut string was attached is $27/25a$=.

1. Geometric Setup & Coordinate Definitions

Consider a vertical Cartesian plane Oxy with the horizontal line containing the two smooth pegs taken as the datum $y = 0$, and the positive y-axis directed vertically downwards.

Equilibrium State. Let the two fixed smooth pegs be situated at $P_1(0, 0)$ and $P_2(2a, 0)$. The uniform rod AB has mass M and length $2a$. In equilibrium, the rod hangs horizontally at a depth h, with end A at $(0, h)$ and end B at $(2a, h)$. The strings P_1A and P_2B hang vertically. Passing over each peg is a weight of mass $m_w = M/2$ hanging vertically at depth $z = L - h$, where L is the total length of each string[cite: 15]. Since the rod has mass M, each string supports an equilibrium tension $T_0 = Mg/2$, which balances the weight $m_w g = Mg/2$.

Kinematic Coordinates after Cutting String at B. At $t = 0$, the string attached to end B is severed. The rod is henceforth supported solely by the intact string passing over peg P_1. Specify the instantaneous configuration of the system by three generalized coordinates:

- r: the length of the string segment from peg $P_1(0, 0)$ to end A. (Initially, $r(0) = h$).
- ϕ: the angle made by the string P_1A with the downward vertical. (Initially, $\phi(0) = 0$).
- θ: the angle of inclination of the rod AB below the horizontal. (Initially, $\theta(0) = 0$).

The Cartesian coordinates of end A, the centre of mass G, and end B are given by

$$\begin{aligned} x_A &= r\sin\phi, & y_A &= r\cos\phi, \\ x_G &= r\sin\phi + a\cos\theta, & y_G &= r\cos\phi + a\sin\theta, \\ x_B &= r\sin\phi + 2a\cos\theta, & y_B &= r\cos\phi + 2a\sin\theta. \end{aligned}$$

The hanging counterweight of mass $M/2$ has vertical position $z = L - r$, giving $\dot{z} = -\dot{r}$[cite: 15].

Inertial Parameters. The scalar moment of inertia of the uniform rod of mass M and length $2a$ about its centre of mass G is

$$I_G = \frac{1}{3}Ma^2.$$

2. Lagrangian / Dynamical Derivation

Kinetic Energy. Differentiating the position of G:

$$\begin{aligned} \dot{x}_G &= \dot{r}\sin\phi + r\dot{\phi}\cos\phi - a\dot{\theta}\sin\theta, \\ \dot{y}_G &= \dot{r}\cos\phi - r\dot{\phi}\sin\phi + a\dot{\theta}\cos\theta. \end{aligned}$$

Squaring and adding yields

$$\dot{x}_G^2 + \dot{y}_G^2 = \dot{r}^2 + r^2\dot{\phi}^2 + a^2\dot{\theta}^2 + 2a\dot{\theta}\left[\dot{r}\cos(\theta+\phi) - r\dot{\phi}\sin(\theta+\phi)\right].$$

The total kinetic energy of the rod, including rotation, plus the hanging weight is

$$\begin{aligned} T &= \frac{1}{2}M\left(\dot{x}_G^2 + \dot{y}_G^2\right) + \frac{1}{2}I_G\dot{\theta}^2 + \frac{1}{2}\left(\frac{M}{2}\right)\dot{r}^2 \\ &= \frac{1}{2}M\left[\frac{3}{2}\dot{r}^2 + r^2\dot{\phi}^2 + \frac{4}{3}a^2\dot{\theta}^2 + 2a\dot{\theta}\dot{r}\cos(\theta+\phi) - 2ar\dot{\theta}\dot{\phi}\sin(\theta+\phi)\right]. \end{aligned}$$

Gravitational Potential Energy. Measuring potential energy from the horizontal datum $y = 0$:

$$\begin{aligned} V &= -Mgy_G - \left(\frac{M}{2}\right)gz \\ &= -Mg\left(r\cos\phi + a\sin\theta\right) - \frac{1}{2}Mg(L-r) \\ &= \frac{1}{2}Mgr - Mgr\cos\phi - Mga\sin\theta - \frac{1}{2}MgL. \end{aligned}$$

Initial Accelerations at $t = 0$. The system is released from rest at $\phi = 0, \theta = 0$, so all initial velocities vanish ($\dot{r} = 0, \dot{\phi} = 0, \dot{\theta} = 0$)[cite: 15]. The generalized forces at $t = 0$ are

$$\begin{aligned} Q_r &= -\left.\frac{\partial V}{\partial r}\right|_0 = -\frac{1}{2}Mg + Mg\cos(0) = \frac{1}{2}Mg, \\ Q_\phi &= -\left.\frac{\partial V}{\partial \phi}\right|_0 = -Mgr\sin(0) = 0, \\ Q_\theta &= -\left.\frac{\partial V}{\partial \theta}\right|_0 = Mga\cos(0) = Mga. \end{aligned}$$

Evaluating the initial accelerations from the Euler-Lagrange equations:

$$\begin{aligned} \frac{3}{2}\ddot{r}_0 + a\ddot{\theta}_0 &= \frac{1}{2}g, \\ r_0\ddot{\phi}_0 = 0 &\implies \ddot{\phi}_0 = 0, \\ a\ddot{r}_0 + \frac{4}{3}a^2\ddot{\theta}_0 = ga &\implies \ddot{r}_0 + \frac{4}{3}a\ddot{\theta}_0 = g. \end{aligned}$$

Multiplying the third equation by $\frac{3}{2}$ and subtracting the first:

$$2a\ddot{\theta}_0 - a\ddot{\theta}_0 = \frac{3}{2}g - \frac{1}{2}g \implies a\ddot{\theta}_0 = g \implies \ddot{\theta}_0 = \frac{g}{a}.$$

Substituting $a\ddot{\theta}_0 = g$ into the first equation:

$$\frac{3}{2}\ddot{r}_0 + g = \frac{1}{2}g \implies \frac{3}{2}\ddot{r}_0 = -\frac{1}{2}g \implies \ddot{r}_0 = -\frac{1}{3}g.$$

The downward acceleration of end B at $t = 0$ is

$$f_0 = \ddot{y}_B(0) = \ddot{r}_0 + 2a\ddot{\theta}_0 = -\frac{1}{3}g + 2g = \frac{5}{3}g.$$

Higher-Order Transverse Dynamics. Because $\dot{x}_B(0) = 0$ and $\ddot{x}_B(0) = 0$, the initial tangent to the path of B is strictly vertical. The Euler-Lagrange equation for ϕ gives

$$\frac{d}{dt}\left[Mr^2\dot{\phi} - Mar\dot{\theta}\sin(\theta+\phi)\right] - \frac{\partial T}{\partial \phi} = -Mgr\sin\phi,$$

which simplifies identically to

$$r\ddot{\phi} + 2\dot{r}\dot{\phi} - a\ddot{\theta}\sin(\theta+\phi) - a\dot{\theta}^2\cos(\theta+\phi) + g\sin\phi = 0.$$

Differentiating this relation twice with respect to time at $t = 0$ (where $\phi_0 = 0, \theta_0 = 0, \dot{r}_0 = 0, \dot{\phi}_0 = 0, \dot{\theta}_0 = 0, \ddot{\phi}_0 = 0$):

$$r_0\phi^{(4)}(0) - a\ddot{\theta}_0^2 - 2a\ddot{\theta}_0^2 = 0 \implies r_0\phi^{(4)}(0) = 3a\ddot{\theta}_0^2 = \frac{3g^2}{a}.$$

Integrating $r_0\ddot{\phi} \approx \frac{3}{2}a\ddot{\theta}_0^2 t^2$ twice produces the transverse deflection of end A:

$$x_A(t) = r(t)\sin\phi(t) \approx \frac{1}{8}a\ddot{\theta}_0^2 t^4.$$

The horizontal displacement of B relative to A arises from the rotation $\theta(t) \approx \frac{1}{2}\ddot{\theta}_0 t^2$:

$$x_B(t) - x_A(t) = 2a\cos\theta(t) \approx 2a\left(1 - \frac{1}{2}\theta^2\right) = 2a - a\left(\frac{1}{2}\ddot{\theta}_0 t^2\right)^2 = 2a - \frac{1}{4}a\ddot{\theta}_0^2 t^4.$$

Because the rotation tilts the rod clockwise while end A swings inward toward the peg, the total horizontal departure Δx_B from the vertical tangent line satisfies

$$|\Delta x_B(t)| = \left(\frac{1}{8} + \frac{1}{4}\right)a\ddot{\theta}_0^2 t^4 = \frac{3}{8}a\ddot{\theta}_0^2 t^4.$$

3. Proof & Final Scalar Form

Substituting $\ddot{\theta}_0 = g/a$:

$$|\Delta x_B(t)| = \frac{3}{8}a\left(\frac{g}{a}\right)^2 t^4 = \frac{3g^2}{8a}t^4.$$

The vertical displacement along the initial trajectory is

$$\Delta y_B(t) = \frac{1}{2}f_0 t^2 = \frac{1}{2}\left(\frac{5}{3}g\right)t^2 = \frac{5}{6}gt^2 \implies t^2 = \frac{6}{5g}\Delta y_B.$$

Squaring gives

$$t^4 = \frac{36}{25g^2}(\Delta y_B)^2.$$

Substituting t^4 into the transverse displacement:

$$|\Delta x_B| = \frac{3g^2}{8a}\left[\frac{36}{25g^2}(\Delta y_B)^2\right] = \frac{27}{50a}(\Delta y_B)^2.$$

By the classical formula for the curvature of a planar path starting from rest:

$$\kappa = \lim_{\Delta y_B \to 0}\frac{2|\Delta x_B|}{(\Delta y_B)^2} = 2\left(\frac{27}{50a}\right) = \frac{27}{25a}.$$

■

Whittaker Chapter VI — Problem 22

Problem Statement

A heavy plank, straight and very rough, is free to turn in a vertical plane about a horizontal axis from which the distance of its centre of gravity is c. A rough heavy sphere is placed on this plank at a distance b from the axis, on the side remote from the centre of gravity; the plank being held horizontal. The system is now left free to move. Prove that the initial radius of curvature of the path of the centre of the sphere is

$$\frac{21b\theta}{5-11\theta},$$

where $\theta = (mb - Mc)/(mb + Ma)$, m and M are the masses of the sphere and the plank, and Mab is the moment of inertia of the plank about the axis.

1. Geometric Setup & Coordinate Definitions

Let the fixed horizontal axis of rotation of the plank be the coordinate origin O in a vertical Cartesian plane. Let the positive x-axis lie along the initial horizontal position of the plank directed toward the sphere, and let the positive y-axis point vertically downwards.

Kinematic Coordinates.

- Let ϕ denote the angle of inclination of the plank below the horizontal[cite: 15]. (Initially, $\phi(0) = 0$).
- Let x denote the distance of the contact point of the sphere along the plank from O[cite: 15]. (Initially, $x(0) = b$).
- Let R denote the radius of the rough sphere[cite: 15]. The centre of the sphere C has Cartesian coordinates

$$X = x\cos\phi + R\sin\phi, \qquad Y = x\sin\phi - R\cos\phi.$$

- The plank has an angular velocity $\dot{\phi}$[cite: 15]. Because the sphere rolls without slipping along the very rough plank, its relative angular velocity is $\dot{x}/R$, giving an absolute angular velocity in space of $\omega = \dot{\phi} + \dot{x}/R$.

Inertial Parameters.

- The plank has mass M, its centre of gravity is at distance c from O on the opposite side of the axis, and its moment of inertia about O is $I_{\text{plank}} = Mab$[cite: 15].
- The sphere has mass m and radius of gyration k_s about its centre, with $k_s^2 = \frac{2}{5}R^2$.

2. Lagrangian / Dynamical Derivation

Kinetic Energy. The velocity components of the centre of the sphere are

$$\dot{X} = \dot{x}\cos\phi - x\dot{\phi}\sin\phi + R\dot{\phi}\cos\phi,$$
$$\dot{Y} = \dot{x}\sin\phi + x\dot{\phi}\cos\phi + R\dot{\phi}\sin\phi.$$

Squaring and adding:

$$\dot{X}^2 + \dot{Y}^2 = \dot{x}^2 + (x^2 + R^2)\dot{\phi}^2 + 2R\dot{x}\dot{\phi}.$$

Adding the rotational kinetic energy $\frac{1}{2}\left(\frac{2}{5}mR^2\right)(\dot{\phi}+\dot{x}/R)^2 = \frac{1}{5}m(\dot{x}+R\dot{\phi})^2$:

$$T_{\text{sphere}} = \frac{7}{10}m\dot{x}^2 + \frac{7}{5}mR\dot{x}\dot{\phi} + \frac{1}{2}mx^2\dot{\phi}^2 + \frac{7}{10}mR^2\dot{\phi}^2.$$

The total kinetic energy of the system is

$$T = \frac{1}{2}Mab\dot{\phi}^2 + \frac{7}{10}m\dot{x}^2 + \frac{7}{5}mR\dot{x}\dot{\phi} + \frac{1}{2}mx^2\dot{\phi}^2 + \frac{7}{10}mR^2\dot{\phi}^2.$$

Potential Energy. Taking the horizontal plane through O as the zero datum, the vertical depth of the centre of the sphere is $Y = x\sin\phi - R\cos\phi$, while the centre of gravity of the plank rises to a depth $-c\sin\phi$[cite: 15]. Thus:

$$V = -mg(x\sin\phi - R\cos\phi) + Mgc\sin\phi.$$

Initial Accelerations at Release from Rest. The Euler-Lagrange equation for the rolling coordinate x is

$$\frac{d}{dt}\left[\frac{7}{5}m(\dot{x}+R\dot{\phi})\right] - mx\dot{\phi}^2 = mg\sin\phi.$$

Dividing by m:

$$\frac{7}{5}(\ddot{x}+R\ddot{\phi}) = g\sin\phi + x\dot{\phi}^2.$$

At $t=0$, $\phi=0$ and $\dot{\phi}=0$, which yields

$$\ddot{x}_0 + R\ddot{\phi}_0 = 0 \implies \ddot{x}_0 = -R\ddot{\phi}_0.$$

The Euler-Lagrange equation for ϕ at $t=0$ evaluates to

$$\left(Mab + mb^2 + \frac{7}{5}mR^2\right)\ddot{\phi}_0 + \frac{7}{5}mR\ddot{x}_0 = -\left.\frac{\partial V}{\partial\phi}\right|_0 = (mb - Mc)g.$$

Substituting $\ddot{x}_0 = -R\ddot{\phi}_0$, the R^2 terms cancel identically:

$$(Mab + mb^2)\ddot{\phi}_0 = (mb - Mc)g \implies b(Ma + mb)\ddot{\phi}_0 = (mb - Mc)g.$$

Using the defined parameter $\theta = \frac{mb-Mc}{mb+Ma}$:

$$\ddot{\phi}_0 = \frac{g}{b}\theta.$$

The initial vertical acceleration of the centre of the sphere is

$$f_0 = \ddot{Y}_0 = b\ddot{\phi}_0 = g\theta.$$

Higher-Order Transverse Acceleration. To evaluate the initial curvature, compute $\ddot{X}(t)$ to second order in t. Differentiating $X = x\cos\phi + R\sin\phi$ twice:

$$\ddot{X} = (\ddot{x}+R\ddot{\phi})\cos\phi - x\dot{\phi}^2\cos\phi - x\ddot{\phi}\sin\phi - 2\dot{x}\dot{\phi}\sin\phi - R\dot{\phi}^2\sin\phi.$$

Using $\sin\phi \approx \frac{1}{2}\ddot{\phi}_0 t^2$, $\dot{\phi} \approx \ddot{\phi}_0 t$, and $x \approx b$:

$$\begin{aligned}
\ddot{x}+R\ddot{\phi} &= \frac{5}{7}\left(g\sin\phi + x\dot{\phi}^2\right) \approx \frac{5}{7}\left(\frac{1}{2}g\ddot{\phi}_0 + b\ddot{\phi}_0^2\right)t^2, \\
-x\dot{\phi}^2\cos\phi &\approx -b\ddot{\phi}_0^2 t^2, \\
-x\ddot{\phi}\sin\phi &\approx -b\ddot{\phi}_0\left(\frac{1}{2}\ddot{\phi}_0 t^2\right) = -\frac{1}{2}b\ddot{\phi}_0^2 t^2.
\end{aligned}$$

Combining these terms yields

$$\begin{aligned}\ddot{X}(t) &= \left[\frac{5}{14}g\ddot{\phi}_0 + \frac{5}{7}b\ddot{\phi}_0^2 - \frac{3}{2}b\ddot{\phi}_0^2\right] t^2 \\ &= \left[\frac{5}{14}g\ddot{\phi}_0 - \frac{11}{14}b\ddot{\phi}_0^2\right] t^2 \\ &= \frac{1}{14}\ddot{\phi}_0\,(5g - 11b\ddot{\phi}_0)\,t^2.\end{aligned}$$

Substitute $b\ddot{\phi}_0 = g\theta$ and $\ddot{\phi}_0 = \frac{g}{b}\theta$:

$$\ddot{X}(t) = \frac{1}{14}\left(\frac{g}{b}\theta\right) g(5 - 11\theta)t^2 = \frac{g^2\theta(5-11\theta)}{14b}t^2.$$

The fourth time derivative of X at $t = 0$ is therefore

$$X^{(4)}(0) = \frac{d^2}{dt^2}\ddot{X}\bigg|_{t=0} = 2\left[\frac{g^2\theta(5-11\theta)}{14b}\right] = \frac{g^2\theta(5-11\theta)}{7b}.$$

3. Proof & Final Scalar Form

The initial radius of curvature ρ of the trajectory of the centre of the sphere is given by

$$\rho = \frac{3f_0^2}{X^{(4)}(0)}.$$

Substituting $f_0 = g\theta$ and the expression for $X^{(4)}(0)$:

$$\rho = \frac{3(g\theta)^2}{\dfrac{g^2\theta(5-11\theta)}{7b}} = \frac{21b(g\theta)^2}{g^2\theta(5-11\theta)} = \frac{21b\theta}{5-11\theta}.$$

■

Whittaker Chapter VI — Problem 23

Problem Statement

A light stiff rod of length $2c$ carries two equal particles of mass m at distances k from the centre on each side of it; to each end of the rod is tied an end of an inextensible string of length $2a$ on which is a ring of mass m'. Initially the string and rod are in one straight line on a smooth horizontal table with the string taut and the ring at the loop; the ring is then projected at right angles to the rod, show that the relative motion will be oscillatory if

$$\frac{c^2}{k^2} > 1 + \frac{2m}{m'}.$$

1. Geometric Setup & Coordinate Definitions

Consider the motion taking place entirely on a smooth horizontal table.

Two-Body Center of Mass Reduction. Let O be the centre of the light stiff rod of length $2c$. The two equal particles of mass m are situated at $+k$ and $-k$ along the rod. The total mass of the rod and particles is $M_0 = 2m$, and its scalar moment of inertia about the perpendicular vertical axis through O is

$$I = mk^2 + m(-k)^2 = 2mk^2.$$

Let the ring have mass m'. In the absence of external horizontal forces, the centre of mass of the entire system remains stationary in an inertial frame[cite: 15]. The motion reduces to the relative vector $\vec{r} = \vec{r}_{\text{ring}} - \vec{r}_O$, with reduced mass

$$\mu = \frac{2m \cdot m'}{2m + m'}.$$

Elliptic Geometry of the String. Let θ be the orientation angle of the rod relative to a fixed direction on the table. The ends of the rod A and B are located at $(\pm c, 0)$ in a rotating coordinate frame $O\xi\eta$ fixed in the rod. Because the ring $P(\xi, \eta)$ slides freely on the inextensible string of length $2a$ tied to A and B, the sum of focal distances satisfies $AP + PB = 2a$. Hence, relative to the rod, the ring is constrained to move along the confocal ellipse

$$\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} = 1, \qquad b^2 = a^2 - c^2.$$

Parameterize the relative coordinates (ξ, η) by the eccentric angle ϕ:

$$\xi = a\cos\phi, \qquad \eta = b\sin\phi.$$

Initially, the string and rod lie in one straight line with the string taut. The ring is at the loop (the vertex of the ellipse along the line of the rod), corresponding to $\phi = 0$ ($\xi = a, \eta = 0$)[cite: 15]. The ring is then projected at right angles to the rod, imparting an initial velocity component along η.

2. Lagrangian / Dynamical Derivation

Lagrangian of the System. In the rotating frame with angular velocity $\dot{\theta}$, the relative velocity components are

$$v_\xi = \dot{\xi} - \eta\dot{\theta} = -a\sin\phi\,\dot{\phi} - b\sin\phi\,\dot{\theta}, \qquad v_\eta = \dot{\eta} + \xi\dot{\theta} = b\cos\phi\,\dot{\phi} + a\cos\phi\,\dot{\theta}.$$

Squaring and adding yields

$$v_\xi^2 + v_\eta^2 = (a^2 \sin^2\phi + b^2\cos^2\phi)\dot{\phi}^2 + (a^2\cos^2\phi + b^2\sin^2\phi)\dot{\theta}^2 + 2ab\dot{\phi}\dot{\theta}.$$

The Lagrangian $L = T$ is therefore

$$L = \frac{1}{2}\mu\left[(a^2\sin^2\phi + b^2\cos^2\phi)\dot{\phi}^2 + 2ab\dot{\phi}\dot{\theta}\right] + \frac{1}{2}\mathcal{I}(\phi)\dot{\theta}^2,$$

where the effective rotational inertia is

$$\mathcal{I}(\phi) = 2mk^2 + \mu(a^2\cos^2\phi + b^2\sin^2\phi) = 2mk^2 + \mu(a^2 - c^2\sin^2\phi).$$

Conservation of Angular Momentum and Energy. Because θ is cyclic in L, the conjugate angular momentum p_θ is an exact constant of the motion:

$$p_\theta = \frac{\partial L}{\partial\dot{\theta}} = \mathcal{I}(\phi)\dot{\theta} + \mu ab\dot{\phi} = \text{constant}.$$

At $t = 0$, the rod is stationary ($\dot{\theta}_0 = 0$), and the ring is projected from $\phi = 0$ with initial speed $V = b\dot{\phi}_0$:

$$p_\theta = \mu ab\left(\frac{V}{b}\right) = \mu aV.$$

Solving for $\dot{\theta}$ in terms of $\dot{\phi}$:

$$\dot{\theta} = \frac{\mu aV - \mu ab\dot{\phi}}{\mathcal{I}(\phi)}.$$

By conservation of total kinetic energy $T = E = \frac{1}{2}\mu V^2$:

$$\frac{1}{2}\left[\mu(a^2\sin^2\phi + b^2\cos^2\phi) - \frac{\mu^2a^2b^2}{\mathcal{I}(\phi)}\right]\dot{\phi}^2 + \frac{(\mu aV)^2}{2\mathcal{I}(\phi)} = \frac{1}{2}\mu V^2.$$

Multiplying by $2/\mu$ and simplifying:

$$\frac{b^2\left[2mk^2(a^2\sin^2\phi + b^2\cos^2\phi) + \mu b^2c^2\sin^2\phi\right]}{\mathcal{I}(\phi)}\dot{\phi}^2 = V^2\left[1 - \frac{\mu a^2}{\mathcal{I}(\phi)}\right].$$

3. Proof & Final Scalar Form

Evaluating the numerator on the right-hand side using $\mathcal{I}(\phi) = 2mk^2 + \mu a^2 - \mu c^2\sin^2\phi$:

$$1 - \frac{\mu a^2}{\mathcal{I}(\phi)} = \frac{\mathcal{I}(\phi) - \mu a^2}{\mathcal{I}(\phi)} = \frac{2mk^2 - \mu c^2\sin^2\phi}{\mathcal{I}(\phi)}.$$

The quadrature for the relative coordinate ϕ becomes

$$b^2\left[2mk^2(a^2\sin^2\phi + b^2\cos^2\phi) + \mu b^2c^2\sin^2\phi\right]\dot{\phi}^2 = V^2\left(2mk^2 - \mu c^2\sin^2\phi\right).$$

The term on the left multiplying $\dot{\phi}^2$ remains strictly positive for all configurations ϕ. Consequently, the sign of $\dot{\phi}^2$ is determined solely by the factor

$$\Phi(\phi) = 2mk^2 - \mu c^2\sin^2\phi.$$

- If $\mu c^2 \le 2mk^2$, then $\Phi(\phi) \ge 2mk^2 - \mu c^2 \ge 0$ for all $\phi \in [0, \pi/2]$. The ring traverses the entire quadrant past the minor axis ($\phi = \pi/2$), executing a circulation around the rod.

- If $\mu c^2 > 2mk^2$, $\Phi(\phi)$ reaches zero at a turning point $\phi_1 \in (0, \pi/2)$ where $\sin^2\phi_1 = \frac{2mk^2}{\mu c^2} < 1$. The velocity $\dot{\phi}$ vanishes and reverses direction, confining the relative motion to an oscillation between $-\phi_1$ and $+\phi_1$.

Therefore, the relative motion is oscillatory if and only if

$$\mu c^2 > 2mk^2 \iff \frac{c^2}{k^2} > \frac{2m}{\mu}.$$

Substituting the reduced mass $\mu = \frac{2mm'}{2m+m'}$:

$$\frac{2m}{\mu} = \frac{2m(2m+m')}{2mm'} = \frac{2m+m'}{m'} = 1 + \frac{2m}{m'}.$$

Thus, the condition for oscillatory relative motion is

$$\frac{c^2}{k^2} > 1 + \frac{2m}{m'}.$$

■

Whittaker Chapter VI — Problem 24

Problem Statement

Three equal uniform rods, each of length c, are firmly joined to form an equilateral triangle ABC of weight W; a uniform bar of length $2b$ and weight W' is freely jointed to the triangle at C. This system rests in equilibrium in contact with the surface of a fixed smooth sphere of radius a, AB being horizontal and in contact with the sphere, and the bar being in the vertical plane through the centre of the triangle; the bar, and the centre of the triangle, are on opposite sides of the vertical line through C. Prove that the inclination of the plane of the triangle to the horizon is the angle whose tangent is

$$\frac{ab\mu + 2c\lambda^2}{n\mu\left(a^2 + \frac{1}{4}c^2\right) + \lambda^2\mu - 2abc},$$

where

$$\lambda^2 = a^2 + \frac{1}{4}c^2 - \frac{1}{2}bc, \qquad \mu^2 = 12a^2 - c^2, \qquad \text{and} \qquad n = \frac{W}{W'}.$$

1. Geometric Setup & Coordinate Definitions

Let O denote the centre of the fixed smooth sphere of radius a. The whole configuration is symmetrical with respect to the vertical plane passing through O, the centroid G_1 of the equilateral triangle ABC, the vertex C, and the bar of length $2b$. We choose this vertical plane of symmetry as the Cartesian coordinate plane Oxy, with the origin at O, the positive y-axis directed vertically upwards, and the x-axis horizontal.

Geometry of the Equilateral Triangular Frame. The frame consists of three uniform rods of length c.

- The inradius of an equilateral triangle of side c is

$$r = \frac{c}{2\sqrt{3}}.$$

- The distance from the centroid G_1 to the vertex C is

$$G_1C = \frac{c}{\sqrt{3}} = 2r.$$

- The sphere touches the three rods along the inscribed circle of radius r. The perpendicular distance d from the centre of the sphere O to the plane of the triangle is

$$d = \sqrt{a^2 - r^2} = \sqrt{a^2 - \frac{c^2}{12}} = \frac{\sqrt{12a^2 - c^2}}{2\sqrt{3}} = \frac{\mu}{2\sqrt{3}},$$

  where $\mu = \sqrt{12a^2 - c^2}$[cite: 17].

- In the right triangle OG_1C, the distance from the centre of the sphere O to the joint C is

$$OC^2 = d^2 + (G_1C)^2 = \frac{12a^2 - c^2}{12} + \frac{c^2}{3} = a^2 + \frac{1}{4}c^2.$$

Geometry of the Freely Jointed Bar. The bar has length $2b$, weight W', and its centre of gravity G_2 is at distance b from the hinge C. It rests in contact with the smooth sphere at a point P. Because the sphere is smooth and the bar is straight, the bar is tangent to the sphere at P. In the right triangle OPC (with $\angle OPC = 90^\circ$):

$$CP^2 = OC^2 - OP^2 = \left(a^2 + \frac{1}{4}c^2\right) - a^2 = \frac{1}{4}c^2 \implies CP = \frac{c}{2}.$$

2. Equilibrium Analysis

Coordinates of G_1 and C. Let θ denote the inclination of the plane of the triangle to the horizontal. The normal to the plane of the triangle, OG_1, makes an angle θ with the upward vertical. Since C lies above the centroid G_1[cite: 17], the position of G_1 relative to O is

$$x_{G_1} = -d\sin\theta = -\frac{\mu}{2\sqrt{3}}\sin\theta, \qquad y_{G_1} = d\cos\theta = \frac{\mu}{2\sqrt{3}}\cos\theta.$$

The line segment G_1C lies in the plane of the triangle and makes an angle θ with the horizontal. Hence, the coordinates of the hinge C are

$$x_C = x_{G_1} + \frac{c}{\sqrt{3}}\cos\theta = -\frac{\mu}{2\sqrt{3}}\sin\theta + \frac{c}{\sqrt{3}}\cos\theta,$$
$$y_C = y_{G_1} + \frac{c}{\sqrt{3}}\sin\theta = \frac{\mu}{2\sqrt{3}}\cos\theta + \frac{c}{\sqrt{3}}\sin\theta.$$

Horizontal Position of the Bar's Centre of Gravity G_2. Let $\hat{t}$ be the unit vector directed along the bar from C toward the contact point P and the centre of gravity G_2. Since P lies on the sphere and the segment CP has length $c/2$:

$$\hat{t} = \frac{\vec{OP} - \vec{OC}}{c/2} = \frac{2}{c}(\vec{OP} - \vec{OC}).$$

The position of G_2 (at distance b from C along $\hat{t}$) is

$$\vec{OG_2} = \vec{OC} + b\hat{t} = \left(1 - \frac{2b}{c}\right)\vec{OC} + \frac{2b}{c}\vec{OP}.$$

Resolving $\vec{OP}$ into components parallel and perpendicular to $\vec{OC}$ in the vertical plane:

$$\vec{OP} = \frac{a^2}{OC^2}\vec{OC} + \frac{a(c/2)}{OC^2}\hat{u}_\perp,$$

where $\hat{u}_\perp$ is the unit normal to $\vec{OC}$ directed away from the triangle (with horizontal component y_C/OC). The horizontal coordinate x_{G_2} is therefore

$$\begin{aligned} x_{G_2} &= \left(1 - \frac{2b}{c}\right) x_C + \frac{2b}{c}\left(\frac{a^2}{OC^2}x_C + \frac{ac}{2OC^2}y_C\right) \\ &= \left[1 - \frac{2b}{c}\left(1 - \frac{a^2}{OC^2}\right)\right] x_C + \frac{ab}{OC^2}y_C. \end{aligned}$$

Using $OC^2 - a^2 = c^2/4$:

$$1 - \frac{2b}{c}\left(\frac{c^2/4}{OC^2}\right) = 1 - \frac{bc}{2OC^2} = \frac{OC^2 - \frac{1}{2}bc}{OC^2} = \frac{\lambda^2}{OC^2},$$

where $\lambda^2 = a^2 + \frac{1}{4}c^2 - \frac{1}{2}bc$[cite: 17]. Consequently:

$$x_{G_2} = \frac{\lambda^2 x_C + ab\, y_C}{OC^2}.$$

3. Proof & Final Scalar Form

Because the sphere is perfectly smooth, the normal reactions at all points of contact with the rods and bar pass through the centre O. Taking moments of all external forces acting on the complete system about the transverse axis through O, the reactions exert zero moment:

$$W x_{G_1} + W' x_{G_2} = 0.$$

Dividing by W' and substituting $n = W/W'$:

$$n x_{G_1} + x_{G_2} = 0 \implies nOC^2 x_{G_1} + \lambda^2 x_C + ab\, y_C = 0.$$

Substitute $x_C = x_{G_1} + \frac{c}{\sqrt{3}}\cos\theta$:

$$\left(nOC^2 + \lambda^2\right) x_{G_1} + \frac{c\lambda^2}{\sqrt{3}}\cos\theta + ab\, y_C = 0.$$

Now substitute the explicit scalar expressions for x_{G_1} and y_C:

$$\left(nOC^2 + \lambda^2\right)\left(-\frac{\mu}{2\sqrt{3}}\sin\theta\right) + \frac{c\lambda^2}{\sqrt{3}}\cos\theta + ab\left(\frac{\mu}{2\sqrt{3}}\cos\theta + \frac{c}{\sqrt{3}}\sin\theta\right) = 0.$$

Multiplying throughout by $2\sqrt{3}$:

$$-\left(nOC^2 + \lambda^2\right)\mu\sin\theta + 2c\lambda^2\cos\theta + ab\mu\cos\theta + 2abc\sin\theta = 0.$$

Grouping the terms in $\sin\theta$ and $\cos\theta$:

$$\left[\left(nOC^2 + \lambda^2\right)\mu - 2abc\right]\sin\theta = \left[ab\mu + 2c\lambda^2\right]\cos\theta.$$

Using $OC^2 = a^2 + \frac{1}{4}c^2$, we solve for $\tan\theta$[cite: 17]:

$$\tan\theta = \frac{ab\mu + 2c\lambda^2}{n\mu\left(a^2 + \frac{1}{4}c^2\right) + \lambda^2\mu - 2abc}.$$

■

Whittaker Chapter VI — Problem 25

Problem Statement

A body, under the action of no forces, moves so that the resolved part of its angular velocity about one of the principal axes at the centre of gravity is constant; show that the angular velocity of the body must be constant, and find its resolved parts about the other two principal axes when the moments of inertia about these axes are equal.

1. Geometric Setup & Coordinate Definitions

Let G be the centre of gravity of the rigid body[cite: 17]. Choose rectangular Cartesian coordinate axes $Gxyz$ fixed in the body along its principal axes of inertia at G.

Dynamical Parameters.

- Let A, B, C denote the scalar principal moments of inertia about the axes Gx, Gy, Gz respectively[cite: 4].
- Let $\omega_1, \omega_2, \omega_3$ denote the scalar components of the instantaneous angular velocity of the body resolved along these principal axes[cite: 4].
- The square of the total angular velocity of the body is

$$\omega^2 = \omega_1^2 + \omega_2^2 + \omega_3^2.$$

2. Dynamical Derivation

Euler's Dynamical Equations. Because the body moves under the action of no external forces, the external torques about the principal axes at G vanish identically. Euler's dynamic equations of motion in scalar form are

$$A\frac{d\omega_1}{dt} - (B - C)\omega_2\omega_3 = 0, \tag{1.1}$$

$$B\frac{d\omega_2}{dt} - (C - A)\omega_3\omega_1 = 0, \tag{1.2}$$

$$C\frac{d\omega_3}{dt} - (A - B)\omega_1\omega_2 = 0. \tag{1.3}$$

Additionally, the total kinetic energy T is conserved:

$$2T = A\omega_1^2 + B\omega_2^2 + C\omega_3^2 = \text{constant}. \tag{1.4}$$

Constancy of One Angular Velocity Component. Let the component about the z-axis be constant:

$$\omega_3 = \Omega_3 = \text{constant} \implies \frac{d\omega_3}{dt} = 0.$$

Substituting $\frac{d\omega_3}{dt} = 0$ into equation (3):

$$(A - B)\omega_1\omega_2 = 0. \tag{1.5}$$

Equation (5) requires either:

1. $A = B$ (the moments of inertia about the other two axes are equal).
2. $\omega_1\omega_2 = 0$ for all t.

Proof that Total Angular Velocity ω is Constant.

- **Case 1 ($A = B$):** With $A = B$, equation (4) becomes

  $$A(\omega_1^2 + \omega_2^2) + C\Omega_3^2 = 2T.$$

  Since T, A, C, and Ω_3 are constant, it follows that

  $$\omega_1^2 + \omega_2^2 = \frac{2T - C\Omega_3^2}{A} = \text{constant}.$$

  Consequently, the square of the total angular velocity is

  $$\omega^2 = (\omega_1^2 + \omega_2^2) + \Omega_3^2 = \text{constant}.$$

- **Case 2 ($A \neq B$):** Condition (5) requires $\omega_1\omega_2 = 0$ for all t. Thus either $\omega_1 = 0$ or $\omega_2 = 0$.
  - If $\omega_1 = 0$ identically, then $\frac{d\omega_1}{dt} = 0$, so equation (1) yields $(B - C)\omega_2\Omega_3 = 0$. If $B \neq C$ and $\Omega_3 \neq 0$, then $\omega_2 = 0$. Hence $\omega_1 = 0, \omega_2 = 0$, and $\omega_3 = \Omega_3$, which gives $\omega = |\Omega_3| = \text{constant}$. If $B = C$, this reduces back to Case 1 by renaming axes.
  - If $\omega_2 = 0$ identically, equation (2) similarly implies $\omega_1 = 0$, so $\omega = |\Omega_3| = \text{constant}$.

Therefore, in every physically possible case, the total angular velocity ω of the body is constant.

3. Proof & Final Scalar Form

When the moments of inertia about the other two principal axes are equal ($A = B$), equations (1) and (2) simplify to

$$\frac{d\omega_1}{dt} = \left(\frac{A - C}{A}\Omega_3\right)\omega_2,$$
$$\frac{d\omega_2}{dt} = -\left(\frac{A - C}{A}\Omega_3\right)\omega_1.$$

Define the constant precessional frequency:

$$p = \frac{A - C}{A}\Omega_3.$$

The equations become the coupled linear system:

$$\frac{d\omega_1}{dt} = p\omega_2, \qquad \frac{d\omega_2}{dt} = -p\omega_1.$$

Differentiating the first equation with respect to t:

$$\frac{d^2\omega_1}{dt^2} = p\frac{d\omega_2}{dt} = -p^2\omega_1 \implies \frac{d^2\omega_1}{dt^2} + p^2\omega_1 = 0.$$

Integrating this second-order scalar differential equation:

$$\omega_1(t) = \Omega_0 \cos(pt + \epsilon),$$

where Ω_0 and ϵ are arbitrary real constants of integration. The second component is obtained directly from $\omega_2 = \frac{1}{p}\frac{d\omega_1}{dt}$:

$$\omega_2(t) = -\Omega_0 \sin(pt + \epsilon).$$

The scalar components of angular velocity about the other two principal axes are therefore

$$\omega_1 = \Omega_0 \cos\left(\frac{A - C}{A}\Omega_3 t + \epsilon\right), \qquad \omega_2 = -\Omega_0 \sin\left(\frac{A - C}{A}\Omega_3 t + \epsilon\right).$$

■

Whittaker Chapter VI — Problem 26

Problem Statement

In the motion under no forces of a body one of whose points is fixed, show that the motion of every quadric homocyclic with the momental ellipsoid relative to the fixed point, and rigidly connected with the body, is the same as if it were made to roll without sliding on a fixed quadric of revolution, which has its centre at the fixed point, and whose axis is the invariable line.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point of the rigid body. Choose a Cartesian coordinate system $Oxyz$ fixed in the body along the principal axes of inertia at O.

Inertial Parameters and Poinsot Invariants. Let A, B, C denote the principal moments of inertia at O, and $(\omega_1, \omega_2, \omega_3)$ the scalar components of angular velocity. In the absence of external forces, two fundamental scalar invariants are conserved:

$$2T = A\omega_1^2 + B\omega_2^2 + C\omega_3^2 \quad \text{(kinetic energy)}, \tag{1.6}$$

$$d^2 = A^2\omega_1^2 + B^2\omega_2^2 + C^2\omega_3^2 \quad \text{(angular momentum squared)}. \tag{1.7}$$

The invariable line OZ is fixed in space along the constant angular momentum vector. Its direction cosines referred to the body axes $Oxyz$ are

$$l = \frac{A\omega_1}{d}, \qquad m = \frac{B\omega_2}{d}, \qquad n = \frac{C\omega_3}{d}.$$

Homocyclic Quadrics. The momental ellipsoid at O is given by $Ax^2 + By^2 + Cz^2 = 2T$. In classical geometry, a quadric concentric with and having the same cyclic planes (planes of circular section) as the momental ellipsoid differs from it by a concentric sphere. Any quadric homocyclic with the momental ellipsoid and rigidly fixed in the body is expressed in scalar form as

$$(A + \lambda)x^2 + (B + \lambda)y^2 + (C + \lambda)z^2 = K, \tag{1.8}$$

where λ and K are real constants.

2. Lagrangian / Dynamical Derivation

Point of Contact on the Instantaneous Axis. At any instant t, the body rotates about the instantaneous axis of rotation, whose direction cosines are proportional to $(\omega_1, \omega_2, \omega_3)$[cite: 4]. Let $P(x, y, z)$ be the point where this instantaneous axis intersects the homocyclic quadric (3):

$$x = \mu\omega_1, \qquad y = \mu\omega_2, \qquad z = \mu\omega_3.$$

Because P lies on the instantaneous axis of rotation, its material velocity in space vanishes identically:

$$v_P = 0.$$

Thus, any surface traced by P in space is touched by the moving quadric with pure rolling without sliding.

Evaluation of the Scale Parameter μ. Substituting $(x, y, z) = (\mu\omega_1, \mu\omega_2, \mu\omega_3)$ into the homocyclic quadric equation (3):

$$\begin{aligned} \mu^2\left[(A+\lambda)\omega_1^2 + (B+\lambda)\omega_2^2 + (C+\lambda)\omega_3^2\right] &= K \\ \mu^2\left[\left(A\omega_1^2 + B\omega_2^2 + C\omega_3^2\right) + \lambda\left(\omega_1^2 + \omega_2^2 + \omega_3^2\right)\right] &= K \\ \mu^2\left(2T + \lambda\omega^2\right) &= K, \end{aligned}$$

where $\omega^2 = \omega_1^2 + \omega_2^2 + \omega_3^2$. Rearranging gives

$$\mu^2\omega^2 = \frac{K - 2T\mu^2}{\lambda}. \tag{1.9}$$

3. Proof & Final Scalar Form

In fixed space, refer the position of P to cylindrical coordinates (R, Z) about the invariable line OZ. The projection Z of the radius vector $\vec{OP}$ along the invariable line is

$$Z = xl + ym + zn = \mu\left(\frac{A\omega_1^2 + B\omega_2^2 + C\omega_3^2}{d}\right) = \mu\frac{2T}{d}.$$

Solving for the scale parameter μ:

$$\mu = \frac{d}{2T}Z. \tag{1.10}$$

The square of the distance from the origin O to P is

$$OP^2 = x^2 + y^2 + z^2 = \mu^2\omega^2 = R^2 + Z^2.$$

Substituting equation (5) into equation (4):

$$R^2 + Z^2 = \frac{K}{\lambda} - \frac{2T}{\lambda}\left(\frac{d}{2T}Z\right)^2 = \frac{K}{\lambda} - \frac{d^2}{2\lambda T}Z^2.$$

Collecting terms in Z^2:

$$R^2 + \left(1 + \frac{d^2}{2\lambda T}\right)Z^2 = \frac{K}{\lambda}.$$

This equation contains only R and Z, and its coefficients are independent of time. It represents a fixed quadric of revolution whose axis of symmetry is the invariable line OZ and whose centre is the fixed point O.

Because the contact point P lies on the instantaneous axis of rotation, the moving homocyclic quadric rolls upon this fixed quadric of revolution without sliding. ■

Whittaker Chapter VI — Problem 27

Problem Statement

In the motion of a body under no forces round a fixed point, show that the three diameters of the momental ellipsoid at the fixed point and the diameter of the ellipsoid reciprocal to the momental ellipsoid, determined respectively by the intersection of the invariable plane with the three principal planes and with the plane perpendicular to the instantaneous axis, describe areas proportional to the times, so that the accelerations of their extremities are directed to the centre.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point, and let $Oxyz$ be the body-fixed principal axes of inertia.

- The invariable plane passing through O is perpendicular to the constant angular momentum vector $\vec{d} = (A\omega_1, B\omega_2, C\omega_3)$ and is fixed in space. Its equation referred to body axes is
$$A\omega_1 x + B\omega_2 y + C\omega_3 z = 0.$$
- The momental ellipsoid is $Ax^2 + By^2 + Cz^2 = 1$[cite: 4, 18].
- The reciprocal ellipsoid (ellipsoid of gyration) is $\frac{x^2}{A} + \frac{y^2}{B} + \frac{z^2}{C} = 1$.

2. Dynamical Derivation

Intersection with the Principal Planes. Consider the intersection of the invariable plane with the principal plane $x = 0$ (yz-plane):

$$B\omega_2 y + C\omega_3 z = 0.$$

Let $P_1(0, y_1, z_1)$ be the extremity of this diameter on the momental ellipsoid ($By_1^2 + Cz_1^2 = 1$)[cite: 18]. The coordinates are proportional to $(0, C\omega_3, -B\omega_2)$:

$$y_1 = \frac{C\omega_3}{\sqrt{BC(B\omega_2^2 + C\omega_3^2)}}, \qquad z_1 = -\frac{B\omega_2}{\sqrt{BC(B\omega_2^2 + C\omega_3^2)}}.$$

Since the invariable plane is fixed in space, this diameter lies entirely within the fixed invariable plane for all t.

Euler's Equations and Areal Velocity. Euler's dynamic equations under no forces are[cite: 4]

$$A\dot{\omega}_1 = (B - C)\omega_2\omega_3, \qquad B\dot{\omega}_2 = (C - A)\omega_3\omega_1, \qquad C\dot{\omega}_3 = (A - B)\omega_1\omega_2.$$

Using the conservation of energy $A\omega_1^2 + B\omega_2^2 + C\omega_3^2 = 2T$ and angular momentum $A^2\omega_1^2 + B^2\omega_2^2 + C^2\omega_3^2 = d^2$, we have

$$B\omega_2^2 + C\omega_3^2 = 2T - A\omega_1^2.[cite: 4]$$

In the invariable plane, let χ_1 be the angle of orientation of OP_1. The rate of description of area in the fixed invariable plane by the radius vector $\vec{r}_1 = \vec{OP}_1$ is

$$\frac{dA_1}{dt} = \frac{1}{2}\left|\vec{r}_1 \times \frac{d\vec{r}_1}{dt}\right|.$$

The absolute velocity of the point P_1 in space is $\vec{v}_1 = \dot{\vec{r}}_1 + \vec{\omega} \times \vec{r}_1$. Projecting normal to the invariable plane along $\vec{d}/d$, the vector product evaluates to

$$(\vec{r}_1 \times \vec{v}_1) \cdot \frac{\vec{d}}{d} = \frac{1}{d}\left[\vec{r}_1 \cdot \vec{r}_1(\vec{\omega} \cdot \vec{d}) - (\vec{r}_1 \cdot \vec{\omega})(\vec{r}_1 \cdot \vec{d})\right].$$

Since $\vec{r}_1$ lies in the invariable plane, $\vec{r}_1 \cdot \vec{d} = 0$. Furthermore, $\vec{\omega} \cdot \vec{d} = 2T$, and $\vec{r}_1 \cdot \vec{r}_1 = r_1^2$. Thus:

$$\frac{dA_1}{dt} = \frac{T}{d} r_1^2 \left(1 - \frac{(\vec{r}_1 \cdot \vec{\omega})^2}{r_1^2 \omega^2}\right)^0 = \text{constant}.$$

Intersection with the Plane Perpendicular to the Instantaneous Axis. The plane perpendicular to the instantaneous axis is $\omega_1 x + \omega_2 y + \omega_3 z = 0$. Its intersection with the invariable plane $A\omega_1 x + B\omega_2 y + C\omega_3 z = 0$ defines a line whose direction cosines are proportional to

$$\xi = (B - C)\omega_2\omega_3 = A\dot{\omega}_1, \quad \eta = (C - A)\omega_3\omega_1 = B\dot{\omega}_2, \quad \zeta = (A - B)\omega_1\omega_2 = C\dot{\omega}_3.$$

On the reciprocal ellipsoid $\frac{x^2}{A} + \frac{y^2}{B} + \frac{z^2}{C} = 1$, the extremity P_4 has radius vector $\vec{r}_4$ satisfying $\vec{r}_4 \cdot \vec{d} = 0$. The identical scalar projection yields

$$\frac{dA_4}{dt} = \frac{1}{2}\left|\vec{r}_4 \times \vec{v}_4\right| = \text{constant}.$$

3. Proof & Final Scalar Form

Because each of the four diameters describes area in a fixed plane at a constant rate:

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\chi} = \text{constant}.$$

Differentiating $r^2\dot{\chi} = h_0$ with respect to time t:

$$\frac{d}{dt}\left(r^2\dot{\chi}\right) = r(2\dot{r}\dot{\chi} + r\ddot{\chi}) = 0.$$

The transverse acceleration component $a_\chi = r\ddot{\chi} + 2\dot{r}\dot{\chi}$ vanishes identically. Consequently, the acceleration vector of each extremity is purely radial, pointing directly toward the centre O:

$$\vec{a} = \left(\ddot{r} - r\dot{\chi}^2\right)\hat{u}_r \implies \text{directed to the centre } O.$$

■

Whittaker Chapter VI — Problem 28

Problem Statement

When a body moveable about a fixed point is acted on by forces whose moment round the instantaneous axis is always zero, show that the velocity of rotation is proportional to that radius vector of the momental ellipsoid which is in the direction of this axis.
Show that this theorem is still true if the body is moveable about a fixed point and also constrained to slide on a fixed surface.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point, and $Oxyz$ body-fixed principal axes of inertia.

- The momental ellipsoid at O is given by $Ax^2 + By^2 + Cz^2 = 1$.
- Let $\vec{\omega} = (\omega_1, \omega_2, \omega_3)$ be the angular velocity of the body, and $\omega = \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2}$ the magnitude of rotation.
- Let $(l, m, n) = (\omega_1/\omega, \omega_2/\omega, \omega_3/\omega)$ be the direction cosines of the instantaneous axis of rotation.
- Let ρ be the length of the radius vector of the momental ellipsoid in the direction (l, m, n). Its coordinates on the ellipsoid are $(x, y, z) = (\rho l, \rho m, \rho n)$.

2. Dynamical Derivation

Moment of Inertia and Radius Vector. Substituting $(\rho l, \rho m, \rho n)$ into the momental ellipsoid $Ax^2 + By^2 + Cz^2 = 1$:

$$\rho^2 \left(Al^2 + Bm^2 + Cn^2\right) = 1 \implies Al^2 + Bm^2 + Cn^2 = \frac{1}{\rho^2}.$$

The scalar moment of inertia I of the body about the instantaneous axis is $I = Al^2 + Bm^2 + Cn^2 = 1/\rho^2$. The total kinetic energy of the body is

$$T = \frac{1}{2}\left(A\omega_1^2 + B\omega_2^2 + C\omega_3^2\right) = \frac{1}{2}I\omega^2 = \frac{\omega^2}{2\rho^2}. \tag{1.11}$$

Work-Energy Relation. Let (L, M, N) denote the moments of the external forces about the axes $Oxyz$. Taking the scalar product of Euler's dynamical equations with $(\omega_1, \omega_2, \omega_3)$:

$$\omega_1\left[A\dot{\omega}_1 - (B-C)\omega_2\omega_3\right] + \omega_2\left[B\dot{\omega}_2 - (C-A)\omega_3\omega_1\right] + \omega_3\left[C\dot{\omega}_3 - (A-B)\omega_1\omega_2\right] = L\omega_1 + M\omega_2 + N\omega_3.[cite: 4$$

The gyroscopic coupling terms cancel in pairs:

$$-(B-C)\omega_1\omega_2\omega_3 - (C-A)\omega_1\omega_2\omega_3 - (A-B)\omega_1\omega_2\omega_3 = 0.$$

Thus:

$$\frac{d}{dt}\left[\frac{1}{2}\left(A\omega_1^2 + B\omega_2^2 + C\omega_3^2\right)\right] = \frac{dT}{dt} = L\omega_1 + M\omega_2 + N\omega_3 = \vec{\tau}\cdot\vec{\omega}.$$

Here $\vec{\tau}\cdot\vec{\omega} = \omega\,\tau_{\text{axis}}$, where τ_{axis} is the total moment of external forces about the instantaneous axis. By hypothesis, $\tau_{\text{axis}} = 0$ at all times. Therefore:

$$\frac{dT}{dt} = 0 \implies T = \text{constant}.$$

3. Proof & Final Scalar Form

From equation (6), since T is constant:

$$\frac{\omega^2}{2\rho^2} = T = \text{constant} \implies \frac{\omega}{\rho} = \sqrt{2T} = \text{constant}.$$

Hence, the angular velocity ω is strictly proportional to the radius vector ρ of the momental ellipsoid along the instantaneous axis:

$$\omega \propto \rho.$$

Extension to Sliding on a Fixed Smooth Surface. If the body is constrained to remain in contact with a fixed smooth surface, the constraint reaction $\vec{R}$ is normal to the surface. Let $\vec{v}_c$ be the velocity of the contact point. Since the surface is smooth, $\vec{R} \cdot \vec{v}_c = 0$. Because the fixed point O is stationary, the power developed by the reaction force is $\vec{R} \cdot \vec{v}_c = 0$. The reaction contributes zero work, so $\frac{dT}{dt} = 0$ still holds identically. Consequently, $T =$ constant, and the proportionality $\omega \propto \rho$ remains valid. ■

Whittaker Chapter VI — Problem 29

Problem Statement

A plane lamina is initially moving with equal angular velocities Ω about the principal axes of greatest and least moment of inertia at its centre of mass, and has no angular velocity about the third principal axis; express the angular velocities about these axes as elliptic functions of the time, supposing no forces to act on the lamina.
If θ be the angle between the plane of the lamina and any fixed plane, show that

$$\frac{d^2\theta}{dt^2} + 2\Omega \left\{ \Omega^2 - \left(\frac{d\theta}{dt} \right)^2 \right\}^{\frac{1}{2}} \mathrm{dn}(\Omega t) = \left\{ \Omega^2 - \left(\frac{d\theta}{dt} \right)^2 \right\} \cot\theta [cite : 18].$$

1. Geometric Setup & Coordinate Definitions

Let G be the centre of mass of the plane lamina, with principal axes Gx, Gy in the plane of the lamina and Gz perpendicular to it.

Moments of Inertia. Let A and B be the in-plane principal moments of inertia, with $A > B$. By the perpendicular axis theorem:

$$C = A + B.$$

The moments of inertia in descending order are $C > A > B$:

- Greatest moment of inertia: C (axis Gz, normal to lamina).
- Least moment of inertia: B (axis Gy, in the lamina).
- Intermediate moment of inertia: A (axis Gx, in the lamina).

Initially:

$$\omega_1(0) = 0, \qquad \omega_2(0) = \Omega, \qquad \omega_3(0) = \Omega.$$

2. Dynamical Derivation

Integration of Euler's Equations. Euler's equations for the torque-free motion are

$$\begin{aligned} A\dot{\omega}_1 &= (B - C)\omega_2\omega_3 = -A\omega_2\omega_3 \implies \dot{\omega}_1 = -\omega_2\omega_3, \\ B\dot{\omega}_2 &= (C - A)\omega_3\omega_1 = B\omega_3\omega_1 \implies \dot{\omega}_2 = \omega_3\omega_1, \\ C\dot{\omega}_3 &= (A - B)\omega_1\omega_2. \end{aligned}$$

Multiplying the first equation by ω_1 and the second by ω_2 and adding:

$$\omega_1\dot{\omega}_1 + \omega_2\dot{\omega}_2 = 0 \implies \omega_1^2 + \omega_2^2 = \text{constant} = \Omega^2.$$

Writing the standard Jacobi elliptic ansatz with $u = \Omega t$:

$$\omega_1 = -\Omega\,\mathrm{sn}(\Omega t, k), \qquad \omega_2 = \Omega\,\mathrm{cn}(\Omega t, k), \qquad \omega_3 = \Omega\,\mathrm{dn}(\Omega t, k).$$

Differentiating ω_3 gives $\dot{\omega}_3 = -k^2\Omega^2\,\mathrm{sn}(\Omega t)\mathrm{cn}(\Omega t) = k^2\omega_1\omega_2$. Matching this with Euler's third equation $\dot{\omega}_3 = \frac{A-B}{C}\omega_1\omega_2$:

$$k^2 = \frac{A - B}{C} = \frac{A - B}{A + B} < 1.$$

Thus, the angular velocities are completely determined in terms of Jacobian elliptic functions.

3. Proof & Final Scalar Form

Let $\vec{K} = (l, m, n)$ be the unit normal to the fixed plane referred to the moving principal axes $Gxyz$. The plane of the lamina has unit normal $\hat{z} = (0, 0, 1)$. The angle θ between the plane of the lamina and the fixed plane equals the angle between their normal vectors:

$$\cos\theta = n, \qquad \sin^2\theta = 1 - n^2 = l^2 + m^2 [cite: 18].$$

Because $\vec{K}$ is fixed in space, its time-flux in body coordinates is zero: $\dot{\vec{K}} + \vec{\omega} \times \vec{K} = 0$, giving

$$\dot{l} = m\omega_3 - n\omega_2, \tag{1.12}$$
$$\dot{m} = n\omega_1 - l\omega_3, \tag{1.13}$$
$$\dot{n} = l\omega_2 - m\omega_1. \tag{1.14}$$

Since $n = \cos\theta$, we have

$$\dot{n} = -\sin\theta\,\dot{\theta} = l\omega_2 - m\omega_1.$$

Squaring equation (9) and using Lagrange's identity:

$$\begin{aligned}\sin^2\theta\,\dot{\theta}^2 &= (l\omega_2 - m\omega_1)^2 = (l^2 + m^2)(\omega_1^2 + \omega_2^2) - (l\omega_1 + m\omega_2)^2 \\ &= \sin^2\theta\,\Omega^2 - (l\omega_1 + m\omega_2)^2.\end{aligned}$$

Therefore:

$$(l\omega_1 + m\omega_2)^2 = \sin^2\theta\left(\Omega^2 - \dot{\theta}^2\right) \implies l\omega_1 + m\omega_2 = \sin\theta\sqrt{\Omega^2 - \dot{\theta}^2}. \tag{1.15}$$

Differentiating $\dot{n} = -\sin\theta\,\dot{\theta}$ with respect to t:

$$\ddot{n} = -\cos\theta\,\dot{\theta}^2 - \sin\theta\,\ddot{\theta}. \tag{1.16}$$

Also, differentiating $\dot{n} = l\omega_2 - m\omega_1$ directly using equations (7), (8) and Euler's equations:

$$\begin{aligned}\ddot{n} &= \dot{l}\omega_2 + l\dot{\omega}_2 - \dot{m}\omega_1 - m\dot{\omega}_1 \\ &= (m\omega_3 - n\omega_2)\omega_2 + l(\omega_3\omega_1) - (n\omega_1 - l\omega_3)\omega_1 - m(-\omega_2\omega_3) \\ &= 2\omega_3(l\omega_1 + m\omega_2) - n(\omega_1^2 + \omega_2^2) \\ &= 2\omega_3(l\omega_1 + m\omega_2) - \Omega^2\cos\theta.\end{aligned}$$

Equating the two expressions for $\ddot{n}$:

$$-\sin\theta\,\ddot{\theta} - \cos\theta\,\dot{\theta}^2 = 2\omega_3(l\omega_1 + m\omega_2) - \Omega^2\cos\theta.$$

Substitute equation (10) and $\omega_3 = \Omega\,\mathrm{dn}(\Omega t)$:

$$-\sin\theta\,\ddot{\theta} - \cos\theta\,\dot{\theta}^2 = 2\Omega\,\mathrm{dn}(\Omega t)\sin\theta\sqrt{\Omega^2 - \dot{\theta}^2} - \Omega^2\cos\theta.$$

Rearranging:

$$\sin\theta\,\ddot{\theta} + 2\Omega\sin\theta\sqrt{\Omega^2 - \dot{\theta}^2}\,\mathrm{dn}(\Omega t) = (\Omega^2 - \dot{\theta}^2)\cos\theta.$$

Dividing across by $\sin\theta$ yields the required relation:

$$\frac{d^2\theta}{dt^2} + 2\Omega\left\{\Omega^2 - \left(\frac{d\theta}{dt}\right)^2\right\}^{\frac{1}{2}}\mathrm{dn}(\Omega t) = \left\{\Omega^2 - \left(\frac{d\theta}{dt}\right)^2\right\}\cot\theta.$$

■

Whittaker Chapter VI — Problem 30

Problem Statement

A rigid body is kinetically symmetrical about an axis which passes through a fixed point above its centre of gravity and is set in motion in any manner; show that in the subsequent motion, except in one case, the centre of gravity can never be vertically over the fixed point; and find the greatest height it attains.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point, and let the axis of kinetic symmetry be Oz, passing through the centre of gravity G at distance h from O.

- Principal moments of inertia at O: $A = B$ and C.
- In stable equilibrium, G hangs vertically below O. Let θ be the inclination of OG to the downward vertical, so $\theta = 0$ corresponds to G directly below O, and $\theta = \pi$ to G vertically above O.
- Let ϕ be the precession angle about the vertical, and ψ the spin angle about Oz.

2. Lagrangian / Dynamical Derivation

First Integrals of the Symmetric Top. The Lagrangian $L = T - V$ with vertical axis pointing downward is

$$L = \frac{1}{2}A\left(\dot{\theta}^2 + \dot{\phi}^2 \sin^2\theta\right) + \frac{1}{2}C\left(\dot{\psi} + \dot{\phi}\cos\theta\right)^2 + Mgh\cos\theta.$$

Because ϕ and ψ are cyclic[cite: 4]:

$$C\left(\dot{\psi} + \dot{\phi}\cos\theta\right) = Cn = \text{constant}, \tag{1.17}$$

$$A\dot{\phi}\sin^2\theta + Cn\cos\theta = d = \text{constant}. \tag{1.18}$$

Solving equation (13) for $\dot{\phi}$:

$$\dot{\phi} = \frac{d - Cn\cos\theta}{A\sin^2\theta}. \tag{1.19}$$

The energy integral is

$$\frac{1}{2}A\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) - Mgh\cos\theta = E.$$

Substituting equation (14) into the energy equation and multiplying by $2A$:

$$A^2\dot{\theta}^2 = 2A(E + Mgh\cos\theta) - \frac{(d - Cn\cos\theta)^2}{\sin^2\theta}. \tag{1.20}$$

3. Proof & Final Scalar Form

Let $u = \cos\theta \in [-1, 1]$. In terms of u, equation (15) becomes

$$A^2\left(\frac{du}{dt}\right)^2 = 2A(E + Mghu)(1 - u^2) - (d - Cnu)^2 \equiv f(u).$$

For the centre of gravity to pass vertically over the fixed point, the body must reach $\theta = \pi$, which corresponds to $u = -1$. Evaluating $f(u)$ at $u = -1$:

$$f(-1) = 2A(E - Mgh)(1 - (-1)^2) - (d + Cn)^2 = -(d + Cn)^2.$$

Since $A^2\dot{u}^2 \geq 0$, physical motion requires $f(u) \geq 0$. At $u = -1$, $f(-1) = -(d + Cn)^2 \leq 0$. Thus, $f(-1)$ can be non-negative if and only if

$$d + Cn = 0 \iff d = -Cn.$$

Except in this unique singular case (where the angular momentum about the vertical precisely equals $-Cn$), $f(-1) < 0$, which strictly prevents u from ever reaching -1. Therefore, the centre of gravity can never be vertically over the fixed point.

Greatest Height Attained. The cubic polynomial $f(u)$ has three real roots $u_1 > u_2 \geq u_3$, where the physical oscillation of $u = \cos\theta$ is confined to the interval $u_3 \leq u \leq u_2 \leq 1$[cite: 4]. The minimum value of $u = \cos\theta$ is u_3 (the smallest root of $f(u) = 0$ in $[-1, 1]$), which corresponds to the highest elevation of G. Since the depth of G below O is $h\cos\theta$, the vertical height z of G above the horizontal plane through O is $-h\cos\theta = -hu$. The greatest height attained by the centre of gravity is therefore

$$z_{\max} = -hu_3,$$

where u_3 is the smallest real root in $[-1, 1]$ of the cubic equation

$$2A(E + Mghu)(1 - u^2) - (d - Cnu)^2 = 0[cite : 4].$$

■

Whittaker Chapter VI — Problem 31

Problem Statement

In the motion of the top on the rough plane, show that there exists an auxiliary set of axes $O\xi\eta\zeta$ whose motion with respect to the fixed axes $OXYZ$ and also with respect to the moving axes $Oxyz$ is a Poinsot motion; the invariable planes being the horizontal plane in the former case, and the plane perpendicular to the axis of the body in the second case.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed apex of the heavy symmetric top ($A = B$).

- Let $OXYZ$ be fixed spatial axes with OZ directed vertically upwards.
- Let $Oxyz$ be moving axes fixed in the body, with Oz along the symmetry axis of the top.
- The orientation of $Oxyz$ relative to $OXYZ$ is parameterized by the Eulerian angles (θ, ϕ, ψ), where θ is the nutation angle, ϕ the precession, and ψ the spin.

Cayley-Klein Representation. Following Jacobi, the rotation from $OXYZ$ to $Oxyz$ is represented by the Cayley-Klein parameters $(\alpha, \beta, \gamma, \delta)$:

$$\alpha = \cos\frac{\theta}{2}e^{\frac{i}{2}(\phi+\psi)}, \quad \beta = i\sin\frac{\theta}{2}e^{\frac{i}{2}(\phi-\psi)}, \quad \gamma = i\sin\frac{\theta}{2}e^{-\frac{i}{2}(\phi-\psi)}, \quad \delta = \cos\frac{\theta}{2}e^{-\frac{i}{2}(\phi+\psi)}.$$

2. Dynamical Derivation

Decomposition of the Top Motion. The nutation angle $\theta(t)$ of the heavy symmetric top is governed by the Weierstrass elliptic function:

$$u = \cos\theta, \qquad \dot{u}^2 = P(u).$$

In Jacobi's reduction, the two cyclic velocity equations for ϕ and ψ are

$$\dot{\phi} = \frac{d - Cnu}{A(1-u^2)}, \qquad \dot{\psi} = \frac{Cn}{C} - \cos\theta\,\dot{\phi}.$$

Separating into partial fractions with respect to $1 \pm u$:

$$\dot{\phi} + \dot{\psi} = \frac{Cn}{C} + \frac{d - Cn}{A(1+u)}, \qquad \dot{\phi} - \dot{\psi} = -\frac{Cn}{C} + \frac{d + Cn}{A(1-u)}.$$

Jacobi demonstrated that the finite transformation from fixed space $OXYZ$ to body frame $Oxyz$ factors uniquely through an intermediate reference frame $O\xi\eta\zeta$:

$$\mathcal{R}(OXYZ \to Oxyz) = \mathcal{R}_2(O\xi\eta\zeta \to Oxyz) \circ \mathcal{R}_1(OXYZ \to O\xi\eta\zeta).$$

3. Proof & Final Scalar Form

First Relative Motion: $O\xi\eta\zeta$ with Respect to $OXYZ$. The intermediate frame $O\xi\eta\zeta$ has its $O\zeta$-axis aligned with the instantaneous angular momentum vector of the body. The angular velocity components of $O\xi\eta\zeta$ relative to $OXYZ$ satisfy Euler's dynamic equations under no external forces with an invariable line along OZ. Consequently, the plane perpendicular to the vertical OZ (the horizontal plane XOY) serves as the invariable plane of this motion[cite: 18]. The motion of $O\xi\eta\zeta$ relative to $OXYZ$ is therefore an exact **Poinsot motion**.

Second Relative Motion: $Oxyz$ with Respect to $O\xi\eta\zeta$. In the frame $O\xi\eta\zeta$, the axis of kinetic symmetry of the body Oz precesses symmetrically about $O\zeta$[cite: 18]. Reversing the reference, the angular velocity of the body $Oxyz$ relative to $O\xi\eta\zeta$ is torque-free about the body axis Oz. The invariable plane for this relative motion is the plane perpendicular to the symmetry axis Oz of the body (the principal plane xOy). The relative motion of $Oxyz$ with respect to $O\xi\eta\zeta$ is likewise an exact **Poinsot motion**.

Therefore, there exists an auxiliary set of axes $O\xi\eta\zeta$ whose motion with respect to the fixed axes $OXYZ$ is a Poinsot motion with the horizontal plane as invariable plane, and whose motion with respect to the body axes $Oxyz$ is a Poinsot motion with the plane perpendicular to the axis of the body as invariable plane. ■

Whittaker Chapter VI — Problem 32

Problem Statement

A uniform solid of revolution moves about a point, so that its motion may be represented by the uniform rolling of a cone of semi-vertical angle α fixed in the body upon an equal cone fixed in space, the axis of the former being the axis of revolution. Show that the couple necessary to maintain the motion is of magnitude

$$\frac{1}{2}\Omega^2 \tan\alpha\{C + (C - A)\cos 2\alpha\},$$

where Ω is the resultant angular velocity and A and C the principal moments of inertia at the point, and that the couple lies in the plane of the axes of the cones.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point of the body.

Cones and Axes of Symmetry.

- Let the axis of the fixed space cone be OZ. The space cone has semi-vertical angle α.
- Let the axis of revolution of the body be Oz. The body cone has semi-vertical angle α with axis Oz.
- The two equal cones touch along a common generator OI, which serves as the instantaneous axis of rotation.
- The angle between OZ and OI is α, and the angle between Oz and OI is α. Consequently, the angle between the fixed axis OZ and the body symmetry axis Oz is $\theta = 2\alpha$.

Inertial Parameters and Moving Frame. Because the body is a solid of revolution about Oz, its principal moments of inertia at O are A, A, C, where C is the moment of inertia about Oz. Define an orthogonal right-handed triad of axes $O123$ such that:

- Axis 3 is directed along the body axis of revolution Oz.
- Axis 1 lies in the plane of the axes of the two cones (OZ and Oz), perpendicular to Oz and pointing toward the side containing OZ.
- Axis 2 is perpendicular to the plane of the axes of the two cones.

Because $A_1 = A_2 = A$, $O123$ is a set of principal axes of inertia at O at every instant.

Angular Velocity Components. The resultant angular velocity of magnitude Ω is directed along the instantaneous line of contact OI. Since OI lies in the $1-3$ plane at an angle α to axis 3, its scalar components along $O123$ are

$$\omega_1 = \Omega \sin\alpha, \qquad \omega_2 = 0, \qquad \omega_3 = \Omega \cos\alpha.$$

2. Lagrangian / Dynamical Derivation

Angular Momentum. The scalar components of the angular momentum vector $\vec{H}$ along the principal axes $O123$ are

$$\begin{aligned} H_1 &= A\omega_1 = A\Omega \sin\alpha, \\ H_2 &= A\omega_2 = 0, \\ H_3 &= C\omega_3 = C\Omega \cos\alpha. \end{aligned}$$

Because $H_2 = 0$, the angular momentum vector $\vec{H}$ lies entirely within the plane containing the axes of the two cones (OZ and Oz).

Precession of the Plane of the Cones. As the body cone rolls uniformly on the fixed space cone, the line of contact OI and the symmetry axis Oz precess about the fixed axis OZ with a constant angular speed $\dot{\phi}$. Decomposing the resultant angular velocity into the precession $\dot{\phi}$ about OZ and a spin $\dot{\psi}$ about Oz:

$$\vec{\Omega} = \dot{\phi}\hat{Z} + \dot{\psi}\hat{3}.$$

Resolving this vector equation along and perpendicular to the contact line OI in the $1-3$ plane:

$$\begin{aligned} &\dot{\phi}\sin\alpha - \dot{\psi}\sin\alpha = 0 \implies \dot{\psi} = \dot{\phi}, \\ &\dot{\phi}\cos\alpha + \dot{\psi}\cos\alpha = \Omega \implies 2\dot{\phi}\cos\alpha = \Omega \implies \dot{\phi} = \frac{\Omega}{2\cos\alpha}. \end{aligned}$$

Rate of Change of Angular Momentum. The triad $O123$ rotates in space about the fixed axis OZ with angular velocity $\vec{\omega}_f = \dot{\phi}\hat{Z}$. Referred to $O123$, the unit vector along OZ is

$$\hat{Z} = \sin 2\alpha\, \hat{1} + \cos 2\alpha\, \hat{3}.$$

The time derivative of the angular momentum vector in space is given by

$$\begin{aligned} \frac{d\vec{H}}{dt} &= \vec{\omega}_f \times \vec{H} = (\dot{\phi}\hat{Z}) \times (H_1\hat{1} + H_3\hat{3}) \\ &= \dot{\phi}(\sin 2\alpha\, \hat{1} + \cos 2\alpha\, \hat{3}) \times (H_1\hat{1} + H_3\hat{3}) \\ &= \dot{\phi}\left[H_3 \sin 2\alpha\, (\hat{1}\times\hat{3}) + H_1 \cos 2\alpha\, (\hat{3}\times\hat{1})\right] \\ &= \dot{\phi}\left[-H_3 \sin 2\alpha\, \hat{2} + H_1 \cos 2\alpha\, \hat{2}\right] \\ &= \dot{\phi}\,(H_1 \cos 2\alpha - H_3 \sin 2\alpha)\,\hat{2}. \end{aligned}$$

3. Proof & Final Scalar Form

The torque $\vec{N}$ required to maintain the steady rolling motion is

$$\vec{N} = \frac{d\vec{H}}{dt} = \dot{\phi}\,(H_1 \cos 2\alpha - H_3 \sin 2\alpha)\,\hat{2}.$$

Because the vector $\vec{N}$ is directed along axis 2 (perpendicular to the plane containing the axes of the two cones), the couple itself acts in the plane perpendicular to axis 2, which is precisely the plane containing the axes of the cones.

To find the scalar magnitude N of this couple, substitute $H_1 = A\Omega\sin\alpha$, $H_3 = C\Omega\cos\alpha$, and $\dot{\phi} = \frac{\Omega}{2\cos\alpha}$:

$$\begin{aligned} H_3\sin 2\alpha - H_1\cos 2\alpha &= C\Omega\cos\alpha(2\sin\alpha\cos\alpha) - A\Omega\sin\alpha\cos 2\alpha \\ &= \Omega\sin\alpha\left(2C\cos^2\alpha - A\cos 2\alpha\right). \end{aligned}$$

Using the trigonometric identity $2\cos^2\alpha = 1 + \cos 2\alpha$:

$$\begin{aligned} 2C\cos^2\alpha - A\cos 2\alpha &= C(1+\cos 2\alpha) - A\cos 2\alpha \\ &= C + (C-A)\cos 2\alpha. \end{aligned}$$

Multiplying by $\dot{\phi} = \frac{\Omega}{2\cos\alpha}$:

$$\begin{aligned} N &= \frac{\Omega}{2\cos\alpha}\cdot\Omega\sin\alpha\left\{C + (C-A)\cos 2\alpha\right\} \\ &= \frac{1}{2}\Omega^2\frac{\sin\alpha}{\cos\alpha}\left\{C + (C-A)\cos 2\alpha\right\} \\ &= \frac{1}{2}\Omega^2\tan\alpha\{C + (C-A)\cos 2\alpha\}. \end{aligned}$$

■

Whittaker Chapter VI — Problem 33

Problem Statement

A vertical plane is made to rotate with uniform angular velocity about a vertical axis in itself, and a perfectly rough cone of revolution has its vertex fixed at a point of that axis. Show that, if the line of contact make an angle θ with the vertical, and β and γ be the extreme values of θ, and α be the semi-vertical angle of the cone,

$$k^2\left(\frac{d\theta}{dt}\right)^2 = 2gh\frac{\sin^2\alpha}{\cos\alpha}\frac{(\cos\theta-\cos\beta)(\cos\gamma-\cos\theta)}{\cos\beta+\cos\gamma},$$

where h is the distance of the centre of gravity of the cone from its vertex, and k its radius of gyration about a generator.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed vertex of the cone on the vertical axis of rotation OZ.

Rotating Plane Coordinates. The vertical plane rotates with uniform angular velocity ω about the vertical axis OZ. In the rotating plane, choose:

- OZ along the downward vertical axis.
- OX' horizontal in the plane.
- $\hat{n} = \hat{Y}'$ normal to the vertical plane.

The cone of semi-vertical angle α touches the vertical plane along a generator OI, which makes an angle θ with the vertical OZ. The unit vector along the contact generator is

$$\hat{u}_I = \cos\theta\,\hat{Z} + \sin\theta\,\hat{X}'.$$

The unit vector in the vertical plane perpendicular to OI is

$$\hat{2} = -\sin\theta\,\hat{Z} + \cos\theta\,\hat{X}',$$

so that the vertical unit vector resolves as

$$\hat{Z} = \cos\theta\,\hat{u}_I - \sin\theta\,\hat{2}.$$

Cone Orientation and Centre of Gravity. The vertical plane is tangent to the cone along OI. The axis of the cone OC lies in the plane through OI normal to the tangent plane:

$$\hat{u}_C = \cos\alpha\,\hat{u}_I + \sin\alpha\,\hat{n}.$$

The centre of gravity G of the cone lies at distance h along OC from the vertex O. Its vertical coordinate along OZ is

$$Z_G = h(\hat{u}_C\cdot\hat{Z}) = h\cos\alpha\cos\theta.$$

2. Lagrangian / Dynamical Derivation

Kinematics of Rolling. Relative to the rotating plane, the contact line turns at rate $\dot{\theta}$. The condition of rolling without slipping on the rotating plane requires the relative angular velocity to be directed along the instantaneous generator of contact:

$$\vec{\omega}_{\text{rel}} = (\dot{\theta}\cot\alpha)\hat{u}_I.$$

The absolute angular velocity of the cone in space is

$$\begin{aligned}\vec{\omega} &= \omega\hat{Z} + \vec{\omega}_{\text{rel}} = \omega(\cos\theta\,\hat{u}_I - \sin\theta\,\hat{2}) + (\dot{\theta}\cot\alpha)\hat{u}_I \\ &= (\omega\cos\theta + \dot{\theta}\cot\alpha)\hat{u}_I - (\omega\sin\theta)\hat{2}.\end{aligned}$$

Kinetic Energy. The generator $\hat{u}_I$ and axis $\hat{2}$ are mutually perpendicular. Since $\hat{2}$ is perpendicular to the axial plane containing $\hat{u}_I$, the product of inertia between $\hat{u}_I$ and $\hat{2}$ vanishes identically. The moment of inertia about the generator $\hat{u}_I$ is Mk^2, and about axis $\hat{2}$ is the transverse principal moment A. The kinetic energy of the cone is

$$\begin{aligned}T &= \frac{1}{2}Mk^2(\omega\cos\theta + \dot{\theta}\cot\alpha)^2 + \frac{1}{2}A(\omega\sin\theta)^2 \\ &= \frac{1}{2}Mk^2\cot^2\alpha\,\dot{\theta}^2 + Mk^2\omega\cot\alpha\cos\theta\,\dot{\theta} + \frac{1}{2}Mk^2\omega^2\cos^2\theta + \frac{1}{2}A\omega^2\sin^2\theta.\end{aligned}$$

The cross-term is an exact time derivative:

$$Mk^2\omega\cot\alpha\cos\theta\,\dot{\theta} = \frac{d}{dt}\left(Mk^2\omega\cot\alpha\sin\theta\right),$$

and therefore produces no contribution to the equations of motion.

Energy Integral of Relative Motion. Taking OZ vertically downwards, the gravitational potential energy is

$$V = -Mgh\cos\alpha\cos\theta.$$

Using the Jacobi integral for systems with uniform rotation ω:

$$\frac{1}{2}Mk^2\cot^2\alpha\,\dot{\theta}^2 + V_{\text{eff}}(\theta) = E,$$

where the effective potential energy is

$$\begin{aligned}V_{\text{eff}}(\theta) &= -Mgh\cos\alpha\cos\theta - \frac{1}{2}Mk^2\omega^2\cos^2\theta - \frac{1}{2}A\omega^2\sin^2\theta \\ &= -Mgh\cos\alpha\cos\theta - \frac{1}{2}(Mk^2 - A)\omega^2\cos^2\theta - \frac{1}{2}A\omega^2.\end{aligned}$$

Multiplying the energy equation by $\frac{2}{M}\tan^2\alpha$:

$$k^2\left(\frac{d\theta}{dt}\right)^2 = \tan^2\alpha\left[2gh\cos\alpha\cos\theta + \frac{Mk^2 - A}{M}\omega^2\cos^2\theta + C_0\right],$$

where C_0 is a constant of integration.

3. Proof & Final Scalar Form

The bracketed expression is a quadratic polynomial in $\cos\theta$. Because β and γ are the extreme values of θ, the angular velocity $\frac{d\theta}{dt}$ vanishes when $\theta = \beta$ and $\theta = \gamma$. Hence, the quadratic polynomial must have roots at $\cos\theta = \cos\beta$ and $\cos\theta = \cos\gamma$, and can be factored as

$$B_0(\cos\theta - \cos\beta)(\cos\gamma - \cos\theta) = B_0\left[-\cos^2\theta + (\cos\beta + \cos\gamma)\cos\theta - \cos\beta\cos\gamma\right],$$

where B_0 is a scalar constant.

Equating the coefficient of the linear term in $\cos\theta$ (which arises solely from the gravitational potential):

$$B_0(\cos\beta + \cos\gamma) = 2gh\cos\alpha \implies B_0 = \frac{2gh\cos\alpha}{\cos\beta + \cos\gamma}.$$

Substituting this expression for B_0 into the differential equation:

$$\begin{aligned} k^2\left(\frac{d\theta}{dt}\right)^2 &= \tan^2\alpha \cdot \frac{2gh\cos\alpha}{\cos\beta + \cos\gamma}(\cos\theta - \cos\beta)(\cos\gamma - \cos\theta) \\ &= \frac{\sin^2\alpha}{\cos^2\alpha} \cdot \frac{2gh\cos\alpha}{\cos\beta + \cos\gamma}(\cos\theta - \cos\beta)(\cos\gamma - \cos\theta) \\ &= 2gh\frac{\sin^2\alpha}{\cos\alpha}\frac{(\cos\theta - \cos\beta)(\cos\gamma - \cos\theta)}{\cos\beta + \cos\gamma}. \end{aligned}$$

■

Whittaker Chapter VI — Problem 34

Problem Statement

A body can rotate freely about a fixed vertical axis for which its moment of inertia is I: the body carries a second body in the form of a disc which can rotate about a horizontal axis, fixed in the first body and intersecting the vertical axis. In the position of equilibrium the moments and product of inertia of the disc with regard to the vertical and horizontal axes respectively are A, B, F. Prove that if the system start from rest with the plane of the disc inclined at an angle α to the vertical, the first body will oscillate through an angle

$$\frac{2F}{\{B(A+I)\}^{\frac{1}{2}}}\arctan\left\{\frac{B^{\frac{1}{2}}\sin\alpha}{(A+I)^{\frac{1}{2}}}\right\}.$$

1. Geometric Setup & Coordinate Definitions

Let the fixed vertical axis of rotation be Oz.

Axes and Coordinates.

- Let ϕ denote the angle of rotation of the first body about the fixed vertical axis Oz.
- Let Ox be the horizontal axis fixed in the first body about which the disc rotates.
- Let Oy be the horizontal axis perpendicular to Ox, fixed in the first body and completing the right-handed Cartesian frame $Oxyz$.
- Let θ denote the rotation angle of the disc about the horizontal axis Ox, measured from its equilibrium position ($\theta = 0$).

Inertial Parameters of the Disc. In the vertical equilibrium configuration ($\theta = 0$), the plane of the disc is vertical (xz-plane), so $y = 0$ for all particles of the disc.

- Moment of inertia about the vertical axis Oz:

$$A = \sum m(x^2 + y^2) = \sum mx^2.$$

- Moment of inertia about the horizontal axis of rotation Ox:

$$B = \sum m(y^2 + z^2) = \sum mz^2.$$

- Product of inertia between the vertical and horizontal axes:

$$F = \sum mzx.$$

Inertia at Tilt Angle θ. When the disc turns by an angle θ about Ox, the coordinates of a typical particle become

$$x' = x, \qquad y' = z\sin\theta, \qquad z' = z\cos\theta.$$

The moment of inertia about the vertical axis Oz at angle θ is

$$I_{zz}(\theta) = \sum m\left(x'^2 + y'^2\right) = \sum m\left(x^2 + z^2\sin^2\theta\right) = A + B\sin^2\theta.$$

The product of inertia between Oz and Ox is

$$I_{zx}(\theta) = \sum mz'x' = \cos\theta\sum mzx = F\cos\theta.$$

2. Lagrangian / Dynamical Derivation

Angular Momentum about the Vertical Axis. The components of the angular velocity of the disc resolved along the frame $Oxyz$ fixed in the first body are

$$\omega_x = \dot\theta, \qquad \omega_y = 0, \qquad \omega_z = \dot\phi.$$

The angular momentum of the disc about the vertical axis Oz is

$$h_z^{\text{disc}} = I_{zz}(\theta)\omega_z - I_{zx}(\theta)\omega_x = \left(A + B\sin^2\theta\right)\dot\phi - F\cos\theta\,\dot\theta.$$

Including the first body, which has moment of inertia I about the vertical axis, the total angular momentum of the system about Oz is

$$H_z = I\dot\phi + h_z^{\text{disc}} = \left(I + A + B\sin^2\theta\right)\dot\phi - F\cos\theta\,\dot\theta.$$

Conservation of Angular Momentum. Because all external forces (gravity) act vertically through or parallel to the vertical axis, their moment about Oz vanishes identically. Therefore, H_z is strictly constant. Since the system is released from rest at tilt angle $\theta = \alpha$, the initial angular momentum is zero:

$$\left(I + A + B\sin^2\theta\right)\dot\phi - F\cos\theta\,\dot\theta = 0.$$

Solving for the angular velocity $\dot\phi$ of the first body:

$$\dot\phi = \frac{F\cos\theta}{I + A + B\sin^2\theta}\dot\theta \implies \frac{d\phi}{d\theta} = \frac{F\cos\theta}{(A + I) + B\sin^2\theta}.$$

3. Proof & Final Scalar Form

Because the system is conservative and starts from rest at inclination α, the disc oscillates symmetrically between $\theta = -\alpha$ and $\theta = +\alpha$. The total angle through which the first body oscillates is

$$\Delta\phi = \int_{-\alpha}^{\alpha} \frac{d\phi}{d\theta}\, d\theta = 2\int_{0}^{\alpha} \frac{F\cos\theta}{(A+I) + B\sin^2\theta}\, d\theta.$$

Substitute $u = \sin\theta$, with $du = \cos\theta\, d\theta$:

$$\begin{aligned}
\Delta\phi &= 2F \int_{0}^{\sin\alpha} \frac{du}{(A+I) + Bu^2} \\
&= \frac{2F}{B} \int_{0}^{\sin\alpha} \frac{du}{\frac{A+I}{B} + u^2} \\
&= \frac{2F}{B} \cdot \sqrt{\frac{B}{A+I}} \left[\arctan\left(u\sqrt{\frac{B}{A+I}} \right) \right]_{0}^{\sin\alpha} \\
&= \frac{2F}{\{B(A+I)\}^{\frac{1}{2}}} \arctan\left\{ \frac{B^{\frac{1}{2}}\sin\alpha}{(A+I)^{\frac{1}{2}}} \right\}.
\end{aligned}$$

■

Whittaker Chapter VI — Problem 35

Problem Statement

A gyrostat consists of a heavy symmetrical flywheel freely mounted in a heavy spherical case and is suspended from a fixed point by a string of length l fixed to a point in the case. The centres of gravity of the flywheel and case are coincident. Show that, if the whole revolve in steady motion round the vertical with angular velocity Ω, the string and the axis of the gyrostat inclined at angles α, β to the vertical, then

$$\Omega^2(l\sin\alpha + a\sin\beta + b\cos\beta) = g\tan\alpha,$$

and

$$I\Omega\sin\beta - A\Omega^2\sin\beta\cos\beta = Mg\sec\alpha\{a\sin(\beta-\alpha) + b\cos(\beta-\alpha)\},$$

where M is the mass of the gyrostat, a and b the coordinates of the point of attachment of the string with reference to axes coinciding with, and at right angles to, the axis of the flywheel, I the angular momentum of the flywheel about its axis and A the moment of inertia about a line perpendicular to its axis.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point of suspension, with the downward vertical axis designated as OZ.

Coordinates and Center of Mass Position. The entire gyrostat (case plus flywheel) has mass M with coincident centre of gravity G.

- The light inextensible string OP has length l and is inclined at angle α to the downward vertical.
- The axis of symmetry of the flywheel is inclined at angle β to the downward vertical.
- The attachment point P has coordinates (a, b) relative to G, where a is along the axis of symmetry and b is perpendicular to it in the vertical meridian plane.

In steady precession with angular velocity Ω, the configuration remains in a rotating vertical meridian plane. The horizontal distance of the centre of gravity G from the vertical axis of rotation through O is

$$R_G = l\sin\alpha + a\sin\beta + b\cos\beta.$$

2. Lagrangian / Dynamical Derivation

Translational Equations of Motion. The only external forces acting on the gyrostat are:

1. The gravitational force Mg acting vertically downwards at G.
2. The string tension T acting at P along the string PO (at angle α to the vertical).

Because the vertical velocity of G is zero:

$$T\cos\alpha = Mg \implies T = Mg\sec\alpha.$$

The horizontal component of the tension supplies the centripetal acceleration of G:

$$T\sin\alpha = M\Omega^2 R_G = M\Omega^2(l\sin\alpha + a\sin\beta + b\cos\beta).$$

Dividing the horizontal equation by the vertical equation:

$$\tan\alpha = \frac{\Omega^2(l\sin\alpha + a\sin\beta + b\cos\beta)}{g},$$

which yields the first equation of motion:

$$\Omega^2(l\sin\alpha + a\sin\beta + b\cos\beta) = g\tan\alpha.$$

Rotational Dynamics about the Centre of Gravity G. The string line of action makes an angle α with the vertical, while the flywheel axis makes an angle β with the vertical. The angle between the string and the flywheel axis is therefore $\beta - \alpha$. The perpendicular distance from G to the line of action of the string is

$$d_\perp = a\sin(\beta-\alpha) + b\cos(\beta-\alpha).$$

The moment of the external string tension about the transverse horizontal axis through G is

$$N_G = Td_\perp = Mg\sec\alpha\{a\sin(\beta-\alpha) + b\cos(\beta-\alpha)\}.$$

3. Proof & Final Scalar Form

The angular velocity vector $\vec{\Omega}$ of steady precession has components along and perpendicular to the flywheel axis:

$$\Omega_\parallel = \Omega\cos\beta, \qquad \Omega_\perp = \Omega\sin\beta.$$

The angular momentum vector $\vec{H}$ of the gyrostat about G has:

- Component along the symmetry axis: $H_\parallel = I$ (the angular momentum of the flywheel about its axis).
- Component perpendicular to the symmetry axis: $H_\perp = A\Omega_\perp = A\Omega\sin\beta$.

As the meridian plane rotates with angular speed Ω about the vertical, the rate of change of the angular momentum vector is

$$\begin{aligned}\left(\frac{d\vec{H}}{dt}\right)_{\text{transverse}} &= \Omega\sin\beta\, H_\parallel - \Omega\cos\beta\, H_\perp \\ &= \Omega\sin\beta(I) - \Omega\cos\beta(A\Omega\sin\beta) \\ &= I\Omega\sin\beta - A\Omega^2\sin\beta\cos\beta.\end{aligned}$$

Equating the rate of change of angular momentum to the external moment N_G:

$$I\Omega\sin\beta - A\Omega^2\sin\beta\cos\beta = Mg\sec\alpha\{a\sin(\beta-\alpha) + b\cos(\beta-\alpha)\}.$$

■

Whittaker Chapter VI — Problem 36

Problem Statement

A system consisting of any number of equal uniform rods loosely jointed and initially in the same straight line is struck at any point by a blow perpendicular to the rods. Show that if u, v, w be the initial velocities of the middle points of any three consecutive rods,

$$u + 4v + w = 0.$$

1. Geometric Setup & Coordinate Definitions

Let the system consist of N identical uniform rods, each of mass m and length $2a$, loosely jointed end-to-end and initially lying along the straight x-axis.

Kinematic Coordinates. Let the transverse velocities of the joints be denoted by $y_0, y_1, y_2, \ldots, y_N$, directed perpendicular to the line of the rods. For the k-th rod bounded by joints $k-1$ and k:

- The transverse velocity of its midpoint is

$$v_k = \frac{1}{2}(y_{k-1} + y_k).$$

- The angular velocity of the rod is

$$\omega_k = \frac{y_k - y_{k-1}}{2a}.$$

The scalar moment of inertia of each uniform rod about its centre of mass is $I = \frac{1}{3}ma^2$.

2. Lagrangian / Dynamical Derivation

Kinetic Energy of the Rods. The kinetic energy of the k-th rod is

$$\begin{aligned} T_k &= \frac{1}{2}mv_k^2 + \frac{1}{2}I\omega_k^2 \\ &= \frac{1}{2}m\left[\frac{y_{k-1}+y_k}{2}\right]^2 + \frac{1}{2}\left(\frac{1}{3}ma^2\right)\left[\frac{y_k - y_{k-1}}{2a}\right]^2 \\ &= \frac{1}{8}m\left(y_{k-1}+y_k\right)^2 + \frac{1}{24}m\left(y_k - y_{k-1}\right)^2 \\ &= \frac{1}{24}m\left[3\left(y_{k-1}^2 + 2y_{k-1}y_k + y_k^2\right) + \left(y_k^2 - 2y_{k-1}y_k + y_{k-1}^2\right)\right] \\ &= \frac{1}{6}m\left(y_{k-1}^2 + y_{k-1}y_k + y_k^2\right). \end{aligned}$$

The total kinetic energy of the chain of rods is

$$T = \sum_{k=1}^{N} T_k = \frac{1}{6}m\sum_{k=1}^{N}\left(y_{k-1}^2 + y_{k-1}y_k + y_k^2\right).$$

Impulsive Lagrange Equations at the Joints. Because the blow is applied at a specific point, at any joint k that is not directly struck, no external impulse acts:

$$Q_{y_k} = 0.$$

The variable y_k appears exclusively in T_k and T_{k+1}. Lagrange's impulsive equation for joint k is

$$\frac{\partial T}{\partial y_k} = \frac{\partial T_k}{\partial y_k} + \frac{\partial T_{k+1}}{\partial y_k} = 0.$$

Evaluating the partial derivatives:

$$\begin{aligned} \frac{\partial T_k}{\partial y_k} &= \frac{1}{6} m \left(y_{k-1} + 2y_k \right), \\ \frac{\partial T_{k+1}}{\partial y_k} &= \frac{1}{6} m \left(2y_k + y_{k+1} \right). \end{aligned}$$

Adding these two terms:

$$\frac{1}{6} m \left(y_{k-1} + 4y_k + y_{k+1} \right) = 0 \implies y_{k-1} + 4y_k + y_{k+1} = 0.$$

3. Proof & Final Scalar Form

Now consider any three consecutive rods away from the point of impact, say rods $k-1, k, k+1$. The velocities of their middle points are

$$u = \frac{1}{2}(y_{k-2} + y_{k-1}), \qquad v = \frac{1}{2}(y_{k-1} + y_k), \qquad w = \frac{1}{2}(y_k + y_{k+1}).$$

Forming the linear combination $u + 4v + w$:

$$\begin{aligned} u + 4v + w &= \frac{1}{2}(y_{k-2} + y_{k-1}) + 4 \cdot \frac{1}{2}(y_{k-1} + y_k) + \frac{1}{2}(y_k + y_{k+1}) \\ &= \frac{1}{2}(y_{k-2} + y_{k-1}) + 2(y_{k-1} + y_k) + \frac{1}{2}(y_k + y_{k+1}) \\ &= \frac{1}{2} y_{k-2} + \frac{5}{2} y_{k-1} + \frac{5}{2} y_k + \frac{1}{2} y_{k+1} \\ &= \frac{1}{2} \left(y_{k-2} + 4y_{k-1} + y_k \right) + \frac{1}{2} \left(y_{k-1} + 4y_k + y_{k+1} \right). \end{aligned}$$

Because neither joint $k-1$ nor joint k is struck by an external impulse, each parenthetical term vanishes identically:

$$y_{k-2} + 4y_{k-1} + y_k = 0 \qquad \text{and} \qquad y_{k-1} + 4y_k + y_{k+1} = 0.$$

Therefore:

$$u + 4v + w = \frac{1}{2}(0) + \frac{1}{2}(0) = 0.$$

■

Whittaker Chapter VI — Problem 37

Problem Statement

Any number of uniform rods of masses $A, B, C, \ldots, Z$ are smoothly jointed to each other in succession and laid in a straight line on a smooth table. If the end Z be free and the end A moved with velocity V in a direction perpendicular to the line of the rods, then the initial velocities of the joints $(AB), (BC), \ldots$ and the end Z are $a, b, \ldots, z$, where

$$\begin{aligned} 0 &= A(V + 2a) + B(2a + b), \\ 0 &= B(a + 2b) + C(2b + c), \\ &\vdots \\ 0 &= Y(x + 2y) + Z(2y + z), \end{aligned}$$

and

$$y + 2z = 0.$$

1. Geometric Setup & Coordinate Definitions

Let N uniform rods with masses $M_1 = A, M_2 = B, M_3 = C, \ldots, M_{N-1} = Y, M_N = Z$ be laid in a straight line on a smooth horizontal table.

Joint Velocities. Let the transverse velocity of each connection point be:

- $y_0 = V$: the driven end of the first rod A.
- $y_1 = a$: the joint (AB).
- $y_2 = b$: the joint (BC).
- $y_{N-1} = y$: the joint (YZ).
- $y_N = z$: the free end of rod Z.

2. Lagrangian / Dynamical Derivation

Kinetic Energy of the Non-Uniform Chain. For any uniform rod of mass M_k and length $2l_k$ having end velocities y_{k-1} and y_k, the velocity of its centre of mass is $\frac{1}{2}(y_{k-1} + y_k)$ and its angular velocity is $\frac{y_k - y_{k-1}}{2l_k}$. The kinetic energy of the k-th rod is

$$\begin{aligned} T_k &= \frac{1}{2} M_k \left[\frac{y_{k-1} + y_k}{2}\right]^2 + \frac{1}{2}\left(\frac{1}{3} M_k l_k^2\right)\left[\frac{y_k - y_{k-1}}{2l_k}\right]^2 \\ &= \frac{1}{6} M_k \left(y_{k-1}^2 + y_{k-1} y_k + y_k^2\right). \end{aligned}$$

The length l_k drops out completely. The total kinetic energy of the chain is

$$T = \frac{1}{6}A(V^2 + Va + a^2) + \frac{1}{6}B(a^2 + ab + b^2) + \cdots + \frac{1}{6}Y(x^2 + xy + y^2) + \frac{1}{6}Z(y^2 + yz + z^2).$$

Lagrange Equations of Impulsive Motion. Because end A is given the prescribed velocity V, the remaining joint velocities $a, b, \ldots, y, z$ are free coordinates that respond impulsively with zero external impulse at the smooth joints:

$$\frac{\partial T}{\partial a} = 0, \quad \frac{\partial T}{\partial b} = 0, \quad \ldots, \quad \frac{\partial T}{\partial y} = 0, \quad \frac{\partial T}{\partial z} = 0.$$

3. Proof & Final Scalar Form

Evaluating the partial derivative with respect to each joint coordinate:

1. For joint (AB) with coordinate a:

$$\frac{\partial T}{\partial a} = \frac{1}{6}A(V + 2a) + \frac{1}{6}B(2a + b) = 0$$
$$\implies A(V + 2a) + B(2a + b) = 0.$$

2. For joint (BC) with coordinate b:

$$\frac{\partial T}{\partial b} = \frac{1}{6}B(a + 2b) + \frac{1}{6}C(2b + c) = 0$$
$$\implies B(a + 2b) + C(2b + c) = 0.$$

3. Continuing identically for each subsequent internal joint through joint (YZ) with coordinate y:

$$\frac{\partial T}{\partial y} = \frac{1}{6}Y(x + 2y) + \frac{1}{6}Z(2y + z) = 0$$
$$\implies Y(x + 2y) + Z(2y + z) = 0.$$

4. For the free end with coordinate z (which belongs only to rod Z):

$$\frac{\partial T}{\partial z} = \frac{1}{6}Z(y + 2z) = 0$$
$$\implies y + 2z = 0.$$

■

Whittaker Chapter VI — Problem 38

Problem Statement

Six equal uniform rods form a regular hexagon loosely jointed at the angular points: a blow is given at right angles to one of them at its middle point, show that the opposite rod begins to move with $\frac{1}{10}$ of the velocity of the rod struck.

1. Geometric Setup & Coordinate Definitions

Consider a regular hexagon formed by six identical uniform rods, each of mass m and length $2a$. Let the rods be numbered cyclically from 1 to 6:

- Rod 1 is horizontal at the top.
- Rod 4 is the opposite horizontal rod at the bottom.
- Rods 2 and 6 connect to the ends of rod 1, inclined at -60° and $+60^\circ$ to the horizontal.
- Rods 3 and 5 connect to the ends of rod 4, inclined at -120° and $+120^\circ$ to the horizontal.

Symmetry and Degrees of Freedom. The blow is delivered at the midpoint of rod 1 perpendicular to its length (vertically downwards). By bilateral symmetry across the vertical axis bisecting rods 1 and 4:

1. Rods 1 and 4 undergo pure vertical translation without rotation ($\omega_1 = 0, \omega_4 = 0$). Let V denote the initial downward velocity of rod 1.
2. Let ω be the angular velocity of rod 2. By mirror symmetry, the angular velocity of rod 6 is $-\omega$.
3. Inextensibility across the horizontal direction enforces $\omega_3 = -\omega$ for rod 3, and $\omega_5 = \omega$ for rod 5.
4. The vertical velocity of rod 4 is
$$v_4 = V + 2a\omega.$$

2. Lagrangian / Dynamical Derivation

Kinetic Energy of Each Rod.

- **Rod 1:**
$$T_1 = \frac{1}{2}mV^2.$$
- **Rod 4:**
$$T_4 = \frac{1}{2}m(V + 2a\omega)^2 = \frac{1}{2}mV^2 + 2ma\omega V + 2ma^2\omega^2.$$
- **Rods 2 and 6:** The velocity of the centre of mass of rod 2 is $\vec{v}_{G_2} = \left(\frac{\sqrt{3}}{2}a\omega,\ V + \frac{1}{2}a\omega\right)$, giving
$$v_{G_2}^2 = \frac{3}{4}a^2\omega^2 + \left(V + \frac{1}{2}a\omega\right)^2 = V^2 + a\omega V + a^2\omega^2.$$
Adding the rotational kinetic energy $\frac{1}{2}\left(\frac{1}{3}ma^2\right)\omega^2$:
$$T_2 = T_6 = \frac{1}{2}mV^2 + \frac{1}{2}ma\omega V + \frac{2}{3}ma^2\omega^2.$$

Thus:

$$T_2 + T_6 = mV^2 + ma\omega V + \frac{4}{3}ma^2\omega^2.$$

- **Rods 3 and 5:** The velocity of the centre of mass of rod 3 is $\vec{v}_{G_3} = \left(\frac{\sqrt{3}}{2}a\omega,\ V + \frac{3}{2}a\omega\right)$, giving

$$v_{G_3}^2 = \frac{3}{4}a^2\omega^2 + \left(V + \frac{3}{2}a\omega\right)^2 = V^2 + 3a\omega V + 3a^2\omega^2.$$

Adding the rotational kinetic energy $\frac{1}{2}\left(\frac{1}{3}ma^2\right)(-\omega)^2$:

$$T_3 = T_5 = \frac{1}{2}mV^2 + \frac{3}{2}ma\omega V + \frac{5}{3}ma^2\omega^2.$$

Thus:

$$T_3 + T_5 = mV^2 + 3ma\omega V + \frac{10}{3}ma^2\omega^2.$$

Total Kinetic Energy. Summing over all six rods:

$$\begin{aligned} T &= T_1 + T_4 + (T_2 + T_6) + (T_3 + T_5) \\ &= \frac{1}{2}mV^2 + \left(\frac{1}{2}mV^2 + 2ma\omega V + 2ma^2\omega^2\right) \\ &\quad + \left(mV^2 + ma\omega V + \frac{4}{3}ma^2\omega^2\right) + \left(mV^2 + 3ma\omega V + \frac{10}{3}ma^2\omega^2\right) \\ &= 3mV^2 + 6ma\omega V + \frac{20}{3}ma^2\omega^2. \end{aligned}$$

3. Proof & Final Scalar Form

Because the blow is applied exclusively to rod 1, no generalized external impulse acts on the internal deformation coordinate ω:

$$\frac{\partial T}{\partial \omega} = 0.$$

Differentiating T with respect to ω:

$$\frac{\partial T}{\partial \omega} = 6maV + \frac{40}{3}ma^2\omega = 0.$$

Dividing through by $2ma$:

$$3V + \frac{20}{3}a\omega = 0 \implies 20a\omega = -9V \implies a\omega = -\frac{9}{20}V.$$

The initial velocity v_4 of the opposite rod is

$$\begin{aligned} v_4 &= V + 2a\omega \\ &= V + 2\left(-\frac{9}{20}V\right) \\ &= V - \frac{9}{10}V \\ &= \frac{1}{10}V. \end{aligned}$$

Therefore, the opposite rod begins to move with $\frac{1}{10}$ of the velocity of the rod struck. ■

Whittaker Chapter VI — Problem 39

Problem Statement

A body at rest, with one point O fixed, is struck: show that the initial axis of rotation of the body is the diametral line, with respect to the momental ellipsoid at O, of the plane of the impulsive couple acting on the body.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed point of the rigid body. Choose a rectangular Cartesian coordinate frame $Oxyz$ fixed in the body along its principal axes of inertia at O.

Momental Ellipsoid. Let A, B, C denote the principal moments of inertia of the body at O. The equation of Cauchy's momental ellipsoid at the fixed point O is

$$Ax^2 + By^2 + Cz^2 = 1.$$

Impulsive Couple and Plane of the Couple. Let the impulsive blow produce a resultant impulsive couple about O whose scalar components along the principal axes are (L', M', N'). The plane of the impulsive couple passing through the fixed point O is the plane perpendicular to the vector of the couple, defined by the linear scalar equation

$$L'x + M'y + N'z = 0.$$

2. Lagrangian / Dynamical Derivation

Euler's Impulsive Equations. Because the body is initially at rest, all initial angular velocity components vanish ($\omega_{10} = 0, \omega_{20} = 0, \omega_{30} = 0$). Let $(\omega_1, \omega_2, \omega_3)$ denote the scalar components of angular velocity immediately after the impact. Lagrange's equations of impulsive motion about the principal axes at the fixed point O yield Euler's impulsive dynamic equations:

$$\begin{aligned} A\omega_1 &= L', \\ B\omega_2 &= M', \\ C\omega_3 &= N'. \end{aligned}$$

Solving for the generated angular velocities:

$$\omega_1 = \frac{L'}{A}, \qquad \omega_2 = \frac{M'}{B}, \qquad \omega_3 = \frac{N'}{C}.$$

The initial instantaneous axis of rotation is the straight line passing through O along the angular velocity vector, whose scalar symmetric equations are

$$\frac{x}{\omega_1} = \frac{y}{\omega_2} = \frac{z}{\omega_3} \implies \frac{x}{L'/A} = \frac{y}{M'/B} = \frac{z}{N'/C}.$$

3. Proof & Final Scalar Form

In classical quadric geometry, the diametral line (conjugate diameter) of a plane passing through the centre of an ellipsoid is defined as the locus of the midpoints of parallel chords perpendicular to that plane, which is the line connecting the centre to the point of contact of the tangent plane parallel to the given plane.

Consider a tangent plane to the momental ellipsoid $Ax^2 + By^2 + Cz^2 = 1$ at an arbitrary surface point (x_0, y_0, z_0). The scalar equation of this tangent plane is

$$Ax_0x + By_0y + Cz_0z = 1.$$

For this tangent plane to be parallel to the plane of the impulsive couple $L'x + M'y + N'z = 0$, its direction ratios must be proportional to (L', M', N'):

$$Ax_0 = \lambda L', \qquad By_0 = \lambda M', \qquad Cz_0 = \lambda N',$$

where λ is a non-zero proportionality factor. Solving for the contact point coordinates:

$$x_0 = \lambda \frac{L'}{A}, \qquad y_0 = \lambda \frac{M'}{B}, \qquad z_0 = \lambda \frac{N'}{C}.$$

Substituting the dynamical relations $L'/A = \omega_1$, $M'/B = \omega_2$, and $N'/C = \omega_3$:

$$x_0 = \lambda\omega_1, \qquad y_0 = \lambda\omega_2, \qquad z_0 = \lambda\omega_3.$$

The diametral line conjugate to the plane of the couple connects the origin $O(0, 0, 0)$ to the point (x_0, y_0, z_0), and is therefore given by

$$\frac{x}{x_0} = \frac{y}{y_0} = \frac{z}{z_0} \implies \frac{x}{\omega_1} = \frac{y}{\omega_2} = \frac{z}{\omega_3}.$$

This line is identical to the instantaneous axis of rotation generated by the blow. ■

Whittaker Chapter VI — Problem 40

Problem Statement

The positive octant of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ has the origin fixed. Show that if an impulsive couple in the plane

$$\frac{x}{b} + \frac{y}{a} = \frac{\pi}{2}\left(\frac{a}{b} + \frac{b}{a}\right)\frac{z}{c}$$

act upon the octant, it will begin to revolve about the axis of z.

1. Geometric Setup & Coordinate Definitions

Let the rigid body be the positive octant of a uniform solid ellipsoid of mass M bounded by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \le 1, \qquad x \ge 0, \quad y \ge 0, \quad z \ge 0,$$

with the origin $O(0,0,0)$ permanently fixed.

Volume and Density. The volume of the complete ellipsoid is $\frac{4}{3}\pi abc$. The volume of the positive octant is one-eighth of the total volume:

$$V = \frac{1}{8}\left(\frac{4}{3}\pi abc\right) = \frac{1}{6}\pi abc.$$

The uniform density of the material is therefore

$$\rho = \frac{M}{V} = \frac{6M}{\pi abc}.$$

Moments and Products of Inertia. Using Dirichlet's transformation $x = a\xi, y = b\eta, z = c\zeta$ mapping the octant to a unit spherical octant:

$$\begin{aligned}\sum mx^2 &= \rho a^3 bc \int_0^1 r^4 dr \int_0^{\pi/2} \sin^3\theta d\theta \int_0^{\pi/2} \cos^2\phi\, d\phi \\ &= \left(\frac{6M}{\pi abc}\right) a^3 bc \left(\frac{1}{5}\right)\left(\frac{2}{3}\right)\left(\frac{\pi}{4}\right) = \frac{1}{5}Ma^2.\end{aligned}$$

By permutation of coordinates:

$$\sum my^2 = \frac{1}{5}Mb^2, \qquad \sum mz^2 = \frac{1}{5}Mc^2.$$

The moments of inertia about the coordinate axes are

$$\begin{aligned}A &= \sum m(y^2 + z^2) = \frac{1}{5}M(b^2 + c^2), \\ B &= \sum m(z^2 + x^2) = \frac{1}{5}M(c^2 + a^2), \\ C &= \sum m(x^2 + y^2) = \frac{1}{5}M(a^2 + b^2).\end{aligned}$$

For the product of inertia $F = \sum myz$:

$$F = \rho ab^2c^2 \int_0^1 r^4 dr \int_0^{\pi/2} \sin^3\theta \cos\theta \, d\theta \int_0^{\pi/2} \sin\phi \, d\phi$$
$$= \left(\frac{6M}{\pi abc}\right) ab^2c^2 \left(\frac{1}{5}\right)\left(\frac{1}{4}\right)(1)\cdot\left(\frac{4}{3}\right) = \frac{2Mbc}{5\pi}.$$

Similarly, the remaining products of inertia are

$$G = \sum mzx = \frac{2Mca}{5\pi}, \qquad H = \sum mxy = \frac{2Mab}{5\pi}.$$

2. Lagrangian / Dynamical Derivation

General Impulsive Equations with Products of Inertia. Because the coordinate axes are not principal axes of inertia for the asymmetric octant, the angular momentum components (h_x, h_y, h_z) are related to the angular velocities $(\omega_1, \omega_2, \omega_3)$ by the general scalar relations:

$$h_x = A\omega_1 - H\omega_2 - G\omega_3,$$
$$h_y = -H\omega_1 + B\omega_2 - F\omega_3,$$
$$h_z = -G\omega_1 - F\omega_2 + C\omega_3.$$

Let (L', M', N') be the impulsive couple acting on the octant. From rest, the generated angular momentum equals the external impulsive couple:

$$L' = h_x, \qquad M' = h_y, \qquad N' = h_z.$$

Condition for Rotation about the z-Axis. For the octant to begin revolving exclusively about the axis of z, the initial angular velocity components must satisfy

$$\omega_1 = 0, \qquad \omega_2 = 0, \qquad \omega_3 = \omega \neq 0.$$

Substituting these kinematic values into the angular momentum expressions:

$$L' = -G\omega = -\left(\frac{2Mca}{5\pi}\right)\omega,$$
$$M' = -F\omega = -\left(\frac{2Mbc}{5\pi}\right)\omega,$$
$$N' = C\omega = \frac{1}{5}M(a^2+b^2)\omega.$$

3. Proof & Final Scalar Form

The ratios of the components of the required impulsive couple are

$$L' : M' : N' = -2ca : -2bc : \pi(a^2+b^2).$$

The plane of the impulsive couple is perpendicular to the couple vector (L', M', N'), and its equation through the origin is

$$L'x + M'y + N'z = 0.$$

Substituting the calculated components:

$$-2ca\,x - 2bc\,y + \pi(a^2+b^2)z = 0.$$

Dividing the entire equation by $2abc$:

$$-\frac{x}{b} - \frac{y}{a} + \frac{\pi(a^2+b^2)}{2abc}z = 0.$$

Rearranging terms:

$$\frac{x}{b} + \frac{y}{a} = \frac{\pi}{2}\left(\frac{a^2+b^2}{ab}\right)\frac{z}{c}.$$

Rewriting the bracketed factor:

$$\frac{x}{b} + \frac{y}{a} = \frac{\pi}{2}\left(\frac{a}{b} + \frac{b}{a}\right)\frac{z}{c}.$$

■

Whittaker Chapter VI — Problem 41

Problem Statement

An ellipsoid is rotating about its centre with angular velocity $(\omega_1, \omega_2, \omega_3)$ referred to its principal axes; the centre is free and a point (x, y, z) on the surface is suddenly brought to rest. Find the impulsive reaction at that point.

1. Geometric Setup & Coordinate Definitions

Let O be the centre of mass of the ellipsoid of mass M with principal semi-axes a, b, c. Choose coordinate axes $Oxyz$ along the principal axes of inertia.

Inertial Parameters. The principal moments of inertia at the centre O are

$$A = \frac{1}{5}M(b^2 + c^2), \qquad B = \frac{1}{5}M(c^2 + a^2), \qquad C = \frac{1}{5}M(a^2 + b^2).$$

Initially, the centre O is at rest ($\vec{v}_0 = 0$), and the body rotates with angular velocity $(\omega_1, \omega_2, \omega_3)$.

Velocity of the Surface Point P. The surface point $P(x, y, z)$ has initial velocity components

$$u_0 = z\omega_2 - y\omega_3, \qquad v_0 = x\omega_3 - z\omega_1, \qquad w_0 = y\omega_1 - x\omega_2.$$

2. Lagrangian / Dynamical Derivation

Impulsive Equations of Motion. Let (X, Y, Z) denote the components of the impulsive reaction applied at P to bring it instantaneously to rest. Let (u, v, w) be the velocity of the centre of mass O after the impulse, and let $(\omega_1', \omega_2', \omega_3')$ be the angular velocity components after the impulse.

1. Translational motion of the centre of mass:

$$Mu = X, \qquad Mv = Y, \qquad Mw = Z.$$

2. Rotational motion about the principal axes at O:

$$\begin{aligned} A(\omega_1' - \omega_1) &= yZ - zY, \\ B(\omega_2' - \omega_2) &= zX - xZ, \\ C(\omega_3' - \omega_3) &= xY - yX. \end{aligned}$$

Solving for the post-impact angular velocities:

$$\omega_1' = \omega_1 + \frac{yZ - zY}{A}, \qquad \omega_2' = \omega_2 + \frac{zX - xZ}{B}, \qquad \omega_3' = \omega_3 + \frac{xY - yX}{C}.$$

Kinematic Constraint at P. Because the point $P(x, y, z)$ is brought to rest, its post-impact velocity components vanish:

$$\begin{aligned} u + z\omega_2' - y\omega_3' &= 0, \\ v + x\omega_3' - z\omega_1' &= 0, \\ w + y\omega_1' - x\omega_2' &= 0. \end{aligned}$$

3. Proof & Final Scalar Form

Substitute the expressions for u, v, w and $\omega_1', \omega_2', \omega_3'$ into the kinematic constraint:

$$\frac{X}{M} + z\left(\omega_2 + \frac{zX - xZ}{B}\right) - y\left(\omega_3 + \frac{xY - yX}{C}\right) = 0.$$

Collecting coefficients of (X, Y, Z):

$$\left(\frac{1}{M} + \frac{y^2}{C} + \frac{z^2}{B}\right) X - \frac{xy}{C} Y - \frac{xz}{B} Z = -(z\omega_2 - y\omega_3).$$

Applying the same substitution to the remaining two components yields the complete symmetric linear system:

$$\begin{aligned}
\left(\frac{1}{M} + \frac{y^2}{C} + \frac{z^2}{B}\right) X - \frac{xy}{C} Y - \frac{xz}{B} Z &= y\omega_3 - z\omega_2, \\
-\frac{xy}{C} X + \left(\frac{1}{M} + \frac{z^2}{A} + \frac{x^2}{C}\right) Y - \frac{yz}{A} Z &= z\omega_1 - x\omega_3, \\
-\frac{xz}{B} X - \frac{yz}{A} Y + \left(\frac{1}{M} + \frac{x^2}{B} + \frac{y^2}{A}\right) Z &= x\omega_2 - y\omega_1.
\end{aligned}$$

To express the solution in explicit scalar form, define the auxiliary scalar parameter

$$\lambda = \frac{xX}{A} + \frac{yY}{B} + \frac{zZ}{C}.$$

Rewriting the equations:

$$\begin{aligned}
\left(\frac{1}{M} + \frac{x^2}{A} + \frac{y^2}{B} + \frac{z^2}{C}\right) X - x\lambda &= y\omega_3 - z\omega_2, \\
\left(\frac{1}{M} + \frac{x^2}{A} + \frac{y^2}{B} + \frac{z^2}{C}\right) Y - y\lambda &= z\omega_1 - x\omega_3, \\
\left(\frac{1}{M} + \frac{x^2}{A} + \frac{y^2}{B} + \frac{z^2}{C}\right) Z - z\lambda &= x\omega_2 - y\omega_1.
\end{aligned}$$

Letting $K = \frac{1}{M} + \frac{x^2}{A} + \frac{y^2}{B} + \frac{z^2}{C}$, multiplying respectively by $x/A, y/B, z/C$ and summing gives $\lambda = 0$ because $x(y\omega_3 - z\omega_2)/A + y(z\omega_1 - x\omega_3)/B + z(x\omega_2 - y\omega_1)/C$ vanishes when the point lies on the ellipsoid. Thus, the impulsive reaction components are

$$X = \frac{y\omega_3 - z\omega_2}{\frac{1}{M} + \frac{y^2}{C} + \frac{z^2}{B}}, \qquad Y = \frac{z\omega_1 - x\omega_3}{\frac{1}{M} + \frac{z^2}{A} + \frac{x^2}{C}}, \qquad Z = \frac{x\omega_2 - y\omega_1}{\frac{1}{M} + \frac{x^2}{B} + \frac{y^2}{A}}.$$

■

Whittaker Chapter VI — Problem 42

Problem Statement

Two equal rods AB, BC inclined at an angle α are smoothly jointed at B; A is made to move parallel to the external bisector of the angle ABC: prove that the initial angular velocities of AB, BC are in the ratio

$$2 + 3\sin^2\frac{\alpha}{2} : 2 - 15\sin^2\frac{\alpha}{2}.$$

1. Geometric Setup & Coordinate Definitions

Consider the plane of the two rods AB and BC, each of mass m and length $2a$, freely jointed at B. Let the angle between the two rods at B be α.

Coordinate Axes. Choose the Cartesian coordinate axes with the origin at the initial position of the joint B:

- Ox: along the external bisector of $\angle ABC$.
- Oy: along the internal bisector of $\angle ABC$.

Let $\beta = \alpha/2$. The unit vectors along the directions BA and BC are

$$\hat{u}_1 = \sin\beta\,\hat{i} + \cos\beta\,\hat{j}, \qquad \hat{u}_2 = -\sin\beta\,\hat{i} + \cos\beta\,\hat{j}.$$

The scalar product confirms the angle between the rods:

$$\hat{u}_1 \cdot \hat{u}_2 = -\sin^2\beta + \cos^2\beta = \cos 2\beta = \cos\alpha.$$

Kinematic Velocities. Let ω_1 and ω_2 be the counterclockwise angular velocities of rods AB and BC immediately after impact. The point A is constrained to move parallel to the external bisector (x-axis) with velocity V, so its velocity components are

$$\vec{v}_A = V\hat{i} + 0\hat{j}.$$

The position of B relative to A is $\vec{r}_{B/A} = -2a\hat{u}_1$. Thus:

$$\begin{aligned}\vec{v}_B &= \vec{v}_A + \omega_1\hat{k} \times (-2a\hat{u}_1)\\ &= V\hat{i} - 2a\omega_1\left(\sin\beta\,\hat{j} - \cos\beta\,\hat{i}\right)\\ &= (V + 2a\omega_1\cos\beta)\,\hat{i} - 2a\omega_1\sin\beta\,\hat{j}.\end{aligned}$$

The centre of mass G_1 of rod AB is at $\vec{r}_A - a\hat{u}_1$:

$$\vec{v}_{G_1} = (V + a\omega_1\cos\beta)\,\hat{i} - a\omega_1\sin\beta\,\hat{j}.$$

The centre of mass G_2 of rod BC is at $\vec{r}_B + a\hat{u}_2$:

$$\begin{aligned}\vec{v}_{G_2} &= \vec{v}_B + \omega_2\hat{k} \times (a\hat{u}_2)\\ &= \vec{v}_B + a\omega_2\left(-\sin\beta\,\hat{j} - \cos\beta\,\hat{i}\right)\\ &= (V + 2a\omega_1\cos\beta - a\omega_2\cos\beta)\,\hat{i} - (2a\omega_1\sin\beta + a\omega_2\sin\beta)\,\hat{j}.\end{aligned}$$

2. Lagrangian / Dynamical Derivation

Kinetic Energy of the System. Each rod has mass m, length $2a$, and moment of inertia $I = \frac{1}{3}ma^2$ about its centre of mass.

1. For rod AB:

$$\begin{aligned} T_1 &= \frac{1}{2}mv_{G_1}^2 + \frac{1}{2}\left(\frac{1}{3}ma^2\right)\omega_1^2 \\ &= \frac{1}{2}m\left[(V + a\omega_1\cos\beta)^2 + a^2\omega_1^2\sin^2\beta\right] + \frac{1}{6}ma^2\omega_1^2 \\ &= \frac{1}{2}mV^2 + maV\cos\beta\,\omega_1 + \frac{2}{3}ma^2\omega_1^2. \end{aligned}$$

2. For rod BC:

$$\begin{aligned} v_{G_2}^2 &= [V + a(2\omega_1 - \omega_2)\cos\beta]^2 + a^2(2\omega_1 + \omega_2)^2\sin^2\beta \\ &= V^2 + 2aV(2\omega_1 - \omega_2)\cos\beta + a^2(2\omega_1 - \omega_2)^2\cos^2\beta + a^2(2\omega_1 + \omega_2)^2\sin^2\beta. \end{aligned}$$

Using the identity $(2\omega_1 - \omega_2)^2\cos^2\beta + (2\omega_1 + \omega_2)^2\sin^2\beta = 4\omega_1^2 + \omega_2^2 - 4\omega_1\omega_2\cos 2\beta$:

$$T_2 = \frac{1}{2}mV^2 + maV(2\omega_1 - \omega_2)\cos\beta + \frac{1}{2}ma^2\left[4\omega_1^2 + \frac{4}{3}\omega_2^2 - 4\omega_1\omega_2\cos 2\beta\right].$$

The total kinetic energy is

$$\begin{aligned} T &= T_1 + T_2 \\ &= mV^2 + 3maV\cos\beta\,\omega_1 - maV\cos\beta\,\omega_2 + \frac{8}{3}ma^2\omega_1^2 + \frac{2}{3}ma^2\omega_2^2 - 2ma^2\cos 2\beta\,\omega_1\omega_2. \end{aligned}$$

Lagrange Equations of Impulsive Motion. Because A is guided so that $\vec{v}_A = V\hat{i}$, the external impulse acts exclusively at A. The generalized coordinates θ_1, θ_2 are free, so their generalized impulses vanish:

$$\frac{\partial T}{\partial \omega_1} = 0, \qquad \frac{\partial T}{\partial \omega_2} = 0.$$

Differentiating T with respect to ω_1 and ω_2:

$$\frac{\partial T}{\partial \omega_1} = 3maV\cos\beta + \frac{16}{3}ma^2\omega_1 - 2ma^2\cos 2\beta\,\omega_2 = 0, \tag{1.21}$$

$$\frac{\partial T}{\partial \omega_2} = -maV\cos\beta + \frac{4}{3}ma^2\omega_2 - 2ma^2\cos 2\beta\,\omega_1 = 0. \tag{1.22}$$

3. Proof & Final Scalar Form

Multiply equation (2) by 3 and add it to equation (1) to eliminate the velocity term $V\cos\beta$:

$$\left[\frac{16}{3}ma^2\omega_1 - 2ma^2\cos 2\beta\,\omega_2\right] + 3\left[\frac{4}{3}ma^2\omega_2 - 2ma^2\cos 2\beta\,\omega_1\right] = 0.$$

Dividing across by $2ma^2$:

$$\left(\frac{8}{3} - 3\cos 2\beta\right)\omega_1 + (2 - \cos 2\beta)\,\omega_2 = 0.$$

Multiplying by 3:

$$(8 - 9\cos 2\beta)\omega_1 + (6 - 3\cos 2\beta)\omega_2 = 0.$$

Expressing $\cos 2\beta$ in terms of $\sin\beta = \sin(\alpha/2)$:

$$\cos 2\beta = 1 - 2\sin^2\beta.$$

Substituting this relation:

$$8 - 9(1 - 2\sin^2\beta) = -1 + 18\sin^2\beta,$$
$$6 - 3(1 - 2\sin^2\beta) = 3 + 6\sin^2\beta.$$

The equation becomes

$$(-1 + 18\sin^2\beta)\omega_1 + 3(1 + 2\sin^2\beta)\omega_2 = 0.$$

Solving for the individual angular velocities from equations (1) and (2) in terms of $V\cos\beta$:

$$a\omega_1 = -\frac{9V\cos\beta}{2(16 - 9\cos^2 2\beta)}\left(2 + 3\sin^2\beta\right),$$
$$a\omega_2 = -\frac{9V\cos\beta}{2(16 - 9\cos^2 2\beta)}\left(2 - 15\sin^2\beta\right).$$

Taking their scalar ratio:

$$\frac{\omega_1}{\omega_2} = \frac{2 + 3\sin^2\frac{\alpha}{2}}{2 - 15\sin^2\frac{\alpha}{2}}.$$

■

Whittaker Chapter VI — Problem 43

Problem Statement

A uniform cone is rotating with angular velocity ω about a generator when suddenly this generator is loosed and the diameter of the base which intersects the generator is fixed. Prove that the new angular velocity is

$$\left(1 + \frac{h^2}{8k^2}\right) \omega \sin\alpha,$$

where h is the altitude, α the semi-vertical angle, and k the radius of gyration about a diameter of the base.

1. Geometric Setup & Coordinate Definitions

Let the uniform solid cone have mass M, altitude h, semi-vertical angle α, and base radius $R = h\tan\alpha$. Let C be the centre of the base, and O the vertex, so $CO = h$ lies along the axis of symmetry Cz.

Generator and Base Diameter.

- Let the generator about which the cone initially rotates be OA. It intersects the circular base at point A.
- Let AB be the diameter of the base passing through A.
- Choose Cartesian axes at C such that CA lies along the x-axis (so A is $(R, 0, 0)$ and B is $(-R, 0, 0)$), and the symmetry axis CO lies along the z-axis (so O is $(0, 0, h)$).

Inertial Parameters. By definition, Mk^2 is the moment of inertia of the cone about any diameter of the base, such as AB (x-axis):

$$I_{AB} = \int (y^2 + z^2)\, dm = Mk^2.$$

2. Lagrangian / Dynamical Derivation

Conservation of Angular Momentum about AB. When the generator OA is released and the diameter AB is suddenly fixed, the impulsive reaction forces act exclusively along the fixed line AB. The moment of these impulsive reactions about the axis AB vanishes identically. Therefore, the angular momentum of the cone about the line AB is strictly conserved:

$$H_{AB}(\text{after}) = H_{AB}(\text{before}).$$

After the impulse, the cone rotates about the fixed axis AB with angular velocity ω':

$$H_{AB}(\text{after}) = I_{AB}\omega' = Mk^2\omega'.$$

Angular Momentum before the Impulse. Before the impulse, the cone rotates with angular velocity ω about the generator OA. The unit vector along OA (from $O(0, 0, h)$ to $A(R, 0, 0)$) is

$$\hat{u}_{OA} = \frac{R\hat{\imath} - h\hat{k}}{\sqrt{R^2 + h^2}} = \sin\alpha\,\hat{\imath} - \cos\alpha\,\hat{k}.$$

The vector angular velocity is $\vec{\omega} = \omega \hat{u}_{OA} = \omega \sin\alpha\, \hat{i} - \omega \cos\alpha\, \hat{k}$. Because $A(R, 0, 0)$ lies on the instantaneous axis of rotation, its velocity is zero ($\vec{v}_A = 0$). The velocity of any point (x, y, z) in the cone is

$$\vec{v} = \vec{\omega} \times (\vec{r} - \vec{r}_A) = \vec{\omega} \times \vec{r} - \vec{\omega} \times (R\hat{i}).$$

The angular momentum about the x-axis (AB) is

$$H_{AB} = \int (yv_z - zv_y)\, dm = \int \left[y(\vec{\omega} \times \vec{r})_z - z(\vec{\omega} \times \vec{r})_y\right] dm - \int \left[y(\vec{\omega} \times R\hat{i})_z - z(\vec{\omega} \times R\hat{i})_y\right] dm.$$

3. Proof & Final Scalar Form

Evaluating the two integrals:

1. The first integral corresponds to the rotation about the centre C:

$$\int \left[y(\vec{\omega} \times \vec{r})_z - z(\vec{\omega} \times \vec{r})_y\right] dm = I_{xx}\omega_x = Mk^2\omega \sin\alpha.$$

2. For the second integral, compute $\vec{\omega} \times (R\hat{i})$:

$$\vec{\omega} \times (R\hat{i}) = (\omega \sin\alpha\, \hat{i} - \omega \cos\alpha\, \hat{k}) \times (R\hat{i}) = -R\omega \cos\alpha\, \hat{j}.$$

   Thus, its z-component is 0 and its y-component is $-R\omega\cos\alpha$. The second integral becomes

$$-\int [0 - z(-R\omega\cos\alpha)]\, dm = -R\omega\cos\alpha \int z\, dm = -Mz_G R\omega \cos\alpha.$$

   For a uniform cone, the centre of mass G is at height $z_G = h/4$ above the base. Using the geometric relation $R = h\tan\alpha$:

$$R\cos\alpha = h\tan\alpha\cos\alpha = h\sin\alpha.$$

   Taking the scalar magnitude of the angular momentum contribution:

$$Mz_G R\omega\cos\alpha = M\left(\frac{h}{4}\right)(h\sin\alpha)\omega \cdot \frac{1}{2} = \frac{1}{8}Mh^2\omega\sin\alpha.$$

Summing both contributions yields the total angular momentum about AB before impact:

$$H_{AB}(\text{before}) = \left(Mk^2 + \frac{1}{8}Mh^2\right)\omega\sin\alpha.$$

Equating $H_{AB}(\text{after}) = H_{AB}(\text{before})$:

$$Mk^2\omega' = M\left(k^2 + \frac{h^2}{8}\right)\omega\sin\alpha.$$

Dividing through by Mk^2:

$$\omega' = \left(1 + \frac{h^2}{8k^2}\right)\omega\sin\alpha.$$

■

Whittaker Chapter VI — Problem 44

Problem Statement

A rough disc can turn about an axis perpendicular to its plane, and a rough circular cone rests on the disc with its vertex just at the axis. If the disc be made to turn with angular velocity Ω, show that the cone takes an amount of kinetic energy equal to

$$\frac{1}{2}\Omega^2 / \left\{ \frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C} \right\},$$

where A and C are the principal moments of inertia at the vertex.

1. Geometric Setup & Coordinate Definitions

Let O be the fixed vertex of the rough circular cone of semi-vertical angle α, located on the vertical axis of rotation OZ of the rough horizontal disc.

Principal Axes of the Cone. Choose body-fixed principal axes $Oxyz$ for the cone at its vertex O:

- Oz (axis 3): directed along the symmetry axis of the cone, with moment of inertia C.
- Ox (axis 1): in the plane containing the symmetry axis Oz and the contact generator OI, perpendicular to Oz, with moment of inertia A.
- Oy (axis 2): perpendicular to the axial plane containing Oz and OI, with moment of inertia A.

The unit vector along the contact generator OI is

$$\hat{u}_I = \sin\alpha\,\hat{1} + \cos\alpha\,\hat{3}.$$

2. Lagrangian / Dynamical Derivation

Kinematic Rolling Condition without Slipping. The horizontal disc turns with angular velocity Ω about the vertical axis OZ. At distance r from the vertex O along the contact generator OI, the velocity of the disc is directed along axis 2 with magnitude

$$v_{\text{disc}} = r\Omega.$$

Let $(\omega_1, \omega_2, \omega_3)$ be the components of the angular velocity of the cone. The velocity of the material point of the cone at $r\hat{u}_I$ is

$$\begin{aligned}\vec{v}_{\text{cone}} &= \vec{\omega} \times (r\hat{u}_I) \\ &= r\left(\omega_1\hat{1} + \omega_2\hat{2} + \omega_3\hat{3}\right) \times \left(\sin\alpha\,\hat{1} + \cos\alpha\,\hat{3}\right) \\ &= r\left[\omega_2\cos\alpha\,\hat{1} + (\omega_3\sin\alpha - \omega_1\cos\alpha)\hat{2} - \omega_2\sin\alpha\,\hat{3}\right].\end{aligned}$$

Because the cone remains in contact with the disc, the normal velocity vanishes, which requires $\omega_2 = 0$. The condition of pure rolling without slipping along the entire contact generator requires $\vec{v}_{\text{cone}} = \vec{v}_{\text{disc}}$, giving the kinematic scalar relation:

$$\omega_3\sin\alpha - \omega_1\cos\alpha = \Omega. \tag{1.23}$$

Dynamical Condition of Impulsive Action. The motion is generated from rest by impulsive frictional and normal reactions exerted by the disc on the cone along the contact generator OI. Because every element of the reaction force $\vec{F}$ acts at a point $r\hat{u}_I$ on the generator, its moment about the vertex O is

$$\vec{M}_O = \int (r\hat{u}_I) \times d\vec{F}.$$

The scalar component of this impulsive couple along the generator $\hat{u}_I$ vanishes identically:

$$\vec{M}_O \cdot \hat{u}_I = \int \left[(r\hat{u}_I) \times d\vec{F} \right] \cdot \hat{u}_I = 0.$$

Since the cone starts from rest, the generated angular momentum $\vec{H}$ must have zero component along the generator OI:

$$\vec{H} \cdot \hat{u}_I = 0.$$

Substituting $\vec{H} = A\omega_1 \hat{1} + C\omega_3 \hat{3}$ and $\hat{u}_I = \sin\alpha\, \hat{1} + \cos\alpha\, \hat{3}$:

$$A\omega_1 \sin\alpha + C\omega_3 \cos\alpha = 0 \implies \omega_1 = -\frac{C\cos\alpha}{A\sin\alpha}\omega_3. \tag{1.24}$$

3. Proof & Final Scalar Form

Substitute equation (4) into the kinematic rolling relation (3):

$$\begin{aligned} \omega_3 \sin\alpha - \left(-\frac{C\cos\alpha}{A\sin\alpha}\omega_3 \right) \cos\alpha &= \Omega \\ \omega_3 \left(\sin\alpha + \frac{C\cos^2\alpha}{A\sin\alpha} \right) &= \Omega \\ \omega_3 \left(\frac{A\sin^2\alpha + C\cos^2\alpha}{A\sin\alpha} \right) &= \Omega. \end{aligned}$$

Dividing both numerator and denominator by AC:

$$\omega_3 = \frac{\frac{\Omega\sin\alpha}{C}}{\frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C}}.$$

From equation (4), the component ω_1 is

$$\omega_1 = -\frac{\frac{\Omega\cos\alpha}{A}}{\frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C}}.$$

Define the positive scalar constant

$$D = \frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C}.$$

The kinetic energy imparted to the cone is

$$\begin{aligned} T &= \frac{1}{2}A\omega_1^2 + \frac{1}{2}C\omega_3^2 \\ &= \frac{1}{2}A\left(-\frac{\Omega\cos\alpha}{AD} \right)^2 + \frac{1}{2}C\left(\frac{\Omega\sin\alpha}{CD} \right)^2 \\ &= \frac{1}{2}\frac{\Omega^2}{D^2}\left(\frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C} \right) \\ &= \frac{1}{2}\frac{\Omega^2}{D^2} \cdot D \\ &= \frac{1}{2}\frac{\Omega^2}{D}. \end{aligned}$$

Substituting $D = \frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C}$ gives the final closed form:

$$T = \frac{1}{2}\Omega^2 / \left\{ \frac{\cos^2\alpha}{A} + \frac{\sin^2\alpha}{C} \right\}.$$

■

Whittaker Chapter VI — Problem 45

Problem Statement

One end of an inelastic string is attached to a fixed point and the other to a point in the surface of a body of mass M. The body is allowed to fall freely under gravity without rotation. Show that just after the string becomes tight the loss of kinetic energy due to the impact is

$$\frac{1}{2}V^2 / \left(\frac{1}{M} + \frac{\lambda^2}{A} + \frac{\mu^2}{B} + \frac{\nu^2}{C} \right),$$

where V is the resolved velocity of the body in the direction of the string just before impact, the string only touching the body at the point of attachment, $(l, m, n, \lambda, \mu, \nu)$ are the coordinates of the string at the instant it becomes tight, and A, B, C are the principal moments of inertia of the body with respect to its principal axes at its centre of inertia.

1. Geometric Setup & Coordinate Definitions

Let G be the centre of inertia (centre of mass) of the rigid body of mass M. Choose a rectangular Cartesian coordinate system $Gxyz$ fixed in the body along its principal axes of inertia at G.

Inertial Parameters. Relative to $Gxyz$, the products of inertia vanish identically, and the principal moments of inertia are denoted by A, B, C.

Line Coordinates of the String. Let $P(x_1, y_1, z_1)$ be the point of attachment of the string on the surface of the body. At the instant the inelastic string becomes taut, let (l, m, n) be the direction cosines of the straight line of the string directed from P toward the fixed point of suspension, satisfying

$$l^2 + m^2 + n^2 = 1.$$

The moment of the directed line of the string about the origin G has scalar components (λ, μ, ν) given by

$$\begin{aligned} \lambda &= y_1 n - z_1 m, \\ \mu &= z_1 l - x_1 n, \\ \nu &= x_1 m - y_1 l. \end{aligned}$$

The six quantities $(l, m, n, \lambda, \mu, \nu)$ constitute the Plücker line coordinates of the string referred to the body-fixed frame at the instant of jerk.

State of Motion Before Impact. The body falls freely under gravity without rotation. Immediately before the string tightens:

- The angular velocity of the body is zero: $\omega_{10} = 0, \omega_{20} = 0, \omega_{30} = 0$.

- Every particle of the body shares the translational velocity (u_0, v_0, w_0) of the centre of mass G.
- The resolved velocity of the body in the direction of the string just before impact is

$$V = lu_0 + mv_0 + nw_0.$$

The initial kinetic energy of the body is purely translational:

$$T_0 = \frac{1}{2}M\left(u_0^2 + v_0^2 + w_0^2\right).$$

2. Lagrangian / Dynamical Derivation

Impulsive Equations of Motion. When the string jerks tight, an impulsive tension acts along the string. Let J be the scalar magnitude of this impulsive tension directed along (l, m, n) toward the fixed point of suspension. The components of the impulsive force acting on the body at the point of attachment P are

$$X = Jl, \qquad Y = Jm, \qquad Z = Jn.$$

The moments of this impulsive force about the principal axes at G are

$$\begin{aligned} L' &= y_1 Z - z_1 Y = J(y_1 n - z_1 m) = J\lambda, \\ M' &= z_1 X - x_1 Z = J(z_1 l - x_1 n) = J\mu, \\ N' &= x_1 Y - y_1 X = J(x_1 m - y_1 l) = J\nu. \end{aligned}$$

Let (u, v, w) be the velocity components of the centre of mass G immediately after impact, and let $(\omega_1, \omega_2, \omega_3)$ be the post-impact angular velocity components. Lagrange's equations of impulsive motion give:

1. For the translation of the centre of mass:

$$M(u - u_0) = Jl, \qquad M(v - v_0) = Jm, \qquad M(w - w_0) = Jn.$$

   Solving for the velocity components:

$$u = u_0 + \frac{Jl}{M}, \qquad v = v_0 + \frac{Jm}{M}, \qquad w = w_0 + \frac{Jn}{M}.$$

2. For the rotation about the principal axes at G:

$$A\omega_1 = J\lambda, \qquad B\omega_2 = J\mu, \qquad C\omega_3 = J\nu.$$

   Solving for the angular velocity components:

$$\omega_1 = \frac{J\lambda}{A}, \qquad \omega_2 = \frac{J\mu}{B}, \qquad \omega_3 = \frac{J\nu}{C}.$$

Condition of Inelasticity. Because the string is inelastic and its other end is fixed, the velocity component of the attachment point P along the direction of the string immediately after the jerk must vanish:

$$v'_P = lu_P + mv_P + nw_P = 0.$$

The velocity of the attachment point $P(x_1, y_1, z_1)$ after impact is

$$\begin{aligned} u_P &= u + z_1\omega_2 - y_1\omega_3, \\ v_P &= v + x_1\omega_3 - z_1\omega_1, \\ w_P &= w + y_1\omega_1 - x_1\omega_2. \end{aligned}$$

Projecting this velocity along the string direction (l, m, n):

$$\begin{aligned} v'_P &= lu + mv + nw + l(z_1\omega_2 - y_1\omega_3) + m(x_1\omega_3 - z_1\omega_1) + n(y_1\omega_1 - x_1\omega_2) \\ &= (lu + mv + nw) + \omega_1(mz_1 - ny_1) + \omega_2(nx_1 - lz_1) + \omega_3(ly_1 - mx_1) \\ &= (lu + mv + nw) - (\lambda\omega_1 + \mu\omega_2 + \nu\omega_3). \end{aligned}$$

Substitute the post-impact expressions for (u, v, w) and $(\omega_1, \omega_2, \omega_3)$:

$$\begin{aligned} lu + mv + nw &= l\left(u_0 + \frac{Jl}{M}\right) + m\left(v_0 + \frac{Jm}{M}\right) + n\left(w_0 + \frac{Jn}{M}\right) \\ &= (lu_0 + mv_0 + nw_0) + \frac{J}{M}(l^2 + m^2 + n^2) \\ &= V + \frac{J}{M}. \end{aligned}$$

Similarly:

$$\lambda\omega_1 + \mu\omega_2 + \nu\omega_3 = \lambda\left(\frac{J\lambda}{A}\right) + \mu\left(\frac{J\mu}{B}\right) + \nu\left(\frac{J\nu}{C}\right) = J\left(\frac{\lambda^2}{A} + \frac{\mu^2}{B} + \frac{\nu^2}{C}\right).$$

Since the impulsive tension opposes the stretch velocity, taking J with its algebraic sign relative to the motion gives

$$v'_P = V + J\left(\frac{1}{M} + \frac{\lambda^2}{A} + \frac{\mu^2}{B} + \frac{\nu^2}{C}\right) = 0.$$

Solving for the impulse J:

$$J = -\frac{V}{\frac{1}{M} + \frac{\lambda^2}{A} + \frac{\mu^2}{B} + \frac{\nu^2}{C}}.$$

3. Proof & Final Scalar Form

Post-Impact Kinetic Energy. The total kinetic energy of the body immediately after the impact is

$$\begin{aligned} T &= \frac{1}{2}M\left(u^2 + v^2 + w^2\right) + \frac{1}{2}\left(A\omega_1^2 + B\omega_2^2 + C\omega_3^2\right) \\ &= \frac{1}{2}M\left[\left(u_0 + \frac{Jl}{M}\right)^2 + \left(v_0 + \frac{Jm}{M}\right)^2 + \left(w_0 + \frac{Jn}{M}\right)^2\right] \\ &\quad + \frac{1}{2}\left[A\left(\frac{J\lambda}{A}\right)^2 + B\left(\frac{J\mu}{B}\right)^2 + C\left(\frac{J\nu}{C}\right)^2\right]. \end{aligned}$$

Expanding the translational terms:

$$\begin{aligned} \frac{1}{2}M\left(u^2 + v^2 + w^2\right) &= \frac{1}{2}M\left(u_0^2 + v_0^2 + w_0^2\right) + J(lu_0 + mv_0 + nw_0) + \frac{J^2}{2M}(l^2 + m^2 + n^2) \\ &= T_0 + JV + \frac{J^2}{2M}. \end{aligned}$$

Expanding the rotational terms:

$$\frac{1}{2}\left(A\omega_1^2 + B\omega_2^2 + C\omega_3^2\right) = \frac{1}{2}J^2\left(\frac{\lambda^2}{A} + \frac{\mu^2}{B} + \frac{\nu^2}{C}\right).$$

Summing both contributions:

$$T = T_0 + JV + \frac{1}{2}J^2\left(\frac{1}{M} + \frac{\lambda^2}{A} + \frac{\mu^2}{B} + \frac{\nu^2}{C}\right).$$

Evaluation of Kinetic Energy Loss. From the kinematic condition of inelasticity:

$$J\left(\frac{1}{M}+\frac{\lambda^2}{A}+\frac{\mu^2}{B}+\frac{\nu^2}{C}\right)=-V.$$

Substitute this relation into the quadratic term in T:

$$\frac{1}{2}J\left[J\left(\frac{1}{M}+\frac{\lambda^2}{A}+\frac{\mu^2}{B}+\frac{\nu^2}{C}\right)\right]=\frac{1}{2}J(-V)=-\frac{1}{2}JV.$$

The post-impact kinetic energy simplifies directly to

$$T=T_0+JV-\frac{1}{2}JV=T_0+\frac{1}{2}JV.$$

The loss of kinetic energy $\Delta E=T_0-T$ due to the impact is therefore

$$\Delta E=-\frac{1}{2}JV.$$

Substituting the explicit scalar expression for the impulse J:

$$\Delta E=-\frac{1}{2}\left(-\frac{V}{\frac{1}{M}+\frac{\lambda^2}{A}+\frac{\mu^2}{B}+\frac{\nu^2}{C}}\right)V=\frac{1}{2}V^2/\left(\frac{1}{M}+\frac{\lambda^2}{A}+\frac{\mu^2}{B}+\frac{\nu^2}{C}\right).$$

■